of natural ecosystems as a design for sustainability. It represents a shift in paradigm from conquering nature to becoming nature.

Ecological Engineering

Unlike industrial ecology, the focus of Ecological Engineering is on the manipulation of natural ecosystems by humans for our purposes, using small amounts of supplemental energy to control systems in which the main energy drives are still coming from non-human sources. It is the design of new ecosystems for human purposes, using the self-organizing principles of natural ecosystems.

Ecological Economics

Integrating ecology and economics in such a way that economic and environmental policies are reinforcing rather than mutually destructive.

Urban ecology

For ecologists, urban ecology is the study of ecology in urban areas, specifically the relationships, interactions, types and numbers of species found in urban habitats. Also, the design of sustainable cities, urban design programs that incorporate political, infrastructure and economic considerations.

Conservation Biology

The application of diverse fields and disciplines to the conservation of biological diversity. Restoration Biology: Application of ecosystem ecology to the restoration of deteriorated landscapes in an attempt to bring it back to its original state as much as possible. Example, prarie grass.

Landscape Ecology

“Landscape ecology is concerned with spatial patterns in the landscape and how they develop, with an emphasis on the role of disturbance, including human impacts” (Smith and Smith). It is a relatively new branch of ecology, that employs Global Information Systems. The goal is to predict the responses of different organisms to changes in landscape, to ultimately facilitate ecosystem management

Organizing principles

All these disciplines require an understanding of the "organizing principles" of ecosystems, i.e., their ecology. This involves the detailed study of the structure and function of ecosystems in their undisturbed state, and using their designs to: determine the resilience of ecosystem functions to human activities. design ecosystems which function in the service of human beings with minimal fossil energy input (ideally none) and minimal waste. design the industrial infrastructure. integrate the value of "goods and services" of natural ecosystems into the global economic system.

What is "Sustainability"?

There are many definitions of this one, depending on your perspective. Here’s ours: Sustainability is a property of a human society in which ecosystems (including humans) are managed such that the conditions supporting present day life on Earth can continue.

50 Levels of studying ecology

Biosphere: The earth’s ecosystem interacting with the physical environment as a whole to maintain a steady state system intermediate in the flow of energy between the high energy input of the sun and the thermal sink of space (merges with atmosphere, lithosphere, hydrosphere…).

Biome: Large scale areas of similar vegetation and climatic characteristics.

Ecosystem: Set of organisms and abiotic components connected by the exchange of matter and energy (forest, lake, coastal ocean). Or, “the smallest units that can sustain life in isolation from all but atmospheric surroundings.”

Community: Interacting populations which significantly affect each other’s distributions and abundance (intertidal, hot spring, wetland).

Population: Group of interacting and interbreeding organisms

- Cell/Organism
- Organelle
- Molecule
- Atom

Oxidation/reduction reactions

The energy that drives all life processes is organized around oxidation/reduction reactions. Ultimately on Earth today, oxygenic photosynthesis, and energy from the sun, fuels the entire biosphere. Oxygenic photosynthesis produces (by the splitting of water as a reducing agent) one of the most powerful oxidants known – oxygen. The biosphere on the contemporary Earth runs largely on the carbon produced by CO2 fixation by oxygenic photosynthesis, and on the free energy difference between O2 and organic carbon, which heterotrophs use to fuel their metabolism.

Autotrophs

The autotrophs synthesize glucose using solar or chemical energy, which is broken down through respiration (either their own or that of the organisms that eat them) to provide the energy necessary for "biological work". Redox reactions are central to all of these energy transformations, and the resulting flows of electrons manifest themselves, collectively, in the form of global biogeochemical cycles.

Role of bacteria

The activities of bacteria keep these cycles moving. For example, the chemosynthetic bacteria oxidize many essential elements in the process of getting the energy required to reduce CO2. Certain anaerobic

bacteria in turn reduce these compounds in the process of anaerobic respiration i.e., they use them as an electron acceptor in the absence of oxygen. This keeps the element cycles cycling maintaining balanced amounts of oxidants and reductants necessary for diverse metabolic processes. This keeps the system from "running down" energetically. Stay tuned for section on Biogeochemical Cycles.

51 Natural history

Lamarck

Began as a botanist; became the professor responsible for 'insects, worms and microscopic animals' at the museum of natural history in Paris. Introduced the vertebrate/invertebrate distinction.Distinguished insects, crustaceans and arachnids. Proposed the first really good system of invertebrates (1801). In that book, he proposed an evolutionary view of life. Proposed 'biology' as the title of the science of life.

Interpreted the ladder or chain of being as a real historical series, depicting the evolution of life.

The chain?

Lamarck realized life couldn't all be ordered in a single chain; we can only do this within each main group. But evolution works exactly that way for him– it 'pushes' living things towards higher and higher positions on the chain. So he needed something other than 'progressive evolution' to explain the variety of life– why it seems to have separate branches.

Branching

The branches arise from adaptation to local conditions. Local conditions alter what the animal does/ how it lives (habits, efforts, activity). This in turn lead it to acquire traits which can be passed on to offspring (when both parents possess them). Nothing new here– but as part of a theory of progressive evolution– that was new.

Needs

On Lamarck's account, it's the animal's felt needs that produce these changes. So evolution arises from an interaction between organism and environment. This gives us some notion of how adaptation can be explained in a natural way. (Reverses the argument from design: instead of the final causation story, 'Animals have trait x because x will serve their needs in that environment,' he says 'Because x will serve their needs, animals change and acquire x.' The environment causes the change– no need for design)

Extinction

For Lamarck, evolution was an alternative to extinction– instead of dying out (in most cases, anyway) ancient animals and plants evolved into different forms that continue today. Lamarck opposed Cuvier's catastrophism. But Cuvier could cite so many examples of extinction, and Lamarck couldn't say much about (for instance) which animals today were the descendents of trilobites, etc. Besides, Lamarck's evolutionary theory was not easy to understand or all that clear: 'sentiment interieure' seemed pretty mysterious...

Cuvier's advantage

Cuvier's anatomical and classificational work was impeccable. He identified four basic body plans for animals– vertebrates, molluscs, arthropods, and radiata (jellyfish & others). With these groups clearly distinguished, the chain of being was no longer part of biology. But links between distinct groups were still

striking– Lamarck cited teeth in fetal baleen whales, for instance, which St. Hilaire took as an instance of 'unity of plan'. A broader sort of unity could restore the chain to some respectability

Unity of plan

St. Hilaire explored the unity of parts and their patterns of development– skeletons in different vertebrates, etc. Established a strong unity of structure for the vertebrates that seemed to go far beyond the demands of Cuvier's correlation of parts. St. Hilaire tried to expand his unity to animals of different embranchements: vertebrates and mollusks. Cuvier was having none of it– his anatomical knowledge won the day for him again.

Von Baer

Development proceeds up the tree of life. All animals begin as eggs. They develop by multiplying, folding in various ways (patterns preserved throughout large groups of animals) to produce the organs, with details being gradually laid down as development continues. Development thus begins in the same way for each embranchement and only later identifies the particular class, family, genus & species of an organism (in that order).

Paleontology

Meanwhile, Smith, Buckland and Cuvier himself (along with Brongniart) were developing paleontology. The order and regular association of particular species of fossil plants and animals with particular formations was becoming well known. Older species differ substantially from known forms. More familiar forms arise gradually as we examine more recent fossils. The transitions were widely regarded as catastrophic (de Beaumont, for example). But Lyell turned the tide on this, showing that the claims of undeniable catastrophes in the record were much exaggerated.

Lyell's mistake

As a part of his general resistance to direction/ substantial change in geology, Lyell maintained there was no real direction of change in the fossil record either. This was greeted with pretty general astonishment, but Lyell claimed the apparent changes were just due to our having bad (misleading) samples…more mammals would soon appear in older formations, etc., as we extend our fossil collections. Within ten years (1840) it was clear that this was completely untenable.

Gradualism and species

Since the environment is steadily changing for Lyell, some plants and animals will become extinct because they're no longer adapted to the environment they're in (and cannot migrate, for some reason). Where do new species come from? This was a much harder question to answer– though Lyell thought it too would be piecemeal and gradual.

But adaptation?

Aside from Lamarck's somewhat vague and hard-to-test story of habits, effort and interior sentiment, we have no account here of how living things became adapted to so many different ways of life. Even Lamarck's account involves effort and needs, so it uses a sort of local final causation to explain how richer, more detailed final causes are worked out. The argument from design remains central here— although it doesn't explain the tree business (recall the whale's tooth).

52 Life in Water

Humans everywhere hold a land-centered perspective of the planet. Consequently, aquatic life is often most profuse where conditions appear most hostile to people, for example, along cold, wave-swept seacoasts, in torrential mountain streams, and in the murky waters where rivers meet the sea.

The hydrologic cycle exchanges water among reservoirs

Of the water in the biosphere, the oceans contain 97% and the polar ice caps and glaciers an additional 2%, leaving less than 1% as freshwater. The turnover of water in the various reservoirs of the hydrologic cycle ranges from only 9 days for the atmosphere to 3,100 years for the oceans.

Biology of aquatic environments

The biology of aquatic environments corresponds broadly to variations in physical factors such as light, temperature, and water movements and to chemical factors such as salinity and oxygen.

The oceans

The oceans form the largest continuous environment on earth. An ocean is generally divided vertically into several depth zones, each with a distinctive assemblage of marine organisms. Limited light penetration restricts photosynthetic organisms to the euphotic, or epipelagic, zone and leads to thermal stratification.

Oceanic temperatures are much more stable than terrestrial temperatures. Tropical seas are more stable physically and chemically; temperate and high-latitude seas are more productive. Highest productivity occurs along coastlines. The open ocean supports large numbers of species and is important to global carbon and oxygen budgets.

Kelp forests

Kelp forests are found mainly at temperate latitudes. Coral reefs are limited to the tropics and subtropics to latitudes between 30° N and S latitudes. Coral reefs are generally one of three types: fringing reefs, barrier reefs, and atolls. Kelp beds share several structural features with terrestrial forests. Both seaweeds and reef-building corals grow only in surface waters, where there is sufficient light to support photosynthesis. Kelp forests are generally limited to areas where temperature ranges from about 10°to 20°C, while reef-building corals are limited to areas with temperatures of about 18° to 29°C. The diversity and productivity of coral reefs rival that of tropical rain forests.

The intertidal zone

The intertidal zone lines the coastlines of the world. It can be divided into several vertical zones: the supratidal, high intertidal, middle intertidal, and low intertidal. The magnitude and timing of the tides is determined by the interaction of the gravitational effects of the sun and moon with the configuration of coastlines and basins. Tidal fluctuation produces steep gradients of physical and chemical conditions within the intertidal zone. Exposure to waves, bottom type, height in the intertidal zone, and biological interactions determine the distribution of most organisms within this zone. Salt marshes, mangrove forests, and estuaries occur at the transitions between freshwater and marine environments and between marine and terrestrial environments. Salt marshes, which are dominated by herbaceous vegetation, are found mainly at temperate and high latitudes. Mangrove forests grow in the tropics and subtropics. Estuaries are extremely dynamic physically, chemically, and biologically. The diversity of species is not as high in estuaries, salt marshes, and mangrove forests as in some other aquatic environments but productivity is exceptional.

Rivers and streams

Rivers and streams drain most of the land area of the earth and reflect the land use in their basins. Rivers and streams are very dynamic systems and can be divided into several distinctive environments: longitudinally, laterally, and vertically. The temperature of rivers follows variation in air temperature but does not reach the extremes occurring in terrestrial habitats. The flow and chemical characteristics of rivers change with climatic regime. Current speed, distance from headwaters, and the nature of bottom sediments are principal determinants of the distributions of stream organisms.

Lakes

Lakes are much like small seas. Most are found in regions worked over by tectonics, volcanism

and glacial activity, the geological forces that produce lake basins. A few lakes contain most of the freshwater in the biosphere. Lake structure parallels that of the oceans but on a much smaller scale. The salinity of lakes, which ranges from very dilute waters to over 200‰, is much more variable than that of the oceans. Lake stratification and mixing vary with latitude. Lake flora and fauna largely reflect geographic location and nutrient content

Potential threats to aquatic systems

Potential threats to all these aquatic systems include over-exploitation of populations and waste dumping. Freshwater environments are particularly vulnerable to the introduction of exotic species. The nature of fish assemblages is being used to assess the "biological integrity" of freshwater communities. The application of this Index of Biological Integrity depends on detailed knowledge of the natural history of regional fish faunas

53 temperature relations

Microclimates

Macroclimate interacts with the local landscape to produce microclimates .The sun's uneven heating of the earth's surface and earth's permanent tilt on its axis produce macroclimate. Macroclimate interacts with the local landscape--mainly altitude, aspect, vegetation, color of the ground, and small-scale structural features such as boulders and burrows--to produce microclimates. For the individual organism, macroclimate may be less significant than microclimate. The physical nature of water limits temperature variation in aquatic environments.

Temperature Ranges

Most species perform best in a fairly narrow range of temperatures. The influence of temperature on the performance of organisms begins at the molecular level, where extreme temperatures impair the functioning of enzymes. Rates of photosynthesis and bacterial activity generally peak in a narrow range of temperatures and are much lower outside of this optimal temperature range. How temperature affects the performance of organisms often corresponds to the current distributions of species and their evolutionary histories.

Regulation of body temperature

Many organisms have evolved ways to compensate for variations in environmental temperature by regulating body temperature. Temperature regulation balances heat gain against heat loss. Plants and ectothermicanimals use morphology and behavior to modify rates of heat exchange with the environment. Birds and mammals rely heavily on metabolic energy to regulate body temperature. The physical nature of the aquatic environment reduces the possibilities for temperature regulation by aquatic organisms. Most endothermic aquatic species are air breathers. Some organisms, mainly flying insects and some large marine fish, improve performance by selectively heating parts of their anatomy. The energetic requirements of thermoregulation may influence the geographic distribution of species.

Survival in extreme temperatures

Many organisms survive extreme temperatures by entering a resting stage. This stage may be as simple as resting in a sheltered spot during the heat of the day or may involve elaborate physiological adjustments. Hummingbirds may enter a state of torpor, a state of low metabolic rate and lowered body temperature, when food is scarce and night temperatures cold. Other animals can go into a state of reduced metabolism that may last several months. If this state occurs mainly in winter, it is called hibernation. If it occurs in summer, it is called estivation. Such reductions in metabolic rate allow these animals to survive extreme environmental conditions during which they must rely entirely on stored energy reserves.

Effects of habitat destruction and climatic warming

Long-term studies of populations of land snails around Basel, Switzerland, have documented local extinctions of these land snails. These extinctions are attributable to habitat destruction and climatic warming. The results of these studies suggest that climatic warming can lead to the local extinction of species. As we face the prospect of climatic warming at a global scale, studies of temperature relations will assume greater importance.

54 Water Relations

Water Availability

The movement of water down concentration gradients in terrestrial and aquatic environments determines the availability of water to organisms. The most familiar relative measure of the water content of air is relative humidity, defined as water vapor density divided by saturation water vapor density multiplied by l00. On land, the tendency of water to move from organisms to the atmosphere can be approximated by the vapor pressure deficit of the air. Vapor pressure deficit is calculated as the difference between the actual water vapor pressure and the saturation water vapor pressure.

Movement of water

In the aquatic environment, water moves down its concentration gradient, from solutions of higher water concentration and lower salt content (hypoosmotic) to solutions of lower water concentration and higher salt content (hyperosmotic). This movement of water creates osmotic pressure. Larger osmotic differences, between organism and environment, generate higher osmotic pressures. In the soil-plant system water flows from areas of higher water potential to areas of lower water potential. The water potential of pure water, which by convention is set at zero, is reduced by adding solute and by matric forces, the tendency of water to cling to soil particles and to plant cell walls. Typically, the water potential of plant fluids is determined by a combination of solute concentrations and matric forces, while the water potential of soils is determined mainly by matric forces. In saline soils, solutes may also influence soil water potential. Water potential, osmotic pressure, and vapor pressure deficit can all be measured in pascals (newtons/m2), a common currency for considering the water relations of diverse organisms in very different environments.

Regulation of internal water on terrain

- **Terrestrial plants and animals regulate their internal water by balancing water acquisition against water loss.** Water regulation by terrestrial animals is summarized by

$$W_{ia} = W_d + W_f + W_a - W_e - W_s,$$

where W_d = drinking, W_f = taken in with food, W_a = absorption from the air, W_e = evaporation and W_s = secretions and excretions. Water regulation by terrestrial plants is summarized by $W_{ip} = Wr + Wa - Wt - Ws$, where Wr = uptake by roots, Wa = absorption from the air, Wt = transpiration, and Ws = secretions and reproductive structures.

Survival in arid climates

Some very different terrestrial plants and animals, such as the camel and saguaro cactus, use similar mechanisms to survive in arid climates. Some organisms, such as scorpions and cicadas, use radically different mechanisms. Comparisons such as these suggest that natural selection is opportunistic

Regulation of internal water in marine environment

Marine and freshwater organisms use complementary mechanisms for water and salt regulation. Marine and freshwater organisms face exactly opposite osmotic challenges. Water regulation in aquatic environments is summarized by:

$$Wi = Wd - Ws + Wo,$$

where **Wd = drinking**, **Ws = secretions and excretions**, **Wo = osmosis**.

Survival in marine environment

An aquatic organism may either gain or lose water through osmosis, depending on the organism and the environment. Many marine invertebrates reduce their water regulation problems by being isosmotic with seawater. Some freshwater invertebrates also reduce the osmotic gradient between themselves and their environment. Sharks, skates, and rays elevate the urea and TMAO content of their body fluids to the point where they are slightly hyperosmotic to seawater Marine bony fish and saltwater mosquito larvae are hypoosmotic relative to their environments, while freshwater bony fish and freshwater mosquito larvae are hyperosmotic. While the strength of environmental challenge varies from one environment to another, and the details of water regulation vary from one organism to another, all organisms in all environments expend energy to maintain their internal pool of water and dissolved substances.

Influence of global-climate change

Scientists who are addressing the question of how global-climate change will influence the distribution of terrestrial vegetation have built a computerized model to predict change in terrestrial biomes across North America. Their model uses what we know about the physiology of major plant growth forms to predict the response of whole biomes. According to the model, climate, the combination of temperature and water availability, determines the distribution of vegetation at large scales, while topography, soils, and fires have local influences. The model gives ecologists a means to explore how various changes in climate may affect the future distributions of biomes.

55 Energy and Nutrient Relations

Three main sources of energy

Organisms use one of three main sources of energy:

- light,
- inorganic molecules, or
- organic molecules.

Photosynthetic plants and algae CO_2 as a source of carbon and light, of wavelengths between 400 and 700 nm, as a source of energy. Light within this band, which is called photosynthetically active radiation, or PAR, accounts for about 45% of the total energy content of the solar spectrum at sea level.

PAR quantification

PAR can be quantified as photosynthetic photon flux density, generally reported as µmol per square meter per second. Among plants, there are three major alternative photosynthetic pathways, C_3, C_4, and CAM. C_4 and CAM plants are more efficient in their use of water than are C_3plants.

Heterotrophs

Heterotrophs use organic molecules both as a source of carbon and as a source of energy. Herbivores, caunivores, and detritivores face fundamentally different trophic problems. Herbivores feed on plant tissues. which often contain a great deal of carbon but little nitrogen. Herbivores must also overcome the physical and chemical defenses of plants.

Detritivores

Detritivores feed on dead plant material, which is even lower in nitrogen than living plant tissues. Carnivores consume prey that are nutritionally rich but very well defended. Chemosynthetic autotrophs, which consist of a highly diverse group of chemosynthetic bacteria, use inorganic molecules as a source of energy. Bacteria are the most trophically diverse organisms in the biosphere.

Rate of uptake of energy

The rate at which organisms can take in energy is limited, either by external or internal constraints. The relationship between photon flux density and plant photosynthetic rate is called photosynthetic response. Herbs and short-lived perennial shrubs from sunny habitats have high maximum photosynthetic rates that level on at high light intensities. The lowest maximum rates of photosynthesis occur among plants from

shady environments. The relationship between food density and animal feeding rate is called the functional response. The shape of the functional response is generally one of three types. The forms of photosynthetic response curves and type 2 animal functional responses are remarkably similar. Energy limitation is a fundamental assumption of optimal foraging theory.

Optimal foraging theory

Optimal foraging theory attempts to model how organisms feed as an optimizin process.

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

Plants appear to allocate energy to roots versus shoots in a way that increases their rate of intake of the resources that limit their growth. Plants in environments with abundant nutrients but little light tend to invest more energy in the growth of stems and leaves and less in roots. In environments rich in light but poor in nutrients, plants tend to invest more energy in the growth of roots.

Trophic diversity of bacteria

The trophic diversity of bacteria, which is critical to the health of the biosphere, can also be used as a tool to address some of our most challenging waste disposal problems. Bacteria can be used to eliminate the huge quantities of sewage produced by human populations, clean up soils and aquifers polluted by petroleum products such as benzene, and eliminate the pollution caused by some kinds of mine waste.

Role of bacteria

The success of these projects requires that ecologists under- stand the energy and nutrient relations of bacteria. Bacteria will likely continue to play a great role as we address some of our most vexing environmental problems.

56 Social Relation

Social relations are important since they often directly and obviously impact the reproductive contribution of individuals to future generations, a key component of evolutionary fitness, the number of offspring, or genes, contributed by an individual to future generations. One of the most fundamental social interactions between individuals takes place during sexual reproduction.

Mate choice

Mate choice by one sex and/or competition for mates among individuals of the same sex can result in selection for particular traits in individuals, a process called sexual selection. Sexual selection results from differences in reproductive rates among individuals as a result of differences in their mating success. Sexual selection is thought to work either through intrasexual selection, where individuals of one sex compete with each other for mates, or intersexual selection, when members of one sex consistently choose mates from among members of the opposite sex on the basis of some particular trait.

The case of male guppies

Experimental evidence supports the hypothesis that the coloration of male guppies in local populations is determined by a dynamic interplay between natural selection exerted by predators, under which lesscolorful males have higher survival, and by female mate choice, which results in higher mating success by more-colorful males.

Scorpionflies

Among scorpionflies, larger males are more likely to successfully defend available arthropod offerings due to their advantages in aggressive encounters and consequently mate mom frequently than smaller males without arthropod offerings. Studies of mating in the wild radish, Raphanus sativus, in greenhouse and field experiments indicate nonrandom mating and suggest interference competition among pollen from different pollen donors.

Evolution of sociality

The evolution of sociality is generally accompanied by cooperative feeding, defense of the social group, and restricted reproductive opportunities. The degree of sociality in a social species ranges from acts as simple as mutual grooming or group protection of young to highly complex, stratified societies such as those found in colonies of ants or termites.

Eusociality

This more complex level of social behavior, which is considered to be the pinnacle of social evolution, is called eusociality. Eusociality is generally thought to include three major characteristics:

(1) individuals of more than one generation living together,
(2) cooperative care of young, and
(3) division of individuals into sterile, or nonreproductive, and reproductive castes.

Examples

Cooperation among green woodhoopoes and African lions appears to be a response to environmental conditions that require cooperation for success. For green woodhoopoes, the scarcity of high quality territories and intense competition between flocks for those territories create conditions that favor staying in the natal territory and helping raise related young and perhaps inheriting the territory at a later date.

African lions

To survive, reproduce, and successfully raise offspring to maturity, African lions must work in cooperative groups of females, which are called prides, and of males, which are called coalitions.

The comparative method for examining eusociality

One of the most valuable tools available to evolutionary biologists is the comparative method. The comparative method examines the characteristics of different species or populations of organisms in a way that attempts to isolate a particular variable or characteristic of interest, such as sociality, while randomizing the influence of confounding, or confusing, variables on the variable of interest. The comparative method has been used to study the evolution of eusociality among a wide variety of animal species including leafcutter ants and naked mole rats, both of which live in social groups in which individuals are divided among castes that engage in very different activities. Compared to other ant species, leafcutter ant colonies have a larger number of castes that engage in a wider variety of behaviors. In contrast to leafcutter ant colonies, where all workers are females, both males and females work in naked mole rat colonies. However, as in leafcutter ant colonies, work in naked mole rat colonies is divided among members according to their size. Many factors have likely contributed to the evolution of eusociality in leafcutter ants and naked mole rats, including kin selection and ecological constraints.

57 Population Genetics and Natural Selection

The rise of modem evolutionary ecology

Darwin and Mendel complemented each other well and their twin visions of the natural world revolutionized biology. The synthesis of the theory of natural selection and genetics gave rise to modem evolutionary ecology. Here we examine four major concepts within the area of population genetics and natural selection.

Genetic and phenotypic variation

Populations include genetic and phenotypic variation among individuals, which are significant aspects of population structure. The first biologists to conduct thorough studies of phenotypic and genotypic variation and to incorporate experiments in their studies, focused on plants. By using plants cloned from the same parent, **Bonnier** was able to control for the effects of genotype on plant form and observe the effects of climate.

The work of different Scientists

- **Kerner** showed that the differences in growth form observed by him and Bonnier were the result of temporary adjustments to environmental variation and not hereditary changes in the experimental plants.
- **Turesson's** studies provided evidence for genetic differences among populations.
- **Clausen, Keck, and Hiesey** explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.
- **Case** determined that the best predictor of chuckwalla, Sauromalus, body length was average winter rainfall.
- **Tracy's** laboratory growth experiments indicated that variation in body size among chuckwalla populations is at least partly determined by genetic differences among populations.

Hardy-Weinberg equilibrium model

The Hardy-Weinberg equilibrium model helps identify evolutionary forces that can change gene frequencies in populations. Because evolution involves changes in gene frequencies in a population, a thorough understanding of evolution must include the area of genetics known as population genetics.

Hardy-Weinberg principle

One of the most fundamental concepts in population genetics, the Hardy-Weinberg principle, states that "in a population mating at random in the absence of evolutionary forces, allele frequencies will remain constant."

For a population in Hardy-Weinberg equilibrium in a situation where there are only two alleles at a particular locus, **p + q = 1.0.** The frequency of genotypes in a population in Hardy-Weinberg equilibrium can be calculated as $\mathbf{(p + q)^2 = (p + q) \times (p + q) = p^2 + 2pq + q^2 = 1.0.}$

The conditions necessary to maintain constent allele frequencies in a population are:

(1) random mating,
(2) no mutations,
(3) large population size,
(4) no immigration, and
(5) equal survival and reproductive rates for all genotypes.

When a population is not in Hardy-Weinberg equilibrium, the Hardy-Weinberg principle helps us to identify the evolutionary forces that may be in play.

Genetic drift

Random processes such as genetic drift can change gene frequencies in populations, especially in small populations. Genetic drift is theoretically most effective at changing gene frequencies in small populations such as those that inhabit islands

Fragmentation of natural ecosystems

One of the greatest concerns associated with fragmentation of natural ecosystems due to human land use is that reducing habitat availability will decrease the size of animal and plant populations to the point where genetic drift will reduce the genetic diversity within natural populations.

The work of different Scientists

Ledig and his colleagues found a significant positive correlation between populafion size and genetic diversity in populations of Chihuahua spruce, a naturally fragmented population of trees living on mountain islands.

Frankham showed that compared to mainland populations, island populations generally include less genetic variation.

Saccheri and his colleagues found that higher heterozygosity (genetic diversity) was associated with lower rates of population extinction through the effects of heterozygosity on larval survival, adult longevity, and egg hatching in populations of the Glanville fritillary butterfly, Melitaea cinxia.

Natural selection

Natural selection, which changes genetic and phenotypic frequencies in populations, can result in adaptation to the environment. The most general postulate of the theory of natural selection is that the environment determines the evolution of the anatomy, physiology, and behaviour of organisms. Some of the clearest demonstrations of natural selection have resulted from studies of populations of Galfapagos finches. Losos, Warhelt, and Schoenerused replicated field experiments to study natural selection for changes in morphology in Anolis lizard populations.

The work of different Scientists

Their results indicate that colonizing populations can adapt rapidly to new environmental conditions.

Studies by Carroll and several colleagues show that soapberry bug populations living on native and introduced host plants have undergone natural selection for traits that favor their survival and reproduction on particular host plant species.

Natural selection

Hundreds of other examples of natural selection have been brought to light during the nearly one and a half century since Darwin published his theory. Still, evolutionary ecology remains a vigorous field of inquiry with plenty of debate, self criticism, and significant work yet to be done.

The earlier research on adaptation of populations to local environmental conditions used transplant experiments to detect genetic differences among populations. More recent research on genetic variation within and among populations has applied techniques developed in molecular biology. Ecologists now have many powerful tools, ranging from classical techniques to modem technologically sophisticated approaches, for assessing the extent of genetic variation within and among populations and meeting the challenge of documenting and conserving biodiversity.

58 Population Distribution and Abundance

Population

Ecologists define a population as a group of individuals of a single species inhabiting an area delimited by natural or human-imposed boundaries. Population studies hold the key to solving practical problems such as saving endangered species, controlling pest populations, or managing fish and game populations. All populations share a number of characteristics including Distribution and abundance.

Physical environment limits the geographic distribution of species

While there are few environments on earth without life, no single species can tolerate the full range of earth's environments. Because all species find some environments too warm, too cold, too saline, and so forth,

the physical environment limits the geographic distribution of species. For instance, there is a close relationship between climate and the distributions of the three largest kangaroos in Australia.

Examples

The tiger beetle Cicindela longilabris is limited to cool boreal and mountain environments. Large- and small-scale variation in temperature and moisture limits the distributions of certain desert plants, such as shrubs in the genus Encelia. However, differences in the physical environment only partially explain the distributions of barnacles within the marine intertidal zone, a reminder that biological factors constitute an important part of an organism's environment.

Distribution patterns of individuals

On small scales, individuals within populations are distributed in patterns that may be random, regular, or clumped. Patterns of distribution can be produced by the social interactions within populations, by the structure of the physical environment, or by a combination of the two. Social organisms tend to be clumped; territorial organisms tend to be regularly spaced.

Clumped distribution

An environment in which resources are patchy also fosters clumped distributions. Aggressive species of stingless bees live in regularly distributed colonies, while the colonies of nonaggressive species are randomly distributed. The distribution of creosote bushes changes as they grow.

On larger scales, individuals within a population are clumped. In North America, populations of both wintering and breeding birds are concentrated in a few hot spots of high population density. Clumped distributions are also shown by plant populations living along steep environmental gradients on mountainsides.

Population density vs increasing organism size

Population density declines with increasing organism size. In general, animal population density declines with increasing body size. This negative relationship holds for animals as varied as terrestrial invertebrates, aquatic invertebrates, birds, poikilothermic vertebrates, and herbivorous mammals. Plant population density also decreases with increasing plant size. However, the biological details underlying the size--density relationship shown by plants are quite different from those underlying the size--density patterns shown by animals. A single species of tree can span a very large range of sizes and densities during its life cycle. The largest trees start life as small seedlings that can live at very high population densities. As trees grow, their population density declines progressively until the mature trees live at low densities.

Influences on the Rarity

Rarity is influenced by geographic range, habitat tolerance, and population size. Rarity of species can be expressed as a combination of extensive versus restricted geographic range, broad versus narrow habitat tolerance, and large versus small population size. The most abundant species and those least threatened by extinction combine large geographic ranges, wide habitat tolerance, and high local population density. All other combinations of geographic range, habitat tolerance, and population size include one or more attributes of rarity. Rare species are vulnerable to extinction. Populations that combine restricted geographic range with narrow habitat tolerance and small population size are the rarest of the rare and are usually the organisms most vulnerable to extinction.

Abundance of organisms

The abundance of organisms and how abundance changes in time and space are among the most fundamental concerns of ecology. To estimate the abundance of species the ecologist must contend with a variety of practical challenges and conceptual subtleties. Mark and recapture methods are useful in the study of populations of active, elusive, or secretive animals.

Mark and recapture techniques

Mark and recapture techniques, which use natural distinguishing marks, are making an important contribution to the study of populations of endangered whales. Ecologists studying organisms, such as corals, algae, and sponges or many types of terrestrial plants that differ a great deal in size and form often estimate abundance as coverage, the area covered by a species. Patterns of distribution and abundance are ultimately determined by underlying population dynamics.

59 Population Dynamics

Survivorship curve

A survivorship curve summarizes the pattern of survival in a population. Patterns of survival can be determined either by following a cohort of individuals of similar age to produce a cohort life table or by determining the age at death of a large number of individuals or the age distribution of a population to produce a static life table.

Life table

Life tables can be used to draw survivorship curves, which generally fall into one of three categories:

(1) type I survivorship, in which there is low mortality among the young but high mortality among older individuals:
(2) type II survivorship, in which there is a fairly constant probability of mortality throughout life; and
(3) type III survivorship, in which there is high mortality among the young and low mortality among older individuals.

Age distribution of a population

The age distribution of a population reflects its history of survival, reproduction, and potential for future growth. Age distributions indicate periods of successful reproduction, high and low survival, and whether the older individuals in a population are replacing themselves or if the population is declining.

Population age structure

Population age structure may be highly complicated in variable environments, such as that of the Galapagos Islands. Populations in highly variable environments may reproduce episodically.

Net reproductive rate (R_0)

A life table combined with a fecundity schedule can be used to estimate net reproductive rate (R_0), geometric rate of increase (λ), generation time (T), and per capita rate of increase (r). Because these population parameters form the core of population dynamics, it is important to understand their derivation as well as their biological meaning.

Net reproductive rate

Net reproductive rate, R0, the average number of off-spring left by an individual in a population, is calculated by multiplying age-specific survivorship rates, l_x, times age-specific birthrates, m_x, and summing the results: $\sum l_x m_x$

Geometric rate of increase

The geometric rate of increase, λ, is calculated as the ratio of population sizes at two successive points in time, Generation time is calculated as: $\mathbf{T= \sum x l_x m_x / R_0}$

The per capita rate of increase

The per capita rate of increase, r, is related to generation time and net reproductive rate as: $\mathbf{r= \ln R_0 / T}$

The per capita rate of increase may be positive, zero, or negative depending on whether a population is growing, stable. Or declining.

Dispersal

Dispersal can increase or decrease local population densities. The contribution of dispersal to

local population density and dynamics is demonstrated by studies of expanding populations of species such as Africanized bees in the Americas and collared doves in Europe.

Effect of climate change

Climate changes can induce massive changes in the ranges of species. As availability of prey changes, predators may disperse, which increases and decreases their local population densities. Stream organisms actively migrating upstream or drifting downstream increase densities of stationary and migrating populations by immigrating and decrease them by emigrating.

Effects of pollutants

Ecologists are using the effects of pollutants on population dynamics to predict the potential ecological impact of these pollutants on populations. Good candidates for indicators of pollution are those aspects of population dynamics that are sensitive to environmental variation

Impact of a wide range of potential pollutants

Based on its environmental sensitivity, per capita rate of increase, r, appears to be an excellent predictor of the impact of a wide range of potential pollutants. The results of this research suggest that population processes and the mechanisms underlying variation in those processes can be used as sensitive predictors of the ecological effects of environmental change.

60 Population Growth

In the presence of abundant resources, populations can grow at geometric or exponential rates, Population growth by organisms with nonoverlapping generations can be described by the geometric model of population growth, Population growth that occurs as a continuous process, as in human or bacterial populations, can be described by the exponential model of population growth.

Examples of exponential growth

Examples of exponential growth from natural populations suggest that this type of growth may be very important to populations during establishment in new environments, during recovery from some form of exploitation, or during exploitation of transient, favorable conditions.

Logistic population growth

As resources are depleted, population growth rate slows and eventually stops; this is known as logistic population growth. As population size increases, population growth eventually slows and then ceases, producing a sigmoidal, or S-shaped, population growth curve. Population growth stops when populations reach a maximum size called the carrying capacity, the number of individuals of a particular population that the environment can support.

Sigmoidal population growth

Sigmoidal population growth can be modelled by the logistic growth equation, a modification of the exponential growth equation that includes a term for environmental resistance. In the logistic model, the rate of population growth decreases as population density increases. Research on laboratory populations indicates that zero population growth at carrying capacity may be attained by many combinations of reduced birthrates and increased death rates.

Effects of environmental limits

The environment limits population growth by changing birth and death rates. The factors affecting population size and growth include biotic factors such as food, disease, and predators and abiotic factors such as rainfall, floods, and temperature. Because the effects of biotic factors, such as disease and predation, ale often influenced by population density, biotic factors are often referred to as density dependent factors. Meanwhile, abiotic factors such as floods and extreme temperature can exert their influences independently of population density and so are often called density-independent factors. As we have already seen, both abiotic and biotic forces have important influences on populations. The significant effects of biotic and abiotic factors on populations have been well demonstrated by studies of Galapagos finches and their major food sources.

The intrinsic rate of increase(rm)

On average, small organisms have higher rates of per capita increase, r, and more variable populations, while large organisms have lower rate a of per capita increase and less variable populations. The intrinsic rate of increase, r_m is the maximum rate of increase for a given species. This rate of increase would occur in the absence of negative environmental influences. The per capita rate of increase, r, which is the realized rate of increase, is generally less than r_m One of the best predictors of r_m is body size. In general, rm decreases with increasing body size, from over l00,000 times from viruses to large vertebrate animals. The present state of the human population can be examined using the conceptual tools of population biology discussed in chapters 9 and 10 and in this chapter. Though humans live on every continent, their population density differs by several orders of magnitude in different regions.

Global population

In 1997, 61.4% of the global population, or about 3.5 billion people, were concentrated in Asia. The remainder of the human population was spread across Africa (12.8%), Europe (11.8%), North America (7.9%), South America (5.7%), Australia (0.3%), and oceanic islands (0.1%).

Sustainable global human population

Population densities in different regions vary from less than 1 person per square kilometer to nearly 1,000 persons per square kilometer. While the populations of some countries are stable, and some are declining, the global population is expected to continue growing well past the year 2000. One of the greatest environmental challenges of the twenty-first century will be to establish a sustainable global human population.

61 Competition

Competition, interactions between individuals that reduce their fitness, is generally divided into intraspecific competition, competition between individuals of the same species, and interspecific competition, competition between individuals of different species.

Forms of Competition

Competition can take the form of interference competition, direct aggressive interactions, or resource competition in which individuals compete through their dependence on the same limiting resources.

Intraspecific competition

Studies of intraspecific competition provide evidence for resource limitation. Experiments with herbaceous plants show that soil nutrients may limit plant growth rates and plant weight and that competition for nutrients increases with plant population density. Plants reflect their competition for resources, including water, light, and nutrients, through the process of self-thinning. Resource competition among leafhoppers also varies with population density and is reflected in reduced survivorship, smaller size, and increased developmental time at higher population densities. Experiments with terrestrial isopods show that even where there is no food shortage, intraspecific competition through interference may be substantial.

Niche

The niche reflects the environmental requirements of species. The niche concept was developed early in the history of ecology and has had a prominent place in the study of interspecific competition because of the competitive exclusion principle:

Same Niche, two species

Two species with identical niches cannot coexist indefinitely. Hutchinson added the concepts of the fundamental niche, the physical conditions under which a species might live in the absence of other species, and the realized niche, the more restricted conditions under which a species actually lives as the result of interactions with other species.While a species' niche is theoretically defined by a very large number of biotic and abiotic factors, Hutchinson's n-dimensional hypervolume, the most important attributes of the niche of most species, can often be summarized by a few variables. For instance, the niches of Galapagos finches are largely determined by their feeding requirements, while the niche of a salt marsh grass can be defined by tidal levels.

Lotka and Volterra

Mathematical and laboratory models provide a theoretical foundation for studying competitive interactions in nature. Lotka and Volterra independently expanded the logistic model of population growth to represent interspecific competition.

Lotka-Volterra competition model

In the Lotka-Volterra competition model, the growth rate of a species depends both upon numbers of conspecifics and numbers of the competing species. In this model, the effect of one species upon another is summarized by competition coefficients. In general, the Lotka-Volterra competition model predicts coexistence of species when interspecific competition is less intense than intraspecific competition. Competitive exclusion in laboratory experiments suggests the potential for competitive exclusion in nature. Even in the laboratory, however, organisms yield results that are much less predictable than the predictions of the Lotka-Volterra competition equations.

Ecological and evolutionary influences on the niches

Competition can have significant ecological and evolutionary influences on the niches of species. Field experiments involving organisms from herbaceous plants to desert rodents have demonstrated that competition can restrict the niches of species to a narrower set of conditions than they would otherwise occupy in the absence of competition.

Natural selection and niches

Theoretically, natural selection may lead to divergence in the niches of competing species, a situation called character displacement. However, stringent requirements for a definitive demonstration have limited the number of documented cases of character displacement in nature. After many decades of theoretical and

experimental work on competition, we can conclude that competition is a common and strong force operating in nature, but not always and not everywhere.

Field experiments

The field experiment is one of the most powerful and important tools at the disposal of the field ecologist.

However, the validity of field experiments depends upon several design features, including

1. a testable hypothesis,
2. knowledge of initial conditions,
3. controls, and
4. replication.

62 Exploitation

Predation, Herbivory, Parasitism, and Disease

Exploitation

The diversity of interactions between herbivores and plants, between predators and prey, and between parasites, parasitoids, pathogens, and hosts can be grouped under the heading of exploitation--interactions between species that enhance the fitness of one individual at the expense of another.

Exploitation weaves populations into a web of relationships that defy easy generalization.

The number of exploitative interactions between species far exceeds the number of species in the biosphere, and the nature of exploitation goes far beyond the typical consumption of one organism by another. For instance, many parasites and pathogens manipulate host behavior to enhance their own fitness at the expense of the host. Spiny-headed worms alter the behavior of a variety of crustacean hosts in a way that increases the probability that the one host species will be eaten by another. A pathogenic fungus manipulates the growth program of its host plant in a way to produce "pseudo flowers," structures aimed at promoting the reproduction of the pathogen. In the process the pathogen usually kills the host plant and always renders it sterile. Predation by one flour beetle species on another can be used as a potent means of interference competition except in the presence of a protozoan parasite, which seems to give a competitive advantage to less predaceous species.

Examples

Predators, parasites, and pathogens influence the distribution, abundance, and structure of prey and host populations. Herbivorous stream insects have been shown to control the density of their algal and bacterial food. The herbivorous moth larva *Cactoblastis cactorum* combined with pathogenic microbes reduced the coverage of prickly pear cactus in Australia from millions of hectares to a few thousand. A parasitic infestation reduced the red fox population in Sweden by 70%, which in turn led to increases in the abundance of several prey species eaten by foxes. This parasitic disease revealed the influence of a predator on its prey populations.

Relationships

Predator-prey, parasite-host, and pathogen-host relationships are dynamic. Populations of a wide variety of predators and prey show highly dynamic fluctuations in abundance ranging from days to decades. A particularly well-studied example of predator-prey cycles is that of snowshoe hares and their predators, which have been shown to result from the combined effects of the snowshoe hares on the quantity and quality of their food and of the predators on the snowshoe hare population.

Predator-prey interactions

Mathematical models of predator-prey interactions by Lotka and Volterra suggest that exploitative interactions themselves can produce population cycles without any influences from outside forces such as weather. Predator-prey cycles have also been observed in a few laboratory populations under restricted circumstances.

To persist in the face of exploitation, hosts and prey need refuges.

The refuges that promote the persistence of hosts and prey include secure places to which the exploiter has limited access. However, living in large groups can be considered as a kind of refuge since it reduces the probability that an individual host or prey will be attacked. It appears that predator satiation is a defensive tactic used by a wide variety of organisms from rain forest trees to temperate insects. Growing to large size can also represent a kind of refuge when the prey species is faced by size-selective predators, Size is used as a refuge by prey species ranging from stream insects and intertidal invertebrates to rhinoceros.

Population control

Predators and parasites have been used to control populations of insects that attack crops or to control invasive weeds. Recent research in Kenya has shown that a crayfish, Procambarus clarkii, controls the snails that act as intermediate hosts for Schistosoma. ahighly pathogenic human parasite. Preliminary results indicate that crayfish successfully control host snails in the artificial impoundments used for livestock watering and domestic water, important sources of infection by Schistosoma.

63 Mutualism

Mutualism

Mutualism, interactions between individuals that benefit both partners, is a common phenomenon in nature that has apparently made important contributions to the evolutionary history of life and continues to make key contributions to the ecological integrity of the biosphere.

Mutualisms types

Mutualisms can be divided into those that are facultative, where species can live without their mutualistic partners, and obligate, where species are so dependent on the mutualistic relationship that they cannot live without their mutualistic partners

Examples

Plants benefit from mutualistie partnerships with a wide variety of bacteria, fungi, and animals. Mutualism provides benefits to plants ranging from nitrogen fixation and enhanced nutrient and water uptake to pollination and seed dispersal.

Mycorrhiza

Ninety percent of terrestrial plants form mutualistic relationships with mycorrhizal fungi, which make substantial contributions to plant performance. Mycorrhizae, which are mostly either vesicular arbuscular mycorrhizae or ectomycorrhizae, are important in increasing plant access to water, nitrogen, phosphorus, and other nutrients. In return for these nutrients, mycorrhizae receive energy-rich root exudates. Experiments have shown that the mutualistic balance sheet between plants and mycorrhizal fungi can be altered by the availability of nutrients. Plant-ant protection mutualisms are found in both tropical and temperate environments.

Ants

In tropical environments, many plants provide ants with food and shelter in exchange for protection from a variety of natural enemies. In temperate environments, mutualistic plants provide ants with food but not shelter in trade for protection

Mutualistic algae

Mutualistic algae called zooxanthellae provide reef-building corals with their principal energy source; in exchange for this energy, corals provide zooxanthellae with nutrients, especially nitrogen, a scarce resource in tropical seas.

Mutualism between corals

Reef-building corals depend upon mutualistic relationships with algae and animals. The coral-centered mutualisms of tropical seas show striking parallels with terrestrial plant-centered mutualisms. The mutualism between corals and zooxanthellae appears to be largely under the control of the coral partner, which chemically solicits the release of organic compounds from zooxanthellae and controls zooxanthellae population growth. Crabs and shrimp protect some coral species from coral predators in exchange for food and shelter.

Evolution of Mutualism

Theory predicts that mutualism will evolve where the benefits of mutualism exceed the costs. **Keeler** built a cost-benefit model for the evolution and persistence of facultative plant-ant protection mutualisms in which the benefits of the mutualism to the plant are represented in terms of the proportion of the plant's energy budget that ants protect from damage by herbivores.

The model assesses the costs of the mutualism to the plant in terms of the proportion of the plant's energy budget invested in extrafloral nectaries and the water, carbohydrates, and amino acids contained in the nectar. The model predicts that the mutualism will be favored where there are high densities of ants and potential herbivores and where the effectiveness of alternative defenses are low.

Humans in Mutualism

Humans have developed a variety of mutualistic relationships with other species, but one of the most spectacular is that between the greater honeyguide and the traditional honey gatherers of Africa. In this apparently ancient mutualism, humans and honeyguides engage in elaborate communication and

cooperation with clear benefit to both partners. The mutualism offers the human side a higher rate of discovery of bees' nests, while the honeyguide gains access to nests that it could not raid without human help. Careful observations have documented that the honeyguide informs the honey gatherers of the direction and distance to bees' nests as well as of their arrival at the nest.

64 Specie Abundance and Diversity

Community

A community is an association of interacting species inhabiting some defined area. Examples of communities include the plant community on a mountainside, the insect community associated with a particular species of tree, or the fish community on a coral reef. Community ecologists often restrict their studies to groups of species that all make their living in a similar way. Animal ecologists call such groups guilds, while plant ecologists use the term life-form. The field of community ecology concerns how the environment influences community structure, including the relative abundance and diversity of species, the subjects of this chapter.

Abundance

Most species are moderately abundant; few are very abundant or extremely rare. Frank Preston (1948) graphed the abundance of species in collections as distributions of species abundance, with each abundance interval twice the preceding one. Preston's graphs were approximately "bell-shaped" curves and are called "lognormal" distributions.

Lognormal distributions

Lognormal distributions, which describe the relative abundance of organisms ranging from algae and terrestrial plants to birds, may result from many random environmental variables acting upon the populations of a large number of species or may be a consequence of how species subdivide resources. Regardless of the underlying mechanisms, the lognormal distribution is one of the best described patterns in community ecology.

Species diversity

A combination of the number of species and their relative abundance defines species diversity.

Two major factors define the diversity of a community:

(1) the number of species in the community, which ecologists usually call species richness, and
(2) the relative abundance of species, or species evenness.

Shannon-Wiener index

One of the most commonly applied indices of species diversity is the Shannon-Wiener index:

$$H' = -\sum_{i=1}^{s} p_i \log_e p_i$$

The relative abundance and diversity of species can also be portrayed using rank-abundance curves.

Species diversity in complex envirnments

Species diversity is higher in complex environments. Robert MacArthur (1958) discovered that five coexisting warbler species feed in different layers of forest vegetation and that the number of warbler species in North American forests increases with increasing forest stature. Various investigators have found that the diversity of forest birds increases with increased foliage height diversity. The niches of algae can be defined by their nutrient requirements.

Heterogeneity

Heterogeneity in physical and chemical conditions across aquatic and terrestrial environments can account for a significant portion of the diversity among planktonic algae and terrestrial plants. Soil characteristics and depth to groundwater strongly influence the nature of local plant communities in the Amazon River basin. Increased nutrient availability correlates with reduced algal and plant diversity.

Disturbance

Intermediate levels of disturbance promote higher diversity. Joseph Connell (1975, 1978) proposed that high diversity is a consequence of continually changing conditions, not of competitive accommodation at equilibrium. He predicted that intermediate levels of disturbance would foster higher levels of diversity. At intermediate levels of disturbance, a wide array of species can colonize open habitats, but there is not enough time for the most effective competitors to exclude the other species

Work of Sousa

Wayne Sousa (1979), who studied the effects of disturbance on the diversity of sessile marine algae and invertebrates growing on intertidal boulders, found support for the intermediate disturbance hypothesis. Diversity in prairie vegetation also appears to be higher in areas receiving intermediate levels of disturbance. The effect of disturbance on diversity appears to depend upon a trade-off between dispersal and competitive abilities.

Human disturbance

Human disturbance is an ancient feature of the biosphere. Human influences touch every portion of the biosphere and have done so for thousands of years. For instance, humans disturbed tropical rain forests in Central America beginning about 11,000 years ago.

Intermediate disturbance hypothesis

The effects of human disturbance fall within the framework of the intermediate disturbance hypothesis. Though intense human disturbance reduces species diversity, moderate levels of disturbance may increase the diversity of some communities such as the European chalk grasslands.

65 Food Webs

A food web summarizes the feeding relations in a community. The earliest work on food webs concentrated on simplified communities in areas such as the Arctic islands. However, researchers such as Charles Elton (1927) soon found that even these so-called simple communities included very complex feeding relations. The level of food web complexity increased substantially, however, as researchers began to study complex communities. Studies of the food webs of tropical freshwater fish communities revealed highly complex networks of trophic interaction that persisted even in the face of various simplifications. A focus on strong interactions can simplify food web structure and identify those interactions responsible for most of the energy flow in communities.

Food web and structure of communities

The feeding activities of a few keystone species may control the structure of communities. Robert Paine (1966) proposed that the feeding activities of a few species have inordinate influences on community structure. He predicted that some predators may increase species diversity by reducing the probability of competitive exclusion. Manipulative studies of predaceous species have identified many keystone species, including starfish and snails in the marine intertidal zone and fish in rivers. On land, birds exert substantial influences on communities of their arthropod prey.

Jane Lubchenko (1978)

Jane Lubchenko (1978) demonstrated that the influence of consumers on community structure depends upon their feeding preferences, their local population density, and the relative competitive abilities of prey species. Keystone species are those that, despite low biomass, exert strong effects on the structure of the communities they inhabit.

Exotic predators and the structure of food webs

Exotic predators can collapse and simplify the structure of food webs. Introduced fishes have devastated the native fishes of Lake Atitlan and Gatun Lake in Central America.

Examples

The influence of the Nile perch on the fish community of Lake Victoria is enmeshed with massive changes in the lake's ecosystem.Introduction of the Nile perch is rapidly reducing the species-rich fish fauna of Lake Victoria to a community dominated by a handful of species.

Humans

Humans have acted as keystone species in communities. People have long manipulated food webs both as a consequence of their own feeding activities and by introducing or deleting species from existing food webs. In addition, many of these manipulations have been focused on keystone species. Hunters in tropical rain forests have been responsible for removing keystone animal species from large areas of the rain forests of Central and South America. Chinese farmers have used ants as keystone predators to control pests in citrus orchards for over 1,700 years.

66 Primary Production and Energy Flow

Flow of energy

We can view a forest, a stream, or an ocean as a system that absorbs, transforms, and stores energy. In this view, physical, chemical, and biological structures and processes are inseparable. When we look at natural

systems in this way we view them as ecosystems. An ecosystem is a biological community plus all of the abiotic factors influencing that community.

Production

- **Primary production**, the fixation of energy by autotrophs, is one of the most important ecosystem processes.
- The **rate of primary production** is the amount of energy fixed over some interval of time.
- **Gross primary production** is the total amount of energy fixed by all the autotrophs in the ecosystem. Net primary production is the amount of energy left over after autotrophs have met their own energetic needs.

Terrestrial primary production

Terrestrial primary production is generally limited by temperature and moisture. The variables most highly correlated with variation in terrestrial primary production are temperature and moisture. Highest rates of terrestrial primary production occur under warm, moist conditions.

Annual Actual Evapo Transpiration (AET)

Temperature and moisture conditions can be combined in a single measure called annual actual evapotranspiration, or AET, which is the total amount of water that evaporates and transpires off a landscape during the course of a year. Annual AET is positively correlated with net primary production in terrestrial ecosystems. However, significant variation in terrestrial primary production results from differences in soil fertility.

Aquatic primary production

Aquatic primary production is generally limited by nutrient availability. One of the best documented patterns in the biosphere is the positive relationship between nutrient availability and rate of primary production in aquatic ecosystems.

Phosphorus concentration usually limits rates of primary production in freshwater ecosystems, while nitrogen concentration usually limits rates of marine primary production.

Exmaples

- Consumers can influence rates of primary production in aquatic and terrestrial ecosystems. Piscivorous fish can indirectly reduce rates of primary production in lakes by reducing the density of plankton-feeding fish..
- Reduced density of planktivorous fish can lead to increased density of herbivorous zooplankton, which can reduce the densities of phytoplankton and rates of primary production. Intense grazing by large mammalian herbivores on the Serengeti increases annual net primary production by inducing compensatory growth in grasses

Energy losses

Energy losses limit the number of trophic levels in ecosystems. Ecosystem ecologists have simplified the trophic structure of ecosystems by arranging species into trophic levels based upon the predominant source of their nutrition. A trophic level is determined by the number of transfers of energy from primary producers to that level.

Energy losses and trophic levels

As energy is transferred from one trophic level to another, energy is lost due to limited assimilation, respiration by consumers, and heat production. As a result of these losses, the quantity of energy in an ecosystem decreases with each successive trophic level, forming a pyramid-shaped distribution of energy among trophic levels. As losses between trophic levels accumulate, eventually there is insufficient energy to support a viable population at a higher trophic level.

Stable isotope analysis

Stable isotope analysis can be used to trace the flow of energy through ecosystems. The ratios of different stable isotopes of important elements such as nitrogen and carbon are generally different in different parts of ecosystems.

Use of Stable isotope analysis

As a consequence, ecologists can use isotopic ratios to study the trophic structure and energy flow through ecosystems. **Stable isotope analysis** has helped **quantify dietary composition of wild populations** and the major sources of energy used by prehistoric human populations.

67 Nutrient Cycling and Retention

Nutrient cycling

The elements organisms require for development, maintenance, and reproduction are called nutrients. Ecologists refer to the use, transformation, movement, and reuse of nutrients in ecosystems as nutrient cycling.

Nutrient cycling is one of the most ecologically significant processes studied by ecosystem ecologists. The carbon, nitrogen, and phosphorus cycles have played especially prominent roles in studies of nutrient cycling.

Decomposition rate

Decomposition rate is influenced by temperature, moisture, and chemical composition of litter and the environment. The rate of decomposition affects the rate at which nutrients, such as nitrogen and phosphorus, are made available to primary producers. Rates of decomposition in terrestrial ecosystems are higher under warm, moist conditions. The rate of decomposition in terrestrial ecosystems increases with nitrogen content and decreases with the lignin content of litter. The chemical composition of litter and the availability of nutrients in the surrounding environment also influence rates of decomposition in aquatic ecosystems.

Cycling of nutrients in ecosystems

- Plants and animals can modify the distribution and cycling of nutrients in ecosystems.
- The dynamics of nutrients in streams are best represented by a spiral rather than a cycle.
- The length of stream required for an atom of a nutrient to complete a cycle is called the spiraling length.
- Stream macroinvertebrates can substantially reduce spiralling length of nutrients in stream ecosystems.
- Animals can also alter the distribution and rate of nutrient cycling in terrestrial ecosystems. Nitrogen-fixing plants increase the quantity and rates of nitrogen cycling in terrestrial ecosystems.

Disturbance and nutrient loss from ecosystems

Disturbance increases nutrient loss from ecosystems. Vegetation exerts substantial control on nutrient retention by terrestrial ecosystems. Vegetative controls on nutrient loss from forest ecosystems appear to be most important in environments that are warm and moist during the growing season

Nutrient enrichment by humans

Vegetative controls appear to be less important in cold and/or dry environments. Nutrient loss by stream ecosystems is highly pulsed and associated with disturbance by flooding. Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems. Nitrate concentration and export by the earth's major rivers correlate directly with human population density. Human disturbance also increases export of phosphorus from aquatic catchments. Nutrient enrichment appears to be reducing the diversity of plants and fungi in terrestrial ecosystems. Land managers around the world use nutrient loading models to predict and manage the impact of land use on aquatic ecosystems.

68 Succession and Stability

Primary succession

Succession is the gradual change in plant and animal communities in an area following disturbance

or the creation of new substrate. Primary succession occurs on newly exposed geological substrates not significantly modified by organisms.

Secondary succession

Secondary succession occurs in areas where disturbance destroys a community without destroying the soil. Succession generally ends with a climax community whose populations remain stable until disrupted by disturbance.

Community changes during succession

Community changes during succession include increases in species diversity and changes in species composition. Primary forest succession around Glacier Bay may require about 1,500 years, while secondary forest succession on the Piedmont Plateau takes about 150 years.

Succession duration

Meanwhile, succession in the intertidal zone requires 1 to 3 years and succession within a desert stream occurs in less than 2 months. Despite the great differences in the time required, all these Successional sequences show increased species diversity over time.

Ecosystem changes during succession

Ecosystem changes during succession include increases in biomass, primary production, respiration, and nutrient retention. Succession at Glacier Bay produces changes in several ecosystem properties, including increased soil depth, organic content, moisture, and nitrogen. Over the same successional sequence, several soil properties show decreases, including soil bulk density, pH, and phosphorus concentration. Succession at the Hubbard Brook Experimental Forest increased nutrient retention by the forest ecosystem.

Predictability of succession

Several ecosystem properties change predictably during succession in Sycamore Creek, Arizona, including biomass, primary production, respiration, and nitrogen retention.

Mechanisms of succession

Mechanisms that drive ecological succession include facilitation, tolerance, and inhibition.
Most studies of succession support the facilitation model, the inhibition model, or some combination of the two.
Both facilitation and inhibition occur during intertidal succession. Facilitation and inhibition also occur during secondary and primary forest succession.

Community stability

Community stability may be due to lack of disturbance or community resistance or resilience in the face of disturbance. Ecologists generally define stability as the persistence of a community or ecosystem in the face of disturbance. Resistance is the ability of a community or ecosystem to maintain structure and/or function in the face of potential disturbance.

Resilience

The ability to bounce back after disturbance is called resilience. A resilient community or ecosystem may be completely disrupted by disturbance but quickly return to its former state. Studies of the Park Grass Experiment suggest that our perception of stability is affected by the scale of measurement. Studies in Sycamore Creek indicate that resilience is sometimes influenced by resource availability and that resistance may result from landscape-level phenomena

Repeat photography

Repeat photography can be used to detect long-term ecological change. Most successional sequences and most community and ecosystem responses to climatic change take place over very long periods of time. Repeat photography has become a valuable tool to help ecologists study these long-term changes.

69 Landscape Ecology

Landscape:

A landscape is a heterogeneous area composed of several ecosystems. The ecosystems making up a landscape generally form a mosaic of visually distinctive patches. These patches are called landscape elements. Landscape ecology is the study of landscape structure and processes.

Landscape structure

Landscape structure includes the size, shape, composition, number, and position of different ecosystems within a landscape. Most questions in landscape ecology require that ecologists quantify landscape structure. Until recently, however, geometry, which means “earth measurement,” could offer only rough approximations of complex landscape structure.

Structure of complex natural shapes

Today, an area of mathematics called fractal geometry can be used to quantify the structure of complex natural shapes. One of the findings of fractal geometry is that the length of the perimeter of complex shapes depends upon the size of the device used to measure the perimeter. One implication of this result is that organisms of different sizes may use the environment in very different ways

Landscape structure influences Ecological processes

Landscape structure influences processes such as the flow of energy, materials, and species between the ecosystems within a landscape. Landscape ecologists have proposed that landscape structure, especially the size, number, and isolation of habitat patches, can influence the movement of organisms between potentially suitable habitats. The group of subpopulations living on such habitat patches make up a metapopulation.

Examples

- Studies of the movements of small mammals in a prairie landscape show that a smaller proportion of individuals moves in more fragmented landscape but that the individuals that do move will move farther. The local population density of the Glanville fritillary butterfly, ***Melitaea cinxia***, is lower on larger and on isolated habitat patches. Small populations of this butterfly and desert bighorn sheep are more vulnerable to local extinction.
- The source of water for lakes in a Wisconsin lake district is determined by their positions in the landscape, which in turn determine their hydrologic and chemical responses to drought.

Landscapes are structured and dynamic

Landscapes are structured and change in response to geological processes, climate, activities of organisms, and fire.

Geological features produced by processes such as volcanism, sedimentation, and erosion interact with climate to provide a primary source of landscape structure. In the Sonoran Desert, plant distributions map clearly onto soils of different ages and form a vegetative mosaic that closely matches soil mosaics. This mosaic will gradually shift as geological processes and climate gradually change the soil mosaic. While geological processes and climate set the basic template for landscape structure, the activities of organisms, from plants to elephants, can be an additional source of landscape structure and change.

Human activity changes the structure of landscapes

Economically motivated human activity changes the structure of landscapes all over the globe. Beavers can quickly change landscape structure and processes over large regions. Fire contributes to the structure of landscapes ranging from tropical savanna to boreal forest. However, fire plays a particularly prominent role in regions with a Mediterranean climate. Because human activity has often altered landscape structure and processes in undesirable ways, there is growing pressure and interest in landscape restoration. Some of the most ambitious current restoration efforts focus on the restoration of riverine landscapes. Rivers and their floodplains form a complex, highly dynamic landscape that includes river riparian forest, marsh, oxbow lake, and wet meadow' ecosystems.

Water management and landscapes

Over the past century, water management by building dams, channelizing rivers, constructing flood levees, and diverting water for irrigation has cut the historic connections between most rivers and their floodplains. The restoration of the Kissimmee River landscape is also one of the largest ecological experiments ever conducted and will be a significant test of the predictive ability of ecological theory.

70 Geographic Ecology

Geographic ecology focuses on large-scale patterns of distribution and diversity of organisms, such as island biogeography, latitudinal patterns of species diversity, and the influences of large-scale regional and historical processes on biological diversity.

Isolation

On islands and habitat patches on continents, species richness increases with area and decreases

with isolation. Larger oceanic islands support more species of most groups of organisms than small islands. Isolated oceanic islands generally contain fewer species than islands near mainland areas. Many habitats on continents are so isolated that they can be considered as islands.

Species richness on habitat islands

Species richness on habitat islands, such as mountain islands in the American Southwest, increases with area and decreases with isolation. Lakes can also be considered as habitat islands. They are aquatic environments isolated from other aquatic environments by land. Fish species richness generally increases with lake area. Species richness is usually negatively correlated with island isolation. However, because organisms differ substantially in dispersal rates, an island that is very isolated for one group of organisms may be completely accessible to another group.

Models of Species richness on islands

Species richness on islands can be modelled as a dynamic balance between immigration and extinction of species.

The equilibrium model of island biogeography proposes that the difference between rates of immigration and extinction determines the species richness on islands.

The equilibrium model of island biogeography

The equilibrium model of island biogeography assumes that rates of species immigration to islands are mainly determined by distance from sources of immigrants. The model assumes that rates of extinction on islands are determined mainly by island size.

Species richness and latitude

Species richness generally increases from middle and high latitudes to the equator. Most groups of organisms are more species-rich in the tropics.

Many factors may contribute to higher tropical species richness, including

(1) time since perturbation,
(2) productivity,
(3) environmental heterogeneity,
(4) favorableness,
(5) niche breadths and interspecific interactions, and
(6) differences in speciation and extinction rates. Several lines of evidence support the hypothesis that differences in surface area play a primary role in determining latitudinal gradients in species richness

Structure of biotas and ecosystems

Long-term historical and regional processes significantly influence the structure of biotas and ecosystems. Much geographic variation in species richness can be explained by historical and regional processes.

Exceptional situations

Some exceptional situations that seem to have resulted from unique historical and regional processes include the exceptional species richness of the Cape floristic region of South Africa, the high species richness of temperate trees in east Asia, and the low bird diversity in beech forests of South America.

A word of advise

The ecologist interested in understanding large-scale patterns of species richness must consider processes occurring over similarly large scales and over long periods of time.

Global positioning systems

Global positioning systems, remote sensing, and geographic information systems are important tools for effective geographic ecology. A global positioning system determines locations on the earth's surface, including latitude, longitude, and altitude, using satellites as reference points.

Remote sensing satellites

Remote sensing satellites are generally fitted with electro-optical sensors that scan several bands of the electromagnetic spectrum. These sensors convert electromagnetic radiation into electrical signals that are in turn converted to digital values by a computer. These digital values can be used to construct an image.

Geographic information systems

Geographic information systems are computer-based systems that store, analyze, and display geographic information. Global positioning systems, remote sensing, and geographic information systems are increasingly valuable parts of the ecologist's tool kit.

Future avenues

Ecologists are using these new tools to study large-scale, dynamic ecological phenomena such as interannual variation in regional terrestrial primary production, dynamics of marine primary production, and potential population responses to climate change.

71-Competition

13.2 Failure of species to coexist in laboratory cultures led to the competitive exclusion principle

Competitive exclusion principle

- Principle: Complete competitors can not coexist. One species must go extinction
- Complete competitors: two species that live in the same place and possess exactly the same ecological requirements.
- Assumptions:
  - Exactly the same resource requirement (no more, no less)
  - Environmental conditions remain constant

72-Competition

13.4 Habitat productivity can influence competition between plant species

Two hypotheses:

1. Plants compete more intensively when mineral nutrients are less abundant in the soil (By Grubb and Tilman)

Plants compete more intensively when nutrients are less. High nutrients are less likely to limit plant population; thus the intraspecific competition is weak.

1. Competition is less intense when water and nutrients are less abundant (Grime and Keddy)

13.4 Habitat productivity can influence competition between plant species

Competition for light is more important than competition for nutrients; limit in water and nutrients would limit the population growth to a certain point that individual plants are widely spread and do not compete for light.

Difference between these hypotheses lies in the relative importance placed on belowground and aboveground competition for resources --Light or nutrient. (Debate)

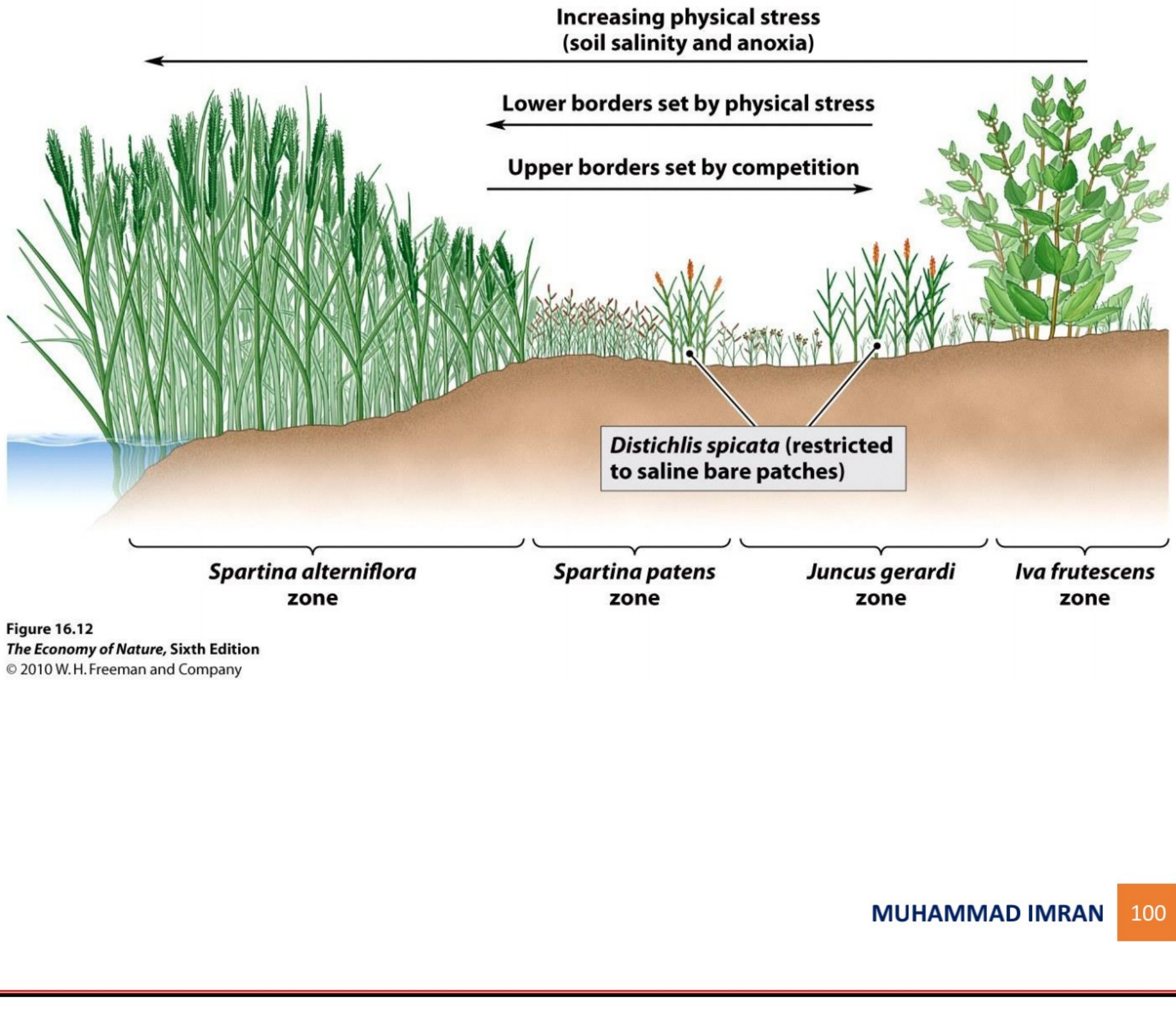

Figure 16.12
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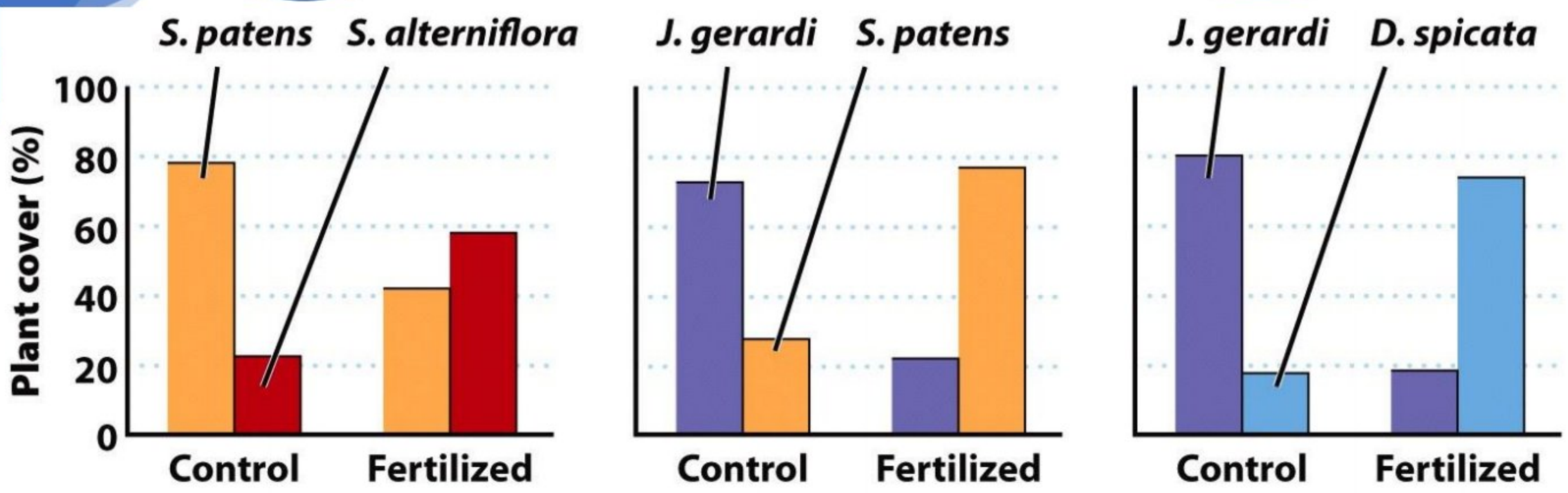

Figure 16.13
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Fertilization alters the outcome of competition by removing nutrient limitation on stress-tolerant plants, expand, away from water.

73-Competition

Interspecific competition reduces the equilibrium level of a population below the carrying capacity

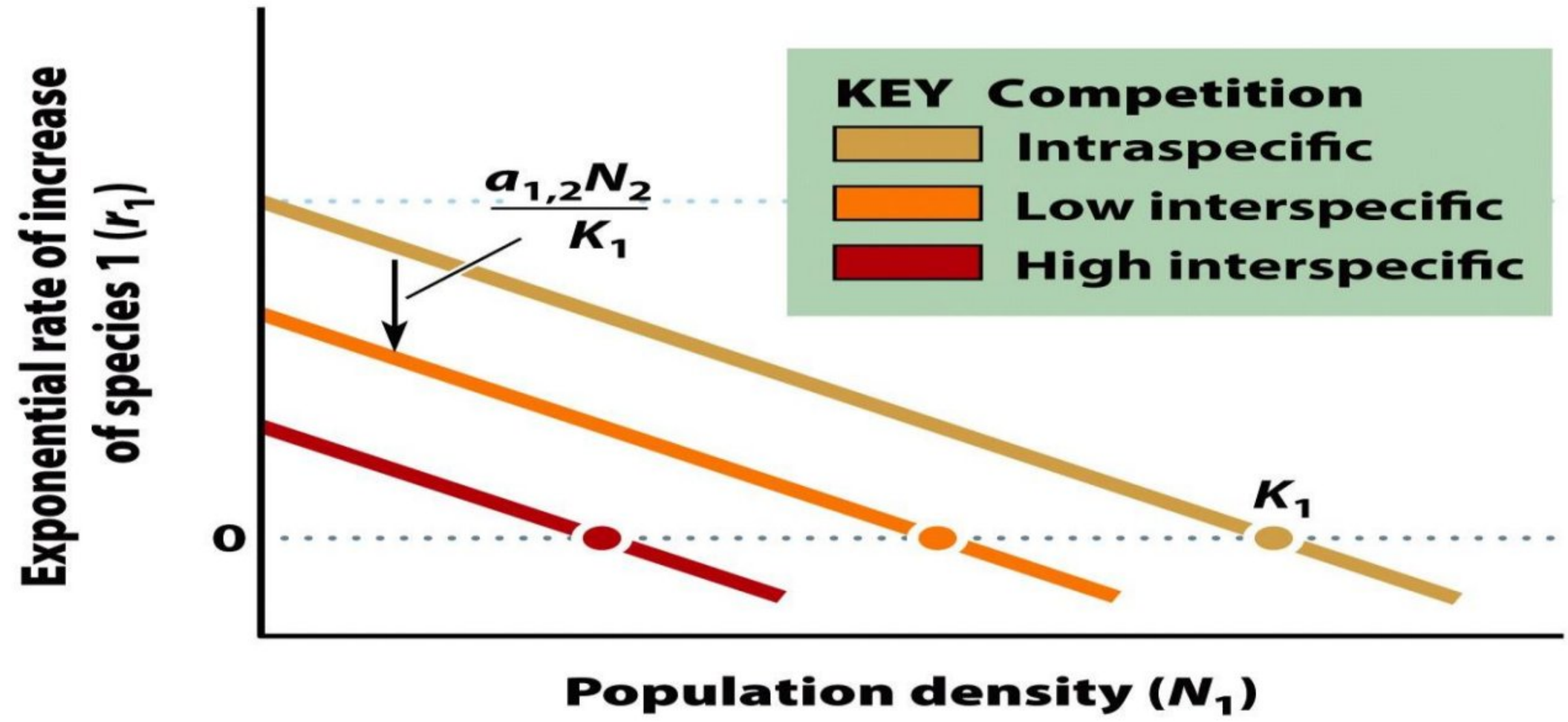

Figure 16.9
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There Are Four Possible Outcomes of Interspecific Competition

- In two situations, one of the species is the superior competitor and wins out over the other

A- In one case, species 1 inhibits the population of species 2 while continuing to increase

B- In one case, species 2 inhibits the population of species 1 while continuing to increase

(c) Species 1 inhibits growth of species 2 and latter goes extinction

(d) Species 2 inhibits growth of species 1 and latter goes extinction

There Are Four Possible Outcomes of Interspecific Competition

- In a third situation, each species, when abundant, inhibits the growth of the other (more than it inhibits its own growth)
  - Eventually one of the two species "wins"
- In a fourth situation, neither species eliminates the other resulting in coexistence
  - Each species inhibits its own population growth more than that of the other species

(e) Unstable situation, both inhibit in a density dependent manner. Depending on initial density, either can make other extinct

(f) Each species inhibits its own population growth more than competitor. Neither can eliminate competitor

74-Competition

13.5 Competition may occur through direct interference

- Exploitation: indirectly influencing each other by consuming the same resources (eat same grass by zebras , compete for water uptake by trees, indirectly)
- Interference: direct influencing each other by preventing others to occupy a habit or access resources

Consumers can influence the outcome of competition

Keystone predator

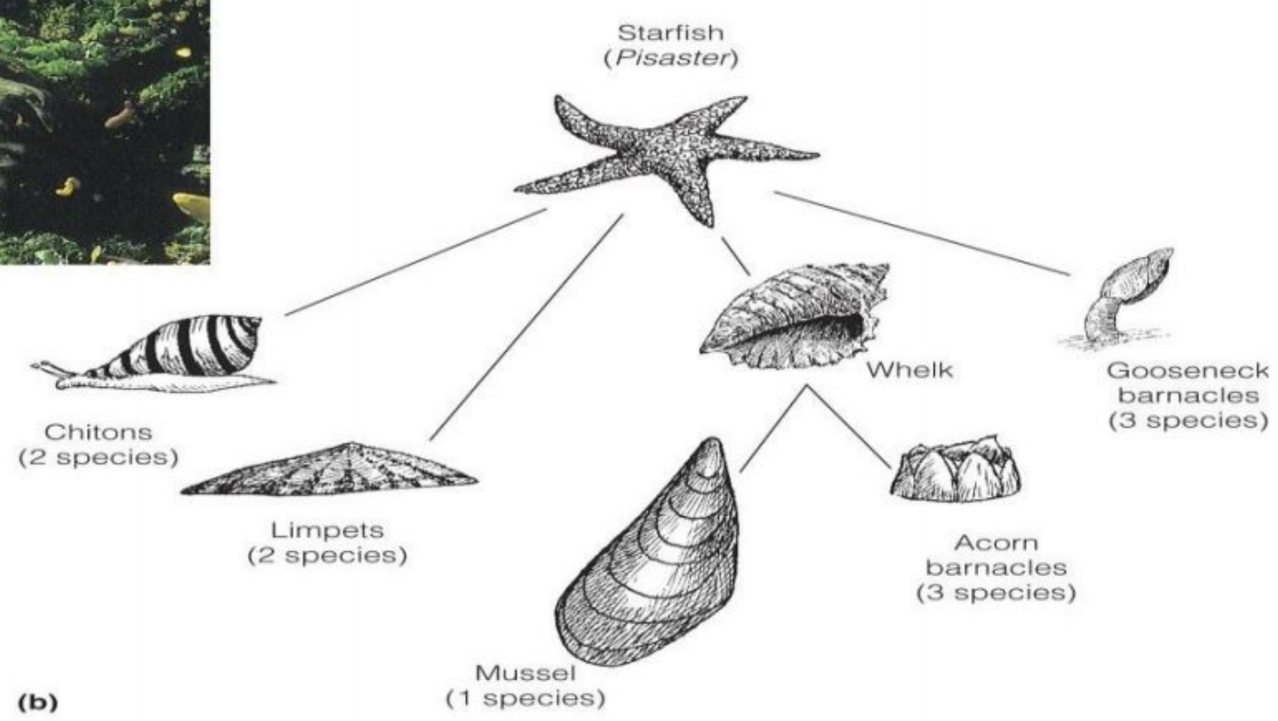
Starfish prey on mussels, barnacles, limpets, and chitons

Remove starfish, what would happen?

Species diversity increase or decrease?

Why?

(a)



(b)

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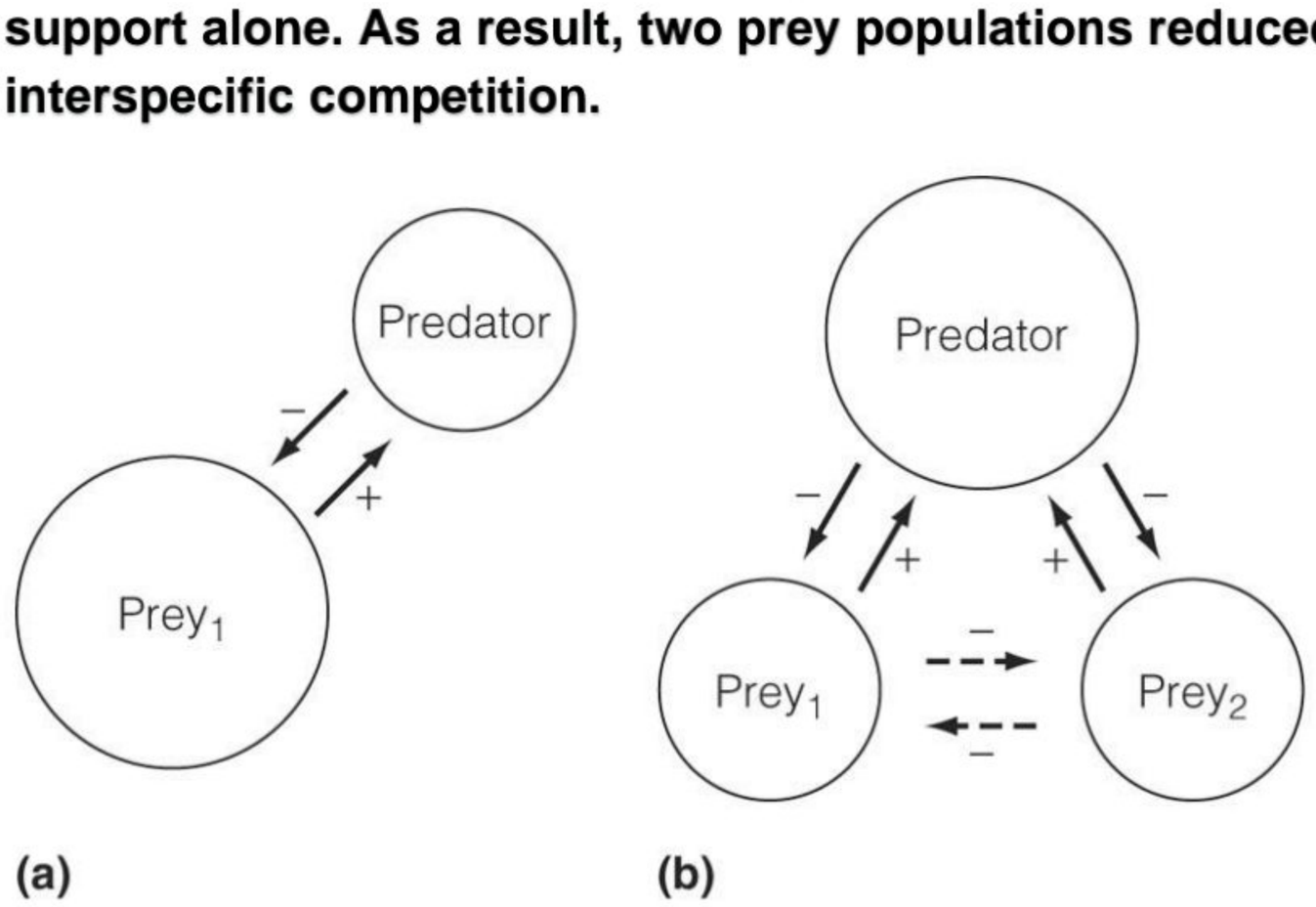
Apparent competition

Combined populations of two prey species support a larger predator population neither can support alone. As a result, two prey populations reduced, gives outward appearance of interspecific competition.



(a) (b)

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Apparent competition

- **In the absence of predator, the population of each prey is regulated by purely intraspecific density-dependent mechanisms**
- **Neither prey species compete, directly or indirectly, with each other**
- **Predator abundance depends on the total abundance of prey**
- **Under these conditions, the combined population abundance of two prey species will support a higher predator density.**

75-Energy and Nutrient Relations

Three main sources of energy

- Organisms use one of three main sources of energy:
  - light,
  - inorganic molecules, or
  - organic molecules.
- Photosynthetic plants and algae CO_2 as a source of carbon and light, of wavelengths between 400 and 700 nm, as a source of energy
- Light within this band, which is called photosynthetically active radiation, or PAR, accounts for about 45% of the total energy content of the solar spectrum at sea level.

PAR quantification

- PAR can be quantified as photosynthetic photon flux density, generally reported as µmol per square meter per second
- Among plants, there are three major alternative photosynthetic pathways, C_3, C_4, and CAM. C_4 and CAM plants are more efficient in their use of water than are C_3plants.

Heterotrophs

- Heterotrophs use organic molecules both as a source of carbon and as a source of energy.
- Herbivores, carnivores, and detritivores face fundamentally different trophic problems.
- Herbivores feed on plant tissues. which often contain a great deal of carbon but little nitrogen.
- Herbivores must also overcome the physical and chemical defences of plants

76-Energy and Nutrient Relations

Detritivores

- Detritivores feed on dead plant material, which is even lower in nitrogen than living plant tissues
- Carnivores consume prey that are nutritionally rich but very well defended.
- Chemosynthetic autotrophs, which consist of a highly diverse group of chemosynthetic bacteria, use inorganic molecules as a source of energy
- Bacteria are the most tropically diverse organisms in the biosphere.

Rate of uptake of energy

- The rate at which organisms can take in energy is limited, either by external or internal constraints.
- The relationship between photon flux density and plant photosynthetic rate is called photosynthetic response.
- Herbs and short-lived perennial shrubs from sunny habitats have high maximum photosynthetic rates that level on at high light intensities.
- The lowest maximum rates of photosynthesis occur among plants from shady environments
- The relationship between food density and animal feeding rate is called the functional response.
- The shape of the functional response is generally one of three types.
- The forms of photosynthetic response curves and type 2 animal functional responses are remarkably similar

77-Rate of uptake of energy

Optimal foraging theory

- Optimal foraging theory attempts to model how organisms feed as an optimizing process.
- Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favour individuals that are more effective at acquiring energy and nutrients.
- Many animals select food in a way that appears to maximize the rate at which they capture energy
- Plants appear to allocate energy to roots versus shoots in a way that increases their rate of intake of the resources that limit their growth.
- Plants in environments with abundant nutrients but little light tend to invest more energy in the growth of stems and leaves and less in roots.
- In environments rich in light but poor in nutrients, plants tend to invest more energy in the growth of roots.

Trophic diversity of bacteria

- The trophic diversity of bacteria, which is critical to the health of the biosphere, can also be used as a tool to address some of our most challenging waste disposal problems
- Bacteria can be used to eliminate the huge quantities of sewage produced by human populations, clean up soils and aquifers polluted by petroleum products such as benzene,
- And eliminate the pollution caused by some kinds of mine waste.

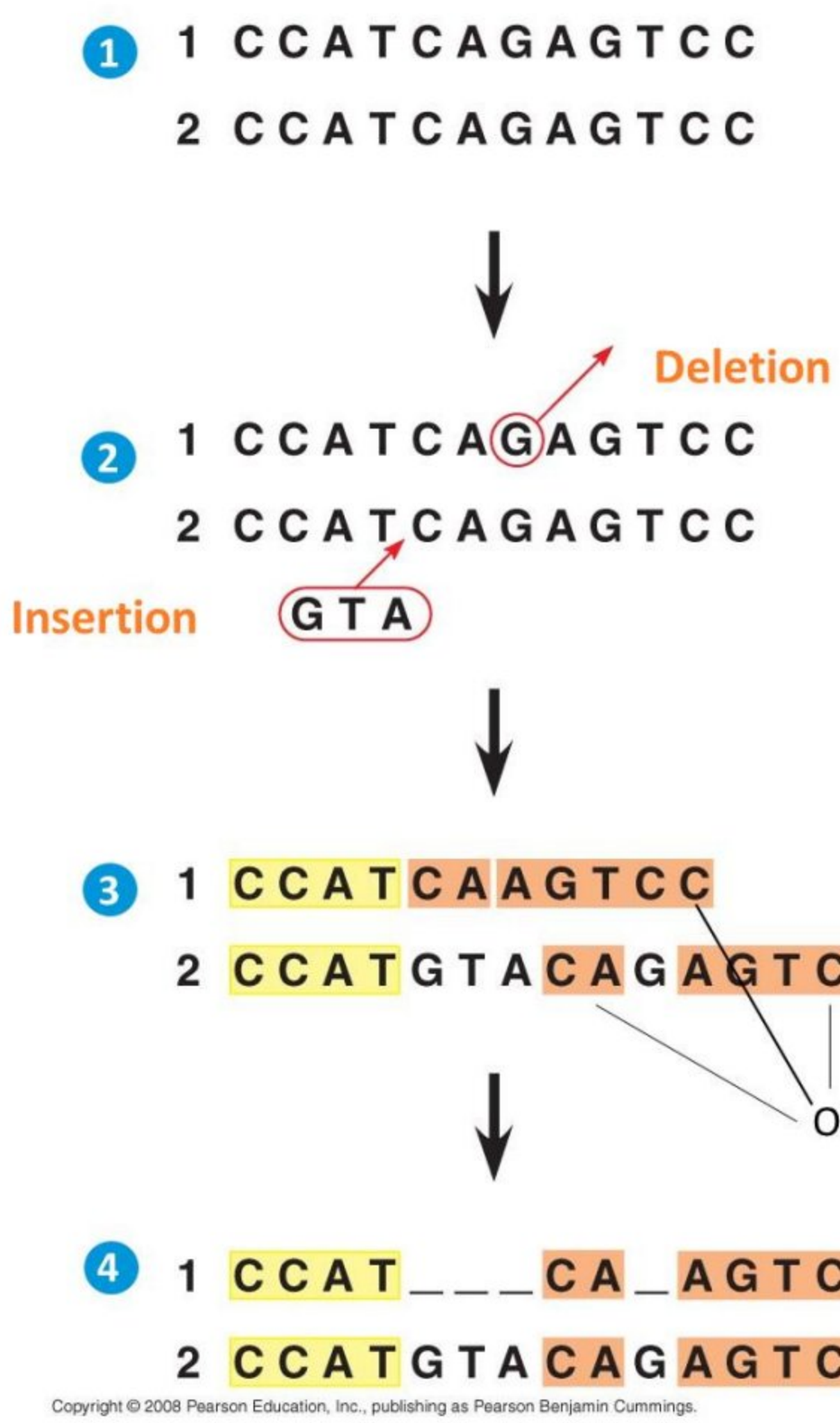
Role of bacteria

- The success of these projects requires that ecologists under- stand the energy and nutrient relations of bacteria.
- Bacteria will likely continue to play a great role as we address some of our most disturbing environmental problems.

78-Evaluating Molecular Homologies

- Systematics use computer programs and mathematical tools when analyzing comparable DNA segments from different organisms

Fig. 26-8



- It is also important to distinguish homology from analogy in molecular similarities
- Mathematical tools help to identify molecular homoplasies, or coincidences

ACGGATAGTCCACTAGGCACTA

TCACCGACAGGTCTTTGACTAG

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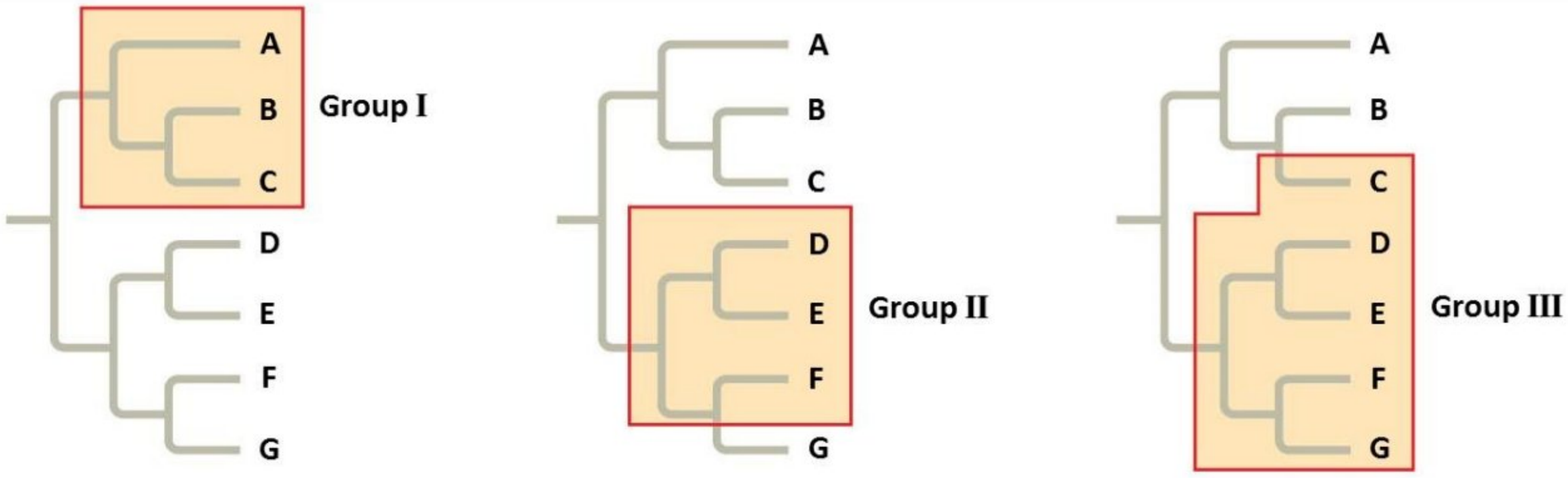
- Molecular systematics uses DNA and other molecular data to determine evolutionary relationships

Shared characters are used to construct phylogenetic trees

- Cladistics groups organisms by common descent
- A **clade** is a group of species that includes an ancestral species and all its descendants
- A valid clade is monophyletic, signifying that it consists of the ancestor species and all its descendants
- A paraphyletic grouping consists of an ancestral species and some, but not all, of the descendants
- A polyphyletic grouping consists of various species that lack a common ancestor

Includes all descendants



(a) Monophyletic group (clade) (b) Paraphyletic group (c) Polyphyletic group

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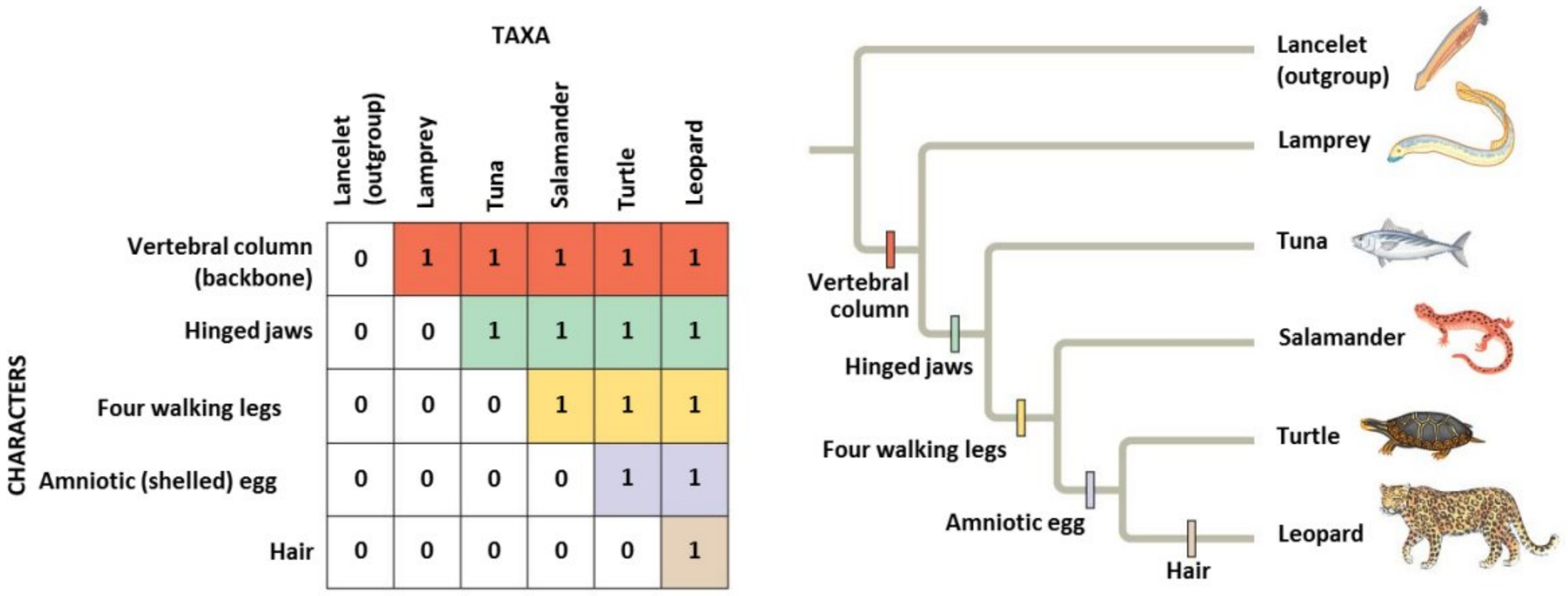
79-Shared Ancestral and Shared Derived Characters

- In comparison with its ancestor, an organism has both shared and different characteristics
- A shared ancestral character is a character that originated in an ancestor of the taxon
- A shared derived character is an evolutionary novelty unique to a particular clade
- A character can be both ancestral and derived, depending on the context, it is useful to know in which clade a shared derived character first appeared



(a) Character table (b) Phylogenetic tree

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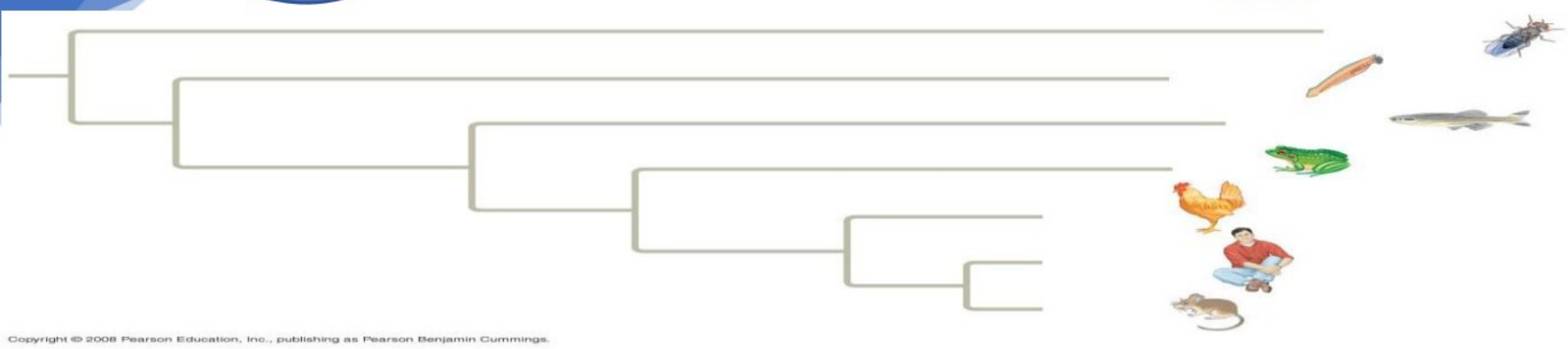
- An out-group is a species or group of species that is closely related to the in-group, the various species being studied
- Systematists compare each in-group species with the out-group to differentiate between shared derived and shared ancestral characteristics

80-Shared Ancestral and Shared Derived Characters

- Homologies shared by the out-group and in-group are ancestral characters that predate the divergence of both groups from a common ancestor

Phylogenetic Trees with Proportional Branch Lengths

- In some trees, the length of a branch can reflect the number of genetic changes that have taken place in a particular DNA sequence in that lineage



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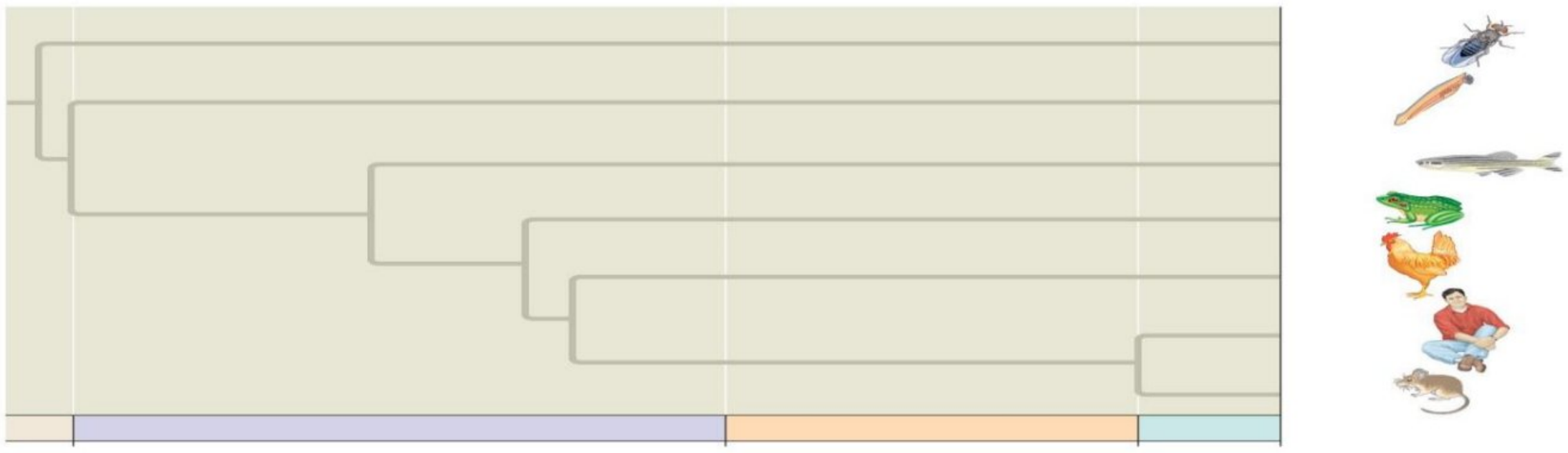
- In other trees, branch length can represent chronological time, and branching points can be determined from the fossil record



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- Systematists can never be sure of finding the best tree in a large data set
- They narrow possibilities by applying the principles of maximum parsimony and maximum likelihood

Maximum Parsimony and Maximum Likelihood

- Maximum parsimony assumes that the tree that requires the fewest evolutionary events (appearances of shared derived characters) is the most likely
- The principle of maximum likelihood states that, given certain rules about how DNA changes over time, a tree can be found that reflects the most likely sequence of evolutionary events

81-Evolution of Definitions of Ecology

- Ecology = from the Greek root OIKOS, “at home”, and OLOGY, “the study of” Haeckle (1870):
- “By ecology we mean the body of knowledge concerning the economy of Nature - the investigation of the total relations of the animal to its inorganic and organic environment.”
- Burdon-Sanderson (1890s): Elevated Ecology to one of the three natural divisions of Biology: Physiology - Morphology - Ecology
- Elton (1927): “Scientific natural history”
- Andrewartha (1961): “The scientific study of the distribution and abundance of organisms”
- Odum (1963): “The structure and function of Nature”

Our Definition

- “Ecology is the scientific study of the processes regulating the distribution and abundance of organisms and the interactions among them, and the study of how these organisms in turn mediate the transport and transformation of energy and matter in the biosphere (i.e., the study of the design of ecosystem structure and function).
- Beyond Fundamental Ecology Applied Ecology:

- Using ecological principles to maintain conditions necessary for the continuation of present day life on earth.

Industrial Ecology

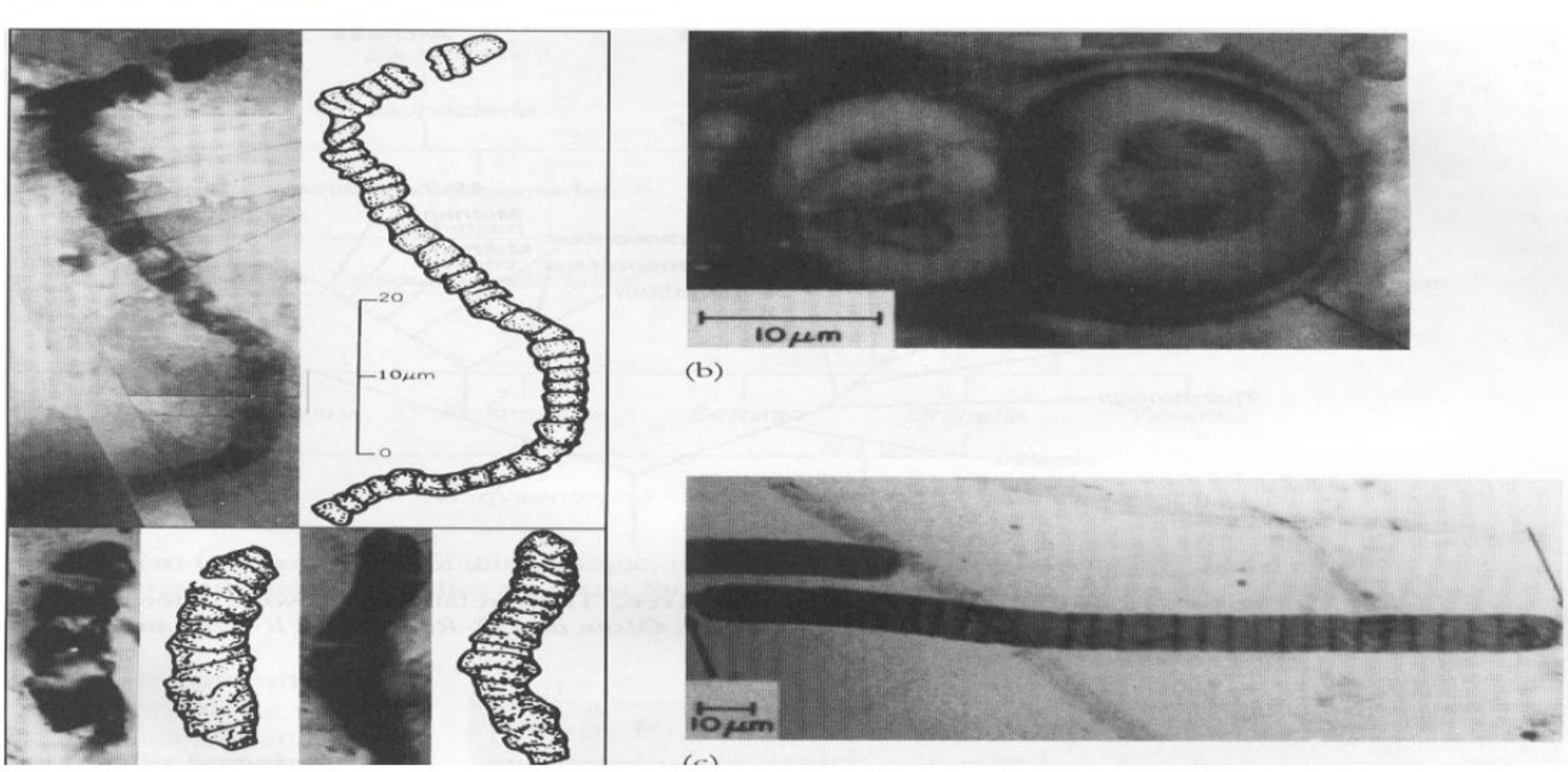
- The design of the industrial infrastructure such that it consists of a series of interlocking "technological ecosystems" interfacing with global natural ecosystems.
- Industrial ecology takes the pattern and processes of natural ecosystems as a design for sustainability. It represents a shift in model from conquering nature to becoming nature.

Ecological Engineering

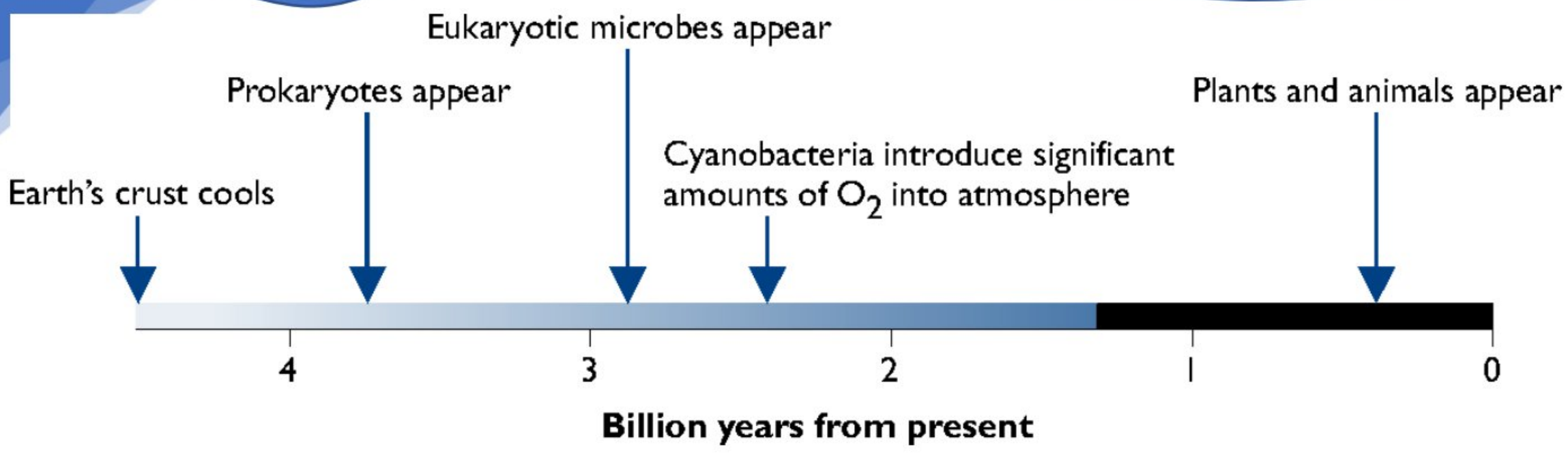
- Unlike industrial ecology, the focus of Ecological Engineering is on the manipulation of natural ecosystems by humans for our purposes, using small amounts of supplemental energy to control systems in which the main energy drives are still coming from non-human sources.
- It is the design of new ecosystems for human purposes, using the self-organizing principles of natural ecosystems.

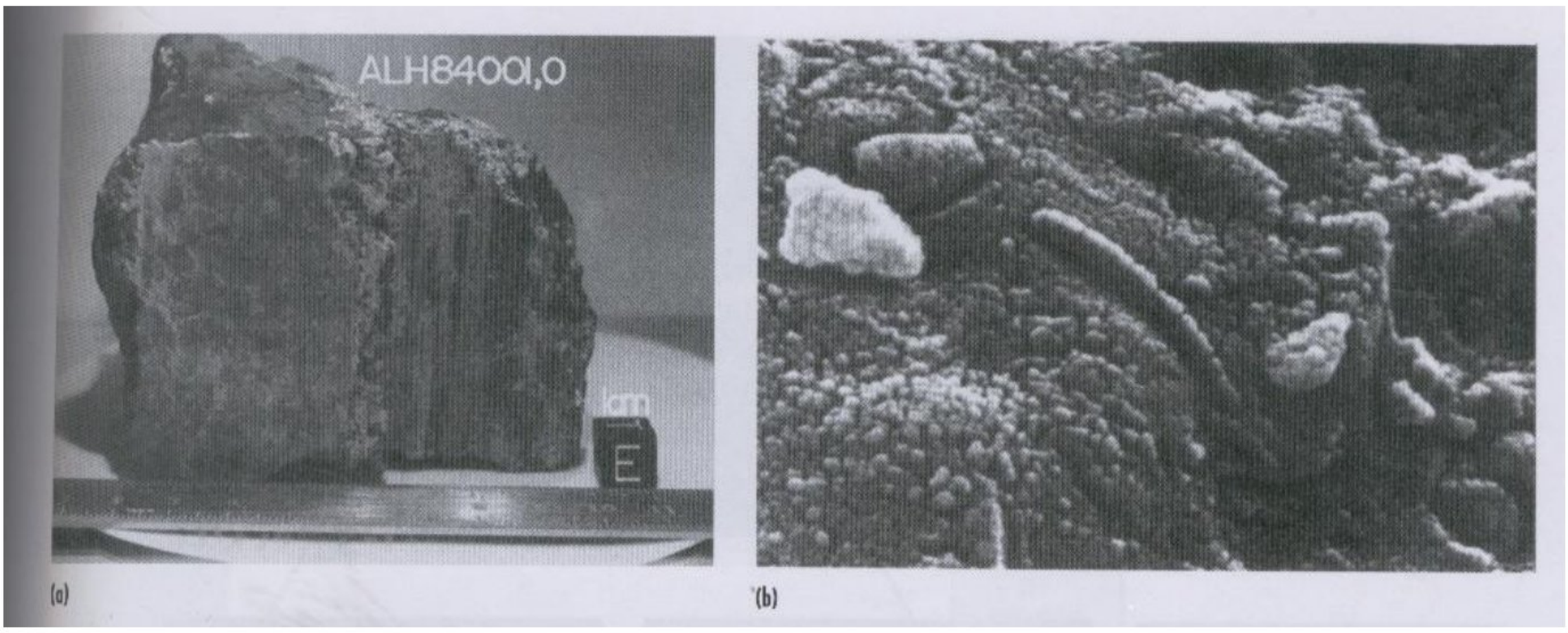
82-Ecological Economics

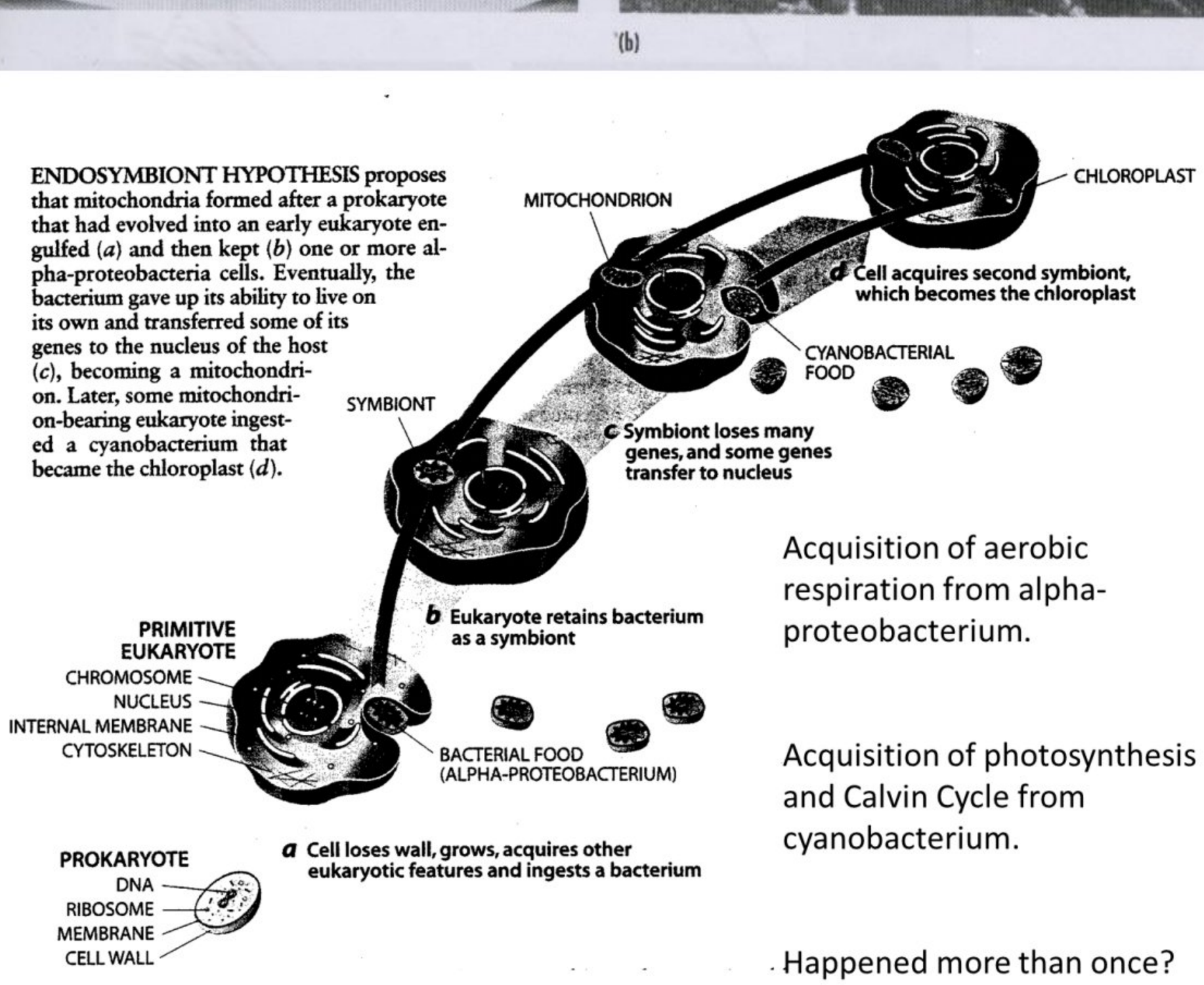
Over 3.5 billion years of "microbes"



Micro-fossils of "cyanobacteria" and contemporary stromatolites.







Acquisition of aerobic respiration from alpha-proteobacterium.

Acquisition of photosynthesis and Calvin Cycle from cyanobacterium.

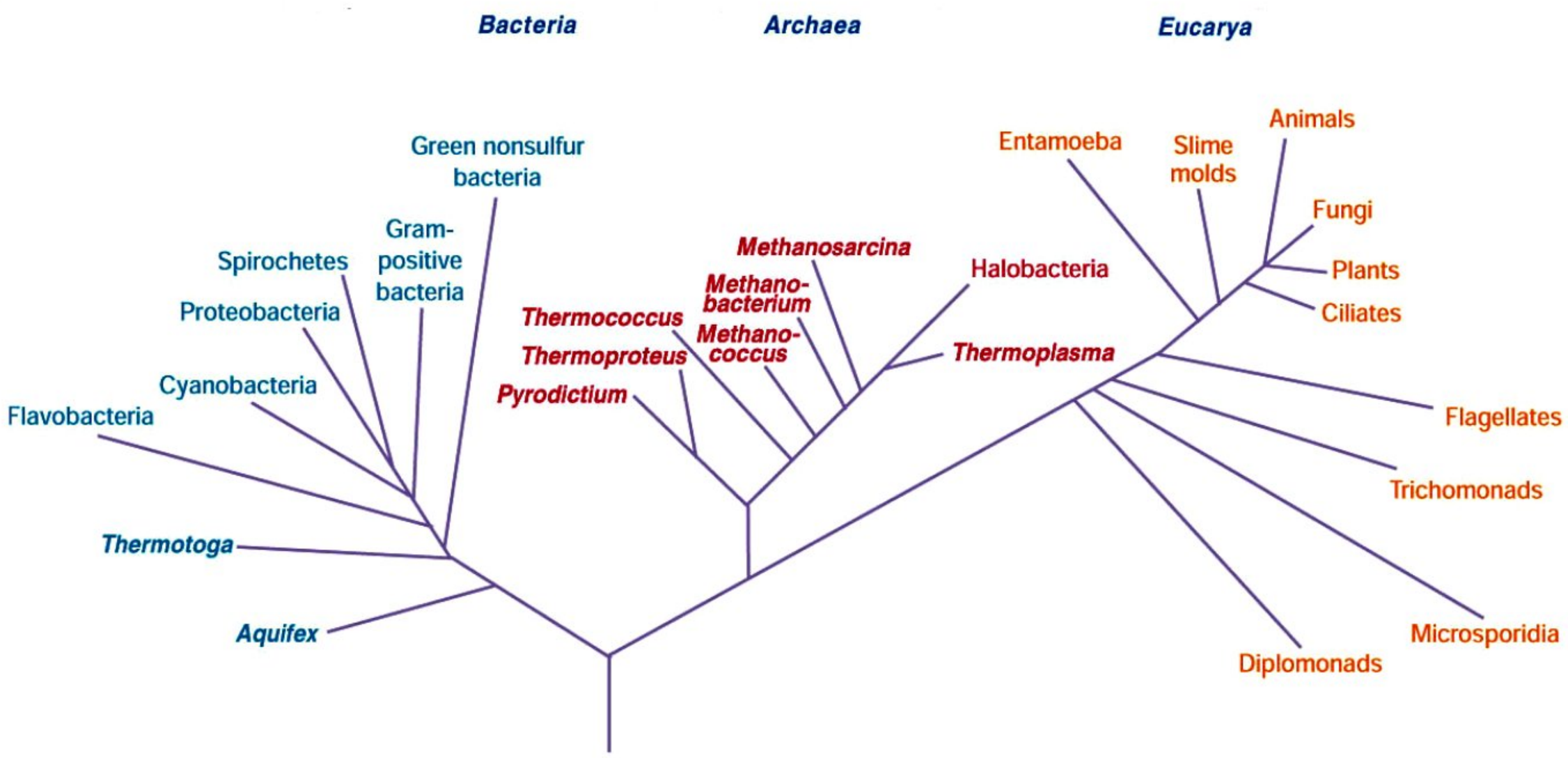
Happened more than once?

Major challenges for endosymbiotic theory

1. Most extant prokaryotes have rigid cell walls and don't do phagocytosis.
2. Hard to explain the nucleus (!) and flagella. (Eukaryotic flagella have a 9+2 arrangement of protein strands, vs. single strand for prokaryotic flagella.)

83-Evolution of Microbes

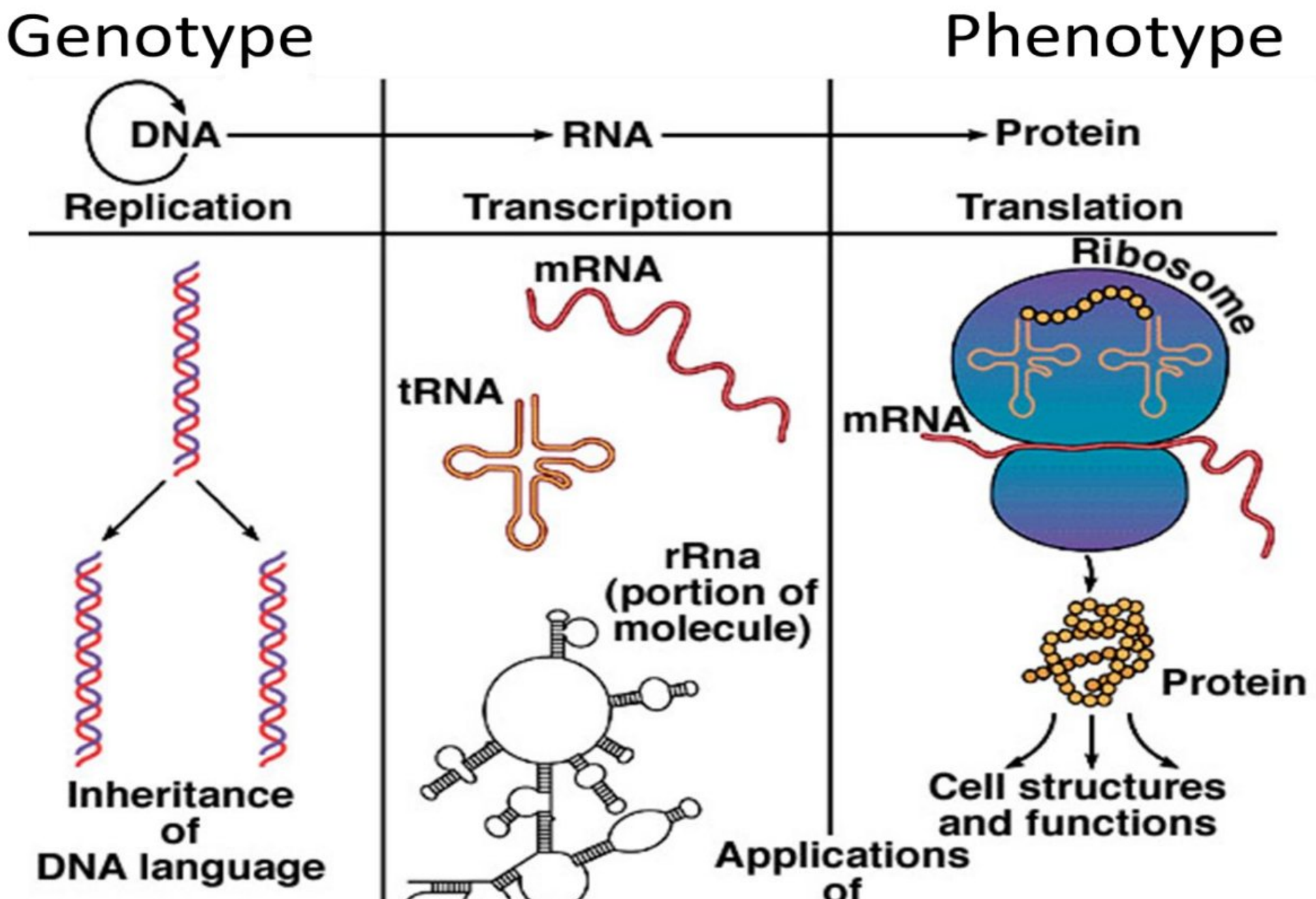
84-Phylogenetic Tree of Life



Rooted Tree based on 16SrRNA and 18SrRNA sequence data.

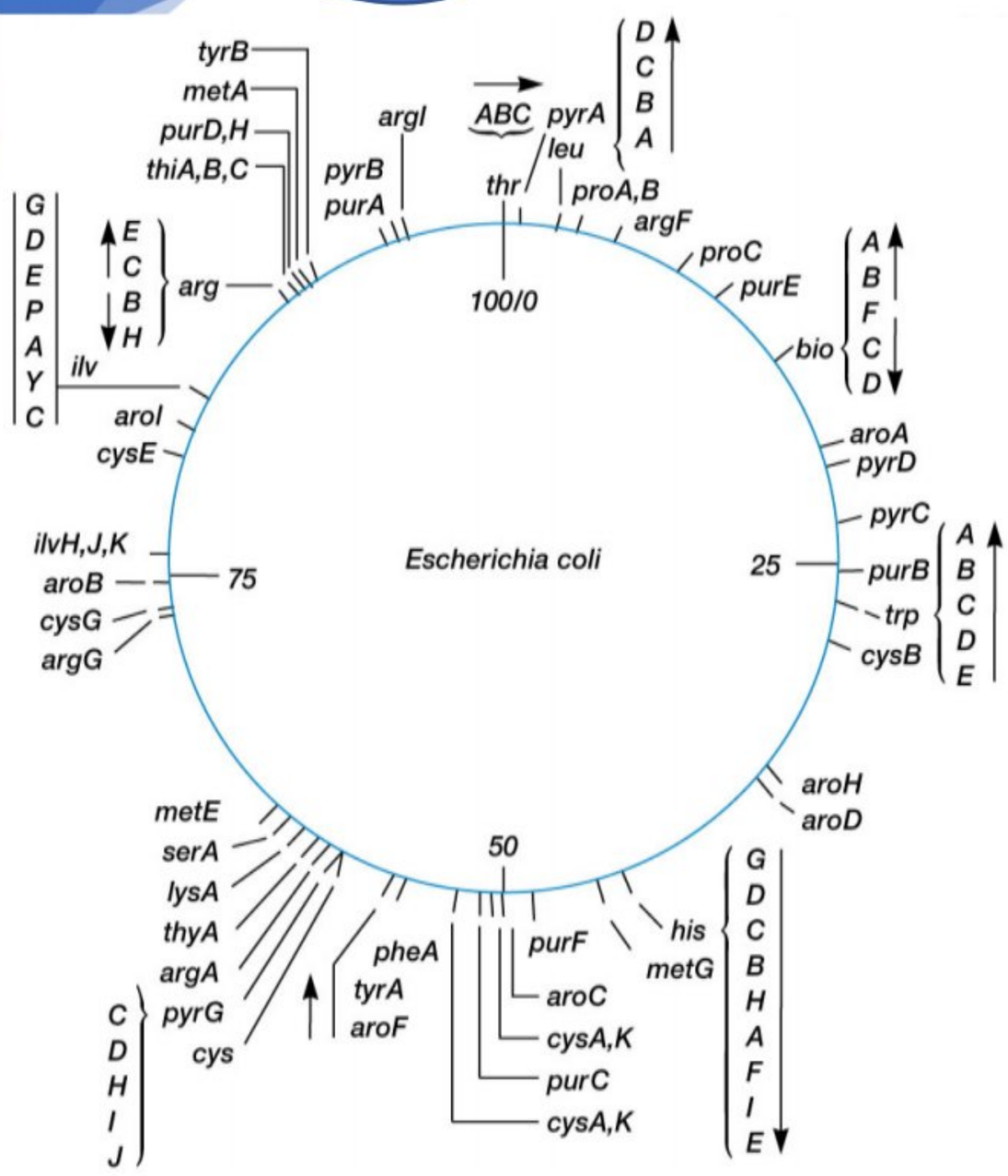
All extant life has evolved; evidence lost by extinction.

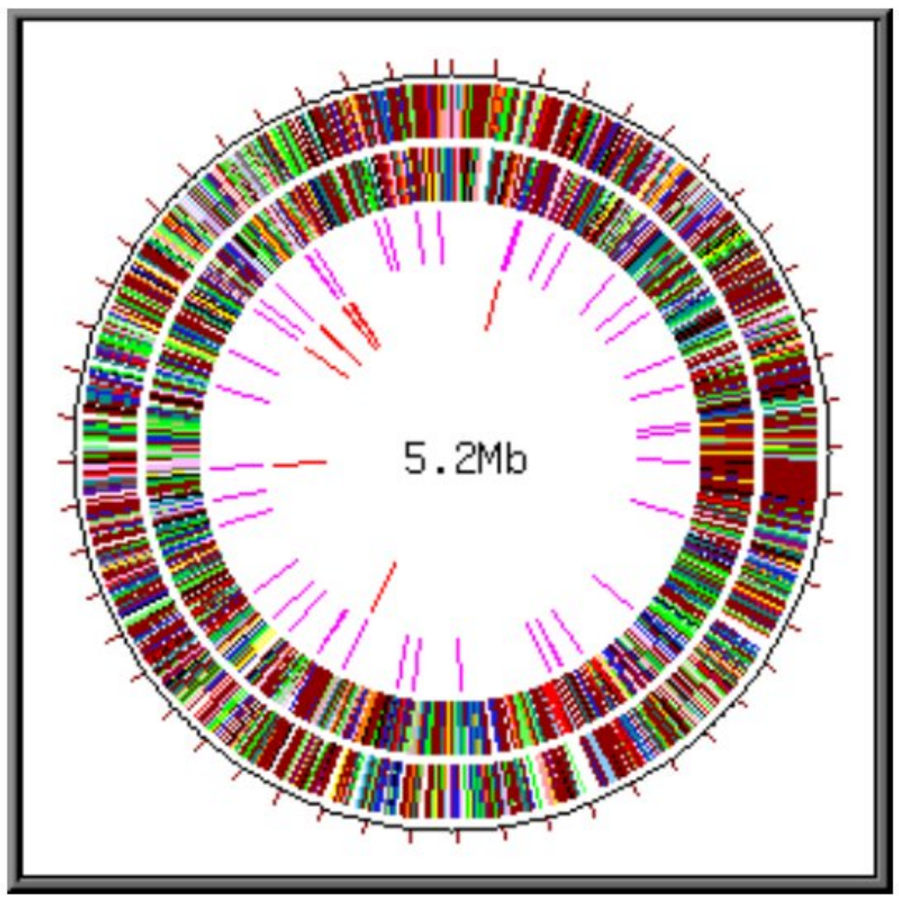
How does evolution work? We need to consider the molecule processes!



Bacterial Genomes

- Plasmids:
  - Size: 2-200 bp
  - Conjugative or not
  - Copy number varies
  - Gene functions vary





Scope of Mutation:

- A mutation is any change in the proper nucleic acid sequence of a specific gene in a cell's genome. It may result from a single base pair mismatch during DNA replication.
- Mutation can create genetic diversity within a population; either beneficial, neutral, bad, or lethal.
- Mutation could result in a new phenotype that is advantageous to successful reproduction of the mutated individual; this depends on particular environmental conditions, called selective pressures.
- Such beneficial mutations stay within a population from generation to generation, and drive the evolution of that species.
- Bad or lethal mutations are often lost from a population over subsequent generations.

Mutation types:

- Macrolesions (large sequence sections)
  - Deleted *a-b-c-d-e-f-g-h → a-b-c-g-h*
  - Inserted *a-b-c-d-e-f-g-h → a-b-c-d-x-y-z-e-f-g-h*
  - Inverted *a-b-c-d-e-f-g-h → a-b-c-f-e-d-g-h*
  - Duplicated *a-b-c-d-e-f-g-h → a-b-c-d-e-f-d-e-f-g-h*
- Microlesions (1 or 2 bp alteration)
  - Point Mutations (Base Substitutions) *ACTG → ATTG*
  - Frameshifts (Insertions or Deletions)

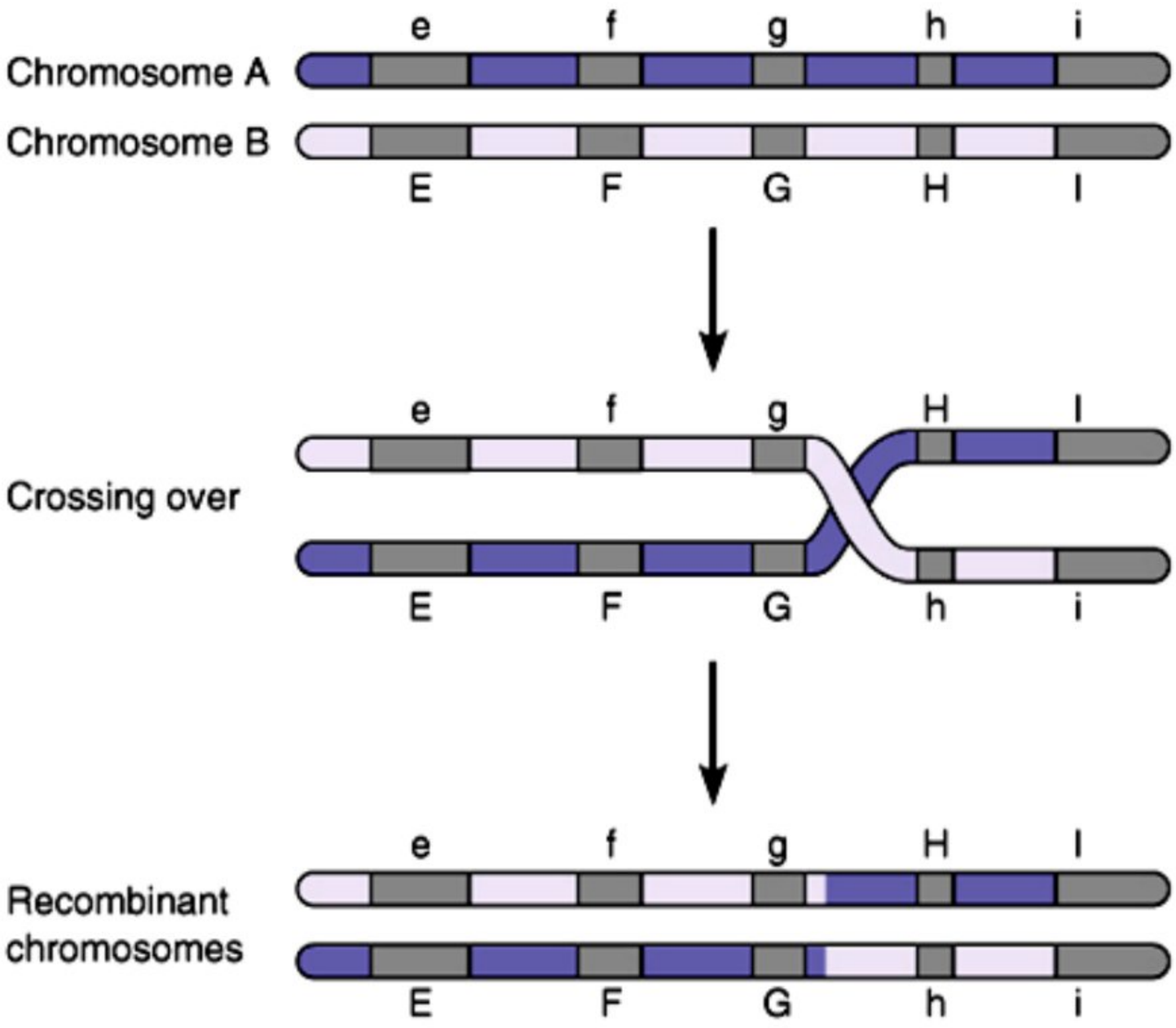
see the cat eat the rat → see thc ate att her at

- Mechanisms of microlesion mutation types
  - Spontaneous (1 per million; most corrected; 1 per billion remain)
  - Chemical mutagens
  - Radiation as mutagens

85-Genetic Recombination

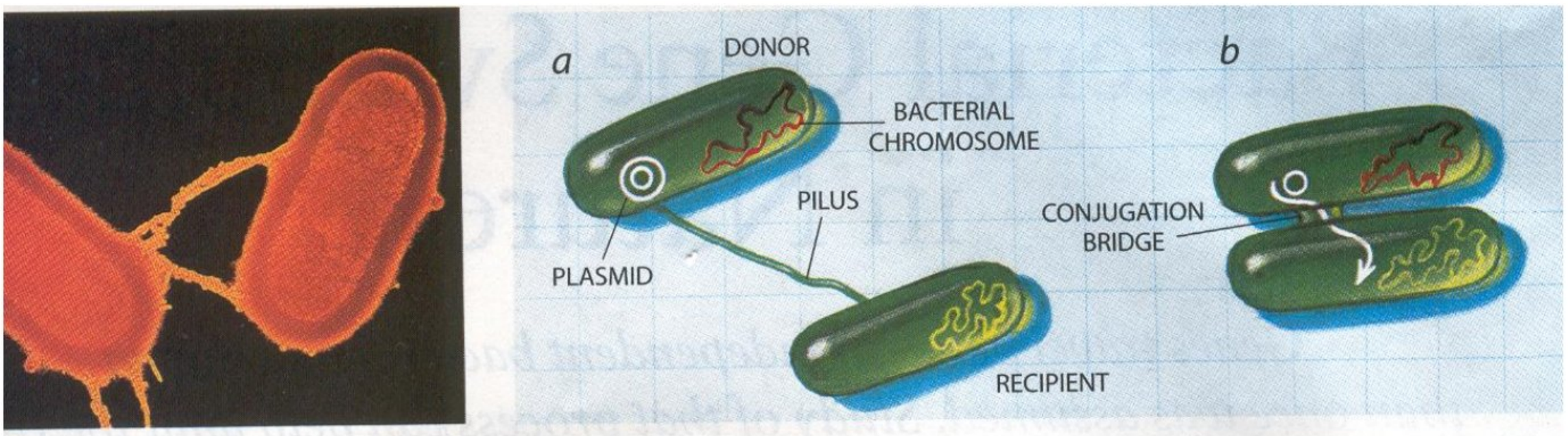
Genetic Recombination:

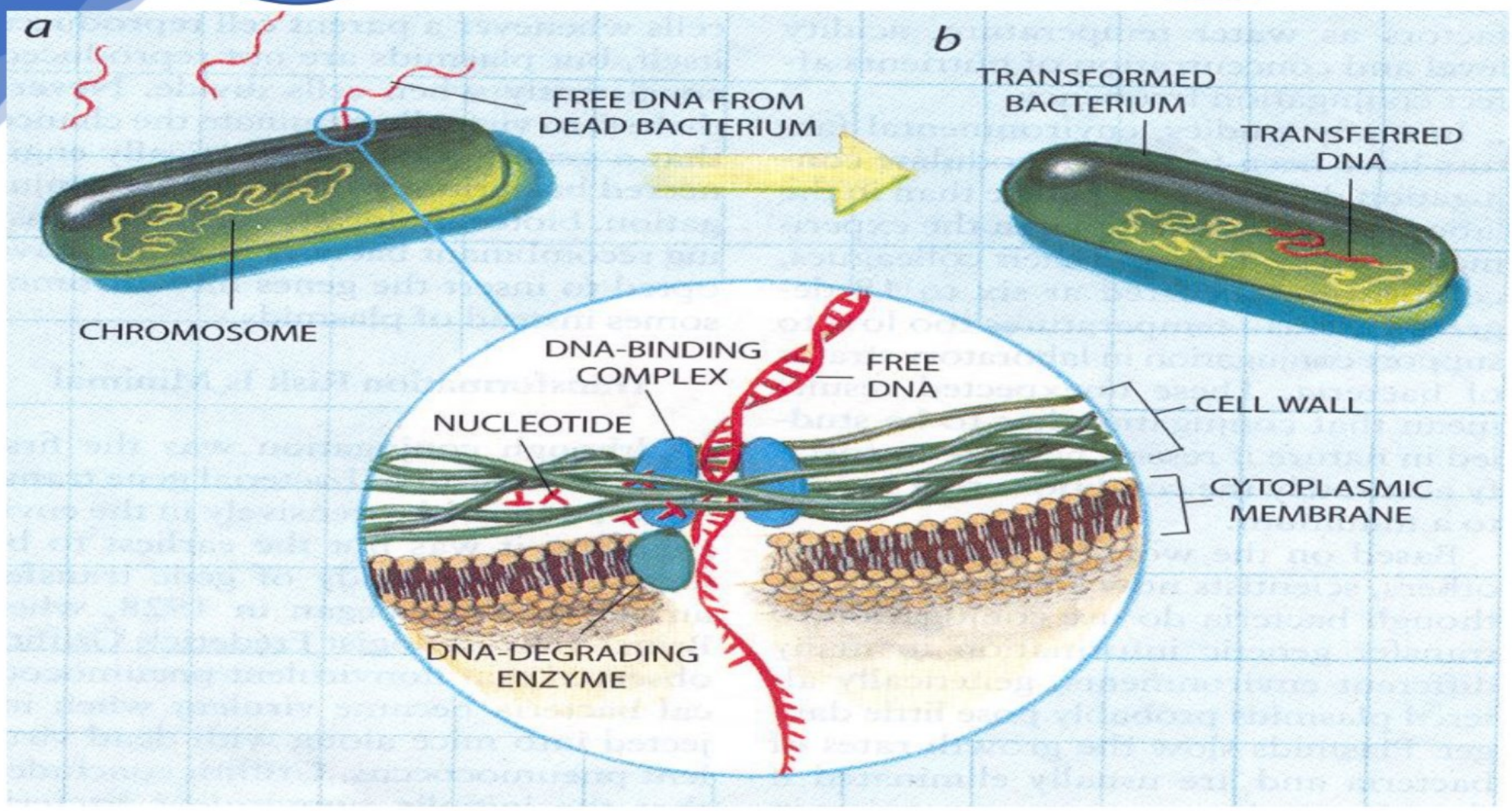
- Two DNA molecules may recombine segments of their molecule in a process called crossing over.
- This is a relatively common event between chromosome copies in eukaryotes during meiosis. (Note the example here.)
- Prokaryote chromosomes, viral DNA, and smaller fragments of "foreign" DNA may recombine, adding new genes (or different alleles) to an individual cell.
- Bacteria can receive a foreign source of DNA for recombination through one of three different mechanisms of Genetic Exchange.

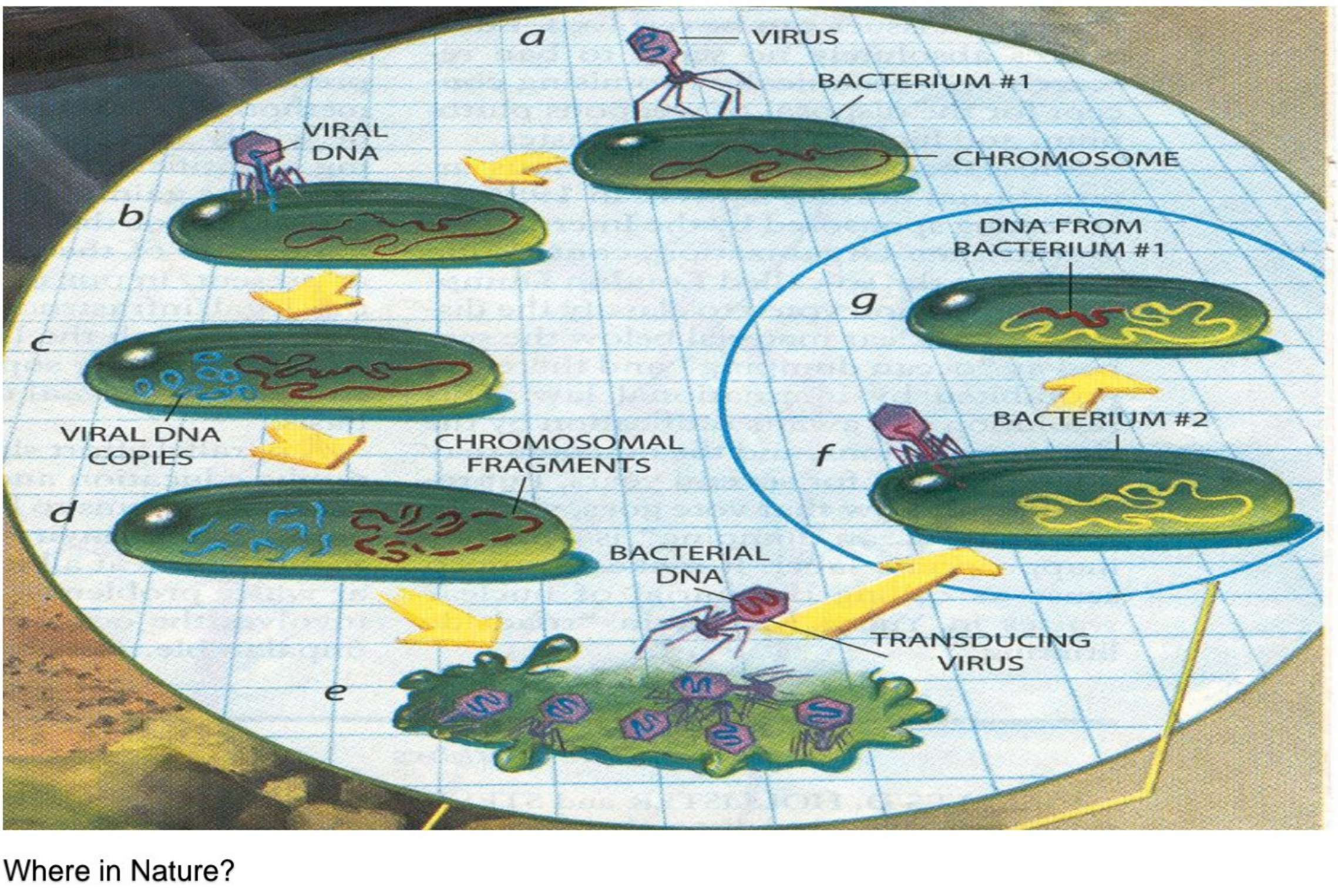


Horizontal Gene Transfer(= lateral gene exchange)

- Conjugation
- Tranformation
- Transduction







Where in Nature?

	TRANSDUCTION	CONJUGATION	TRANSFORMATION
TERRESTRIAL ENVIRONMENTS	Soil, plant surfaces	Soil, plant surfaces	Soil
WATERY ENVIRONMENTS	Lakes, oceans, rivers, sewage in treatment facilities	Lakes, oceans, marine sediments, rivers, epilithon (slime layer) on river stones, sewage in treatment facilities	Marine sediments, rivers, epilithon on river stones
INSIDE ORGANISMS	Shellfish, mice	Plants, insects, chickens, mice, humans	Plants, insects, mice

Summary of Prokaryote Evolution Mechanisms

- Mutation (micro or macro) changes genotype and possibly phenotype.
- Mobile genetic elements (insertions sequences and transposons) may rearrange genes between and within DNA molecules and this may cause mutations.
- Horizontal gene transfer (conjugation, transformation, transduction) may result in recombination of completely new genes.
- Selective pressures in the environment determine if a new phenotype becomes dominant in a population.
- Many changes in genotype are neutral or benign to phenotype and survival; these "cryptic" changes over time may result in genetic drift, i.e. a harmless variation of a gene randomly becomes dominant.

86-Evolutionary Biodiversity in Proterozoic Eon

Endosymbiosis and the Origin of Eukaryotic Cells

- Formed from several prokaryotic cells
    - In a symbiotic relationship
    - Symbiosis,
        - involving a prolonged association of two or more dissimilar organisms,
    - is quite common today
- In many cases both symbionts benefit from the association
    - as occurs in lichens,
        - once thought to be plants
        - but actually symbiotic fungi and algae

Evidence for Endosymbiosis

- Supporting evidence for endosymbiosis
    - comes from studies of living eukaryotic cells
    - containing internal structures called organelles,
        - such as mitochondria and plastids,
    - which contain their own genetic material
- In addition, prokaryotic cells
    - synthesize proteins as a single system,
- whereas eukaryotic cells
    - are a combination of protein-synthesizing systems

Organelles Capable of Protein Synthesis

- That is, some of the organelles
    - within eukaryotic cells are capable of protein synthesis
- These organelles
    - with their own genetic material

- and protein-synthesizing capabilities

- are thought to have been free-living bacteria
  - that entered into a symbiotic relationship,
  - eventually giving rise to eukaryotic cells

Multi-celled Organisms

- Obviously multi-celled organisms
  - are made up of many cells,
  - perhaps billions,
  - as opposed to a single cell as in prokaryotes
- In addition, multi-celled organisms
  - have cells specialized to perform specific functions
  - such as respiration,
  - food gathering,
  - and reproduction

87-Dawn of Multicelled Organisms

- We know from the fossil record
  - that multi-celled organisms were present during the Proterozoic,
  - but we do not know exactly when they appeared
- What seem to be some kind of multi-celled algae appear
  - in the 2.1-billion-year-old fossils
    - from the Negaunee Iron Formation in Michigan
  - as carbonaceous filaments
    - from 1.8 billion-year-old rocks in China
  - as somewhat younger carbonaceous impressions of filaments and spherical forms

Multi-celled Algae?

- Carbonaceous impressions
  - in Proterozoic rocks, Montana
- These may be impressions of multi-celled algae

The Multi-celled Advantage?

- Is there any particular advantage to being multi-celled?
- For something on the order of 1.5 billion years
  - all organisms were single-celled
  - and life seems to have thrived
- In fact, single-celled organisms
  - are quite good at what they do

- but what they do is very limited

- single celled organisms
  - can not grow very large, because as size increases proportionately less of a cell is exposed to the external environment in relation to its volume
  - and the proportion of surface area decreases
- Transferring materials from the exterior to the interior becomes less efficient
- Multi-celled organisms live longer,
  - since cells can be replaced and more offspring can be produced
- Cells have increased functional efficiency
  - when they are specialized into organs with specific capabilities

88-Other Proterozoic Animal Fossils

- A jellyfish-like impression is present in rocks 200 m below the Ediacara Hills
- Wormlike and algae fossils come from 700 to 900 million-year-old rocks in China but the identity and age of these "fossils" has been questioned

Wormlike Fossils from China

- Wormlike fossils from Late Proterozoic rocks in China

Soft Bodies

- All known Proterozoic animals were soft-bodied, but there is some evidence that the earliest stages in the origin of skeletons was underway
- Even some Ediacaran animals may have had a chitinous carapace and others appear to have had areas of calcium carbonate
- The odd creature known as Kimberella from the latest Proterozoic of Russia had a tough outer covering similar to that of some present-day marine invertebrates

Latest Proterozoic Kimberella

- Kimberella, an animal from latest Proterozoic rocks in Russia
  - Exactly what *Kimberella* was remains uncertain
  - Some think it was a sluglike creature
  - whereas others think it was more like a mollusk

Durable Skeletons

- Latest Proterozoic fossils
  - of minute scraps of shell-like material and small tooth like denticles and spicules, presumably from sponges indicate that several animals with skeletons or at least partial skeletons existed
- More durable skeletons of
  - silica, calcium carbonate, and chitin (a complex organic substance) did not appear in abundance until the beginning of the Phanerozoic Eon 545 million years ago = Cambrian age

89-Exploitation

Predation, Herbivory, Parasites and Disease

- The diversity of interactions between herbivores and plants, between predators and prey, and between parasites, pathogens, and hosts can be grouped under the heading of exploitation--interactions between species that enhance the fitness of one individual at the expense of another.
- Exploitation weaves populations into a web of relationships that defy easy generalization.
- The number of exploitative interactions between species far exceeds the number of species in the biosphere, and the nature of exploitation goes far beyond the typical consumption of one organism by another.
- For instance, many parasites and pathogens manipulate host behaviour to enhance their own fitness at the expense of the host.
- Spiny-headed worms alter the behaviour of a variety of crustacean hosts in a way that increases the probability that the one host species will be eaten by another.
- A pathogenic fungus manipulates the growth program of its host plant in a way to produce "pseudoflowers," structures aimed at promoting the reproduction of the pathogen.
- In the process the pathogen usually kills the host plant and always renders it sterile.
- Predation by one flour beetle species on another can be used as a potent means of interference competition except in the presence of a protozoan parasite, which seems to give a competitive advantage to less predaceous species.

Examples

- Predators, parasites, and pathogens influence the distribution, abundance, and structure of prey and host populations.
- Herbivorous stream insects have been shown to control the density of their algal and bacterial food.
- The herbivorous moth larva *Cactoblastis cactorum* combined with pathogenic microbes reduced the coverage of prickly pear cactus in Australia from millions of hectares to a few thousand.
- A parasitic infestation reduced the red fox population in Sweden by 70%, which in turn led to increases in the abundance of several prey species eaten by foxes.
- This parasitic disease revealed the influence of a predator on its prey populations.

90-Ecological Relationships Ecological Relationships of Predator-prey, parasite-host, and pathogen-host

Relationships

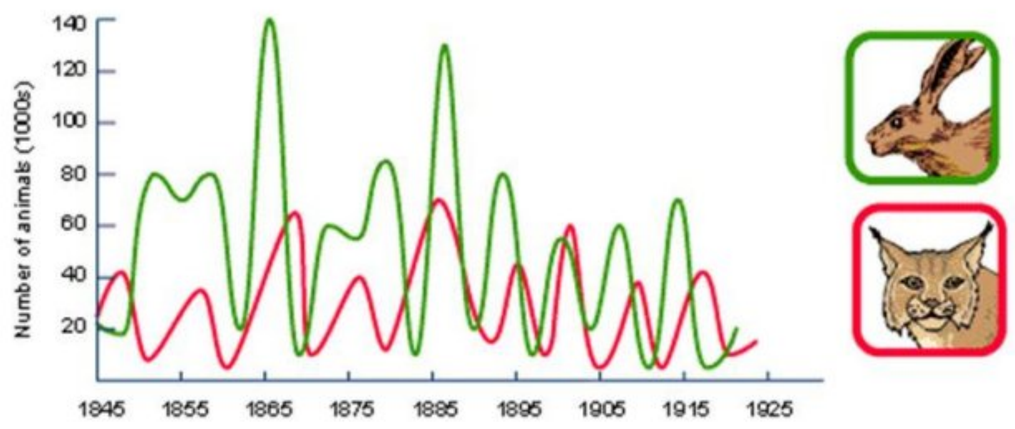
- Predator-prey, parasite-host, and pathogen-host relationships are dynamic.
- Populations of a wide variety of predators and prey show highly dynamic fluctuations in abundance ranging from days to decades.
- A particularly well-studied example of predator-prey cycles is that of snowshoe hares and their predators, which have been shown to result from the combined effects of the snowshoe hares on the quantity and quality of their food and of the predators on the snowshoe hare population.

1. Predator Prey Relationships

Learning Objectives

- Be able to describe the relationship between predators and their prey.

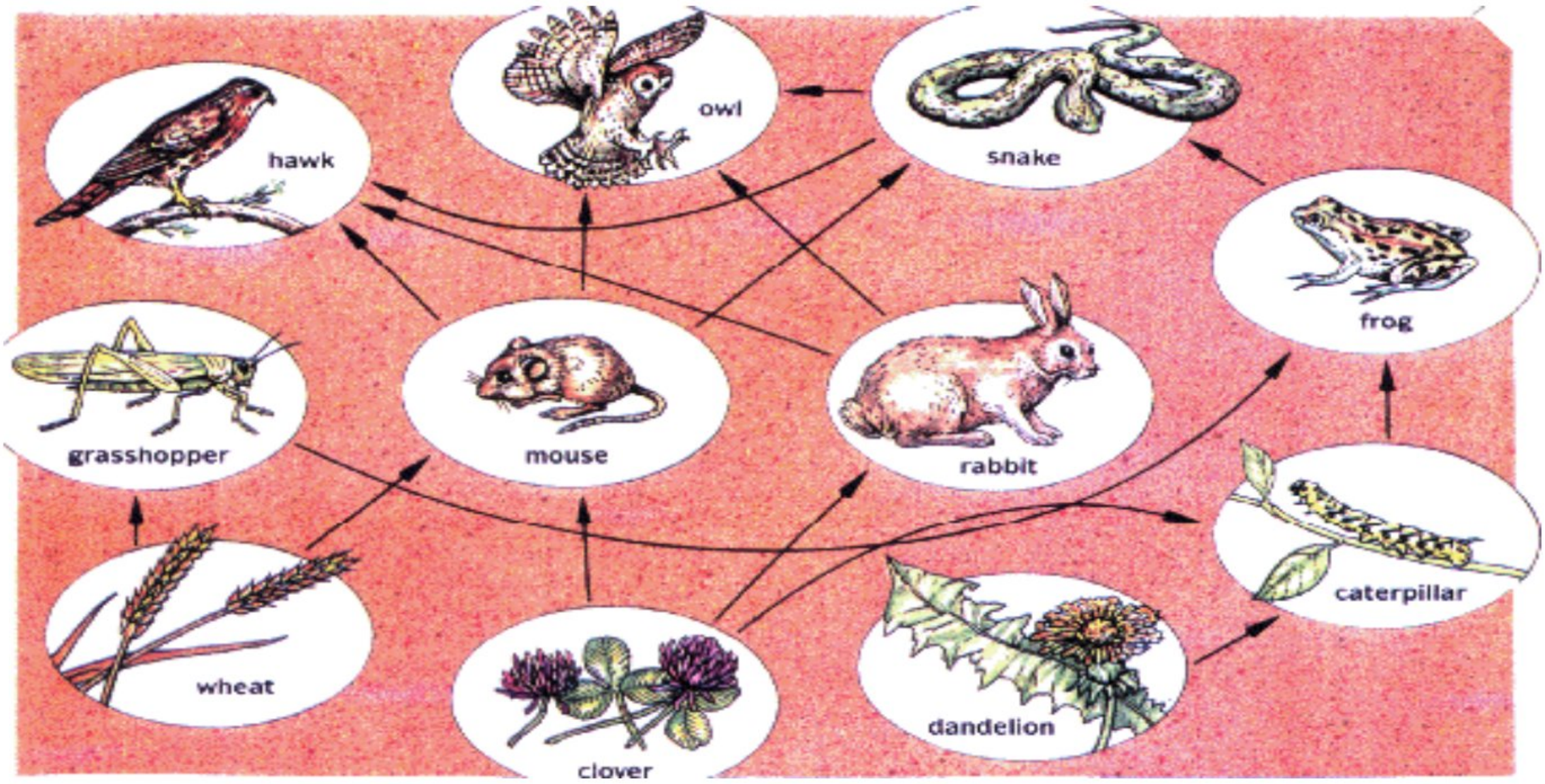
- Be able to explain the shape of a predator – prey graph.



2. Food web - Interdependence

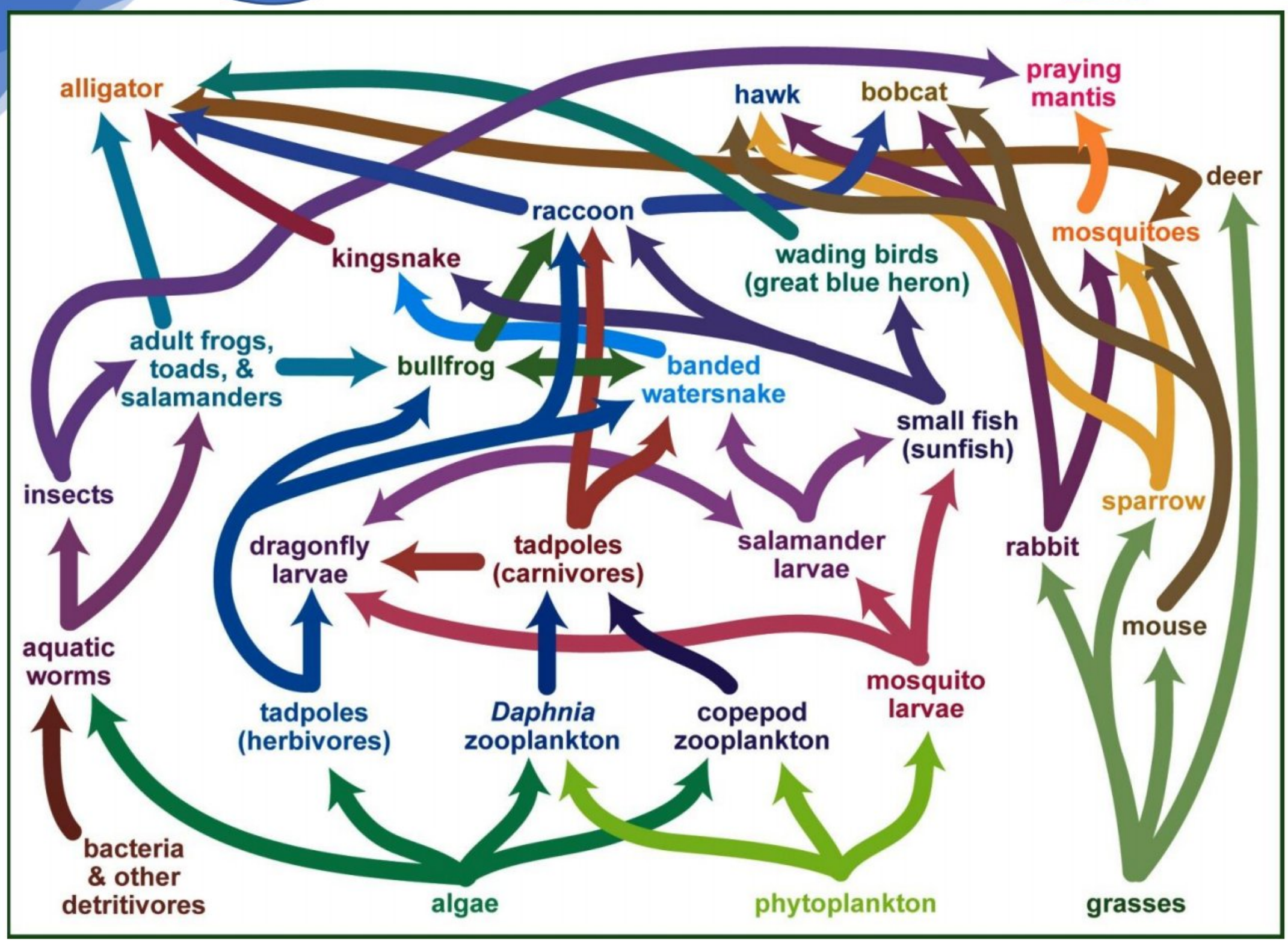
Starter

If the mouse is considered a pest and baited, find as many effects on the food web as you can, in 3 minutes.



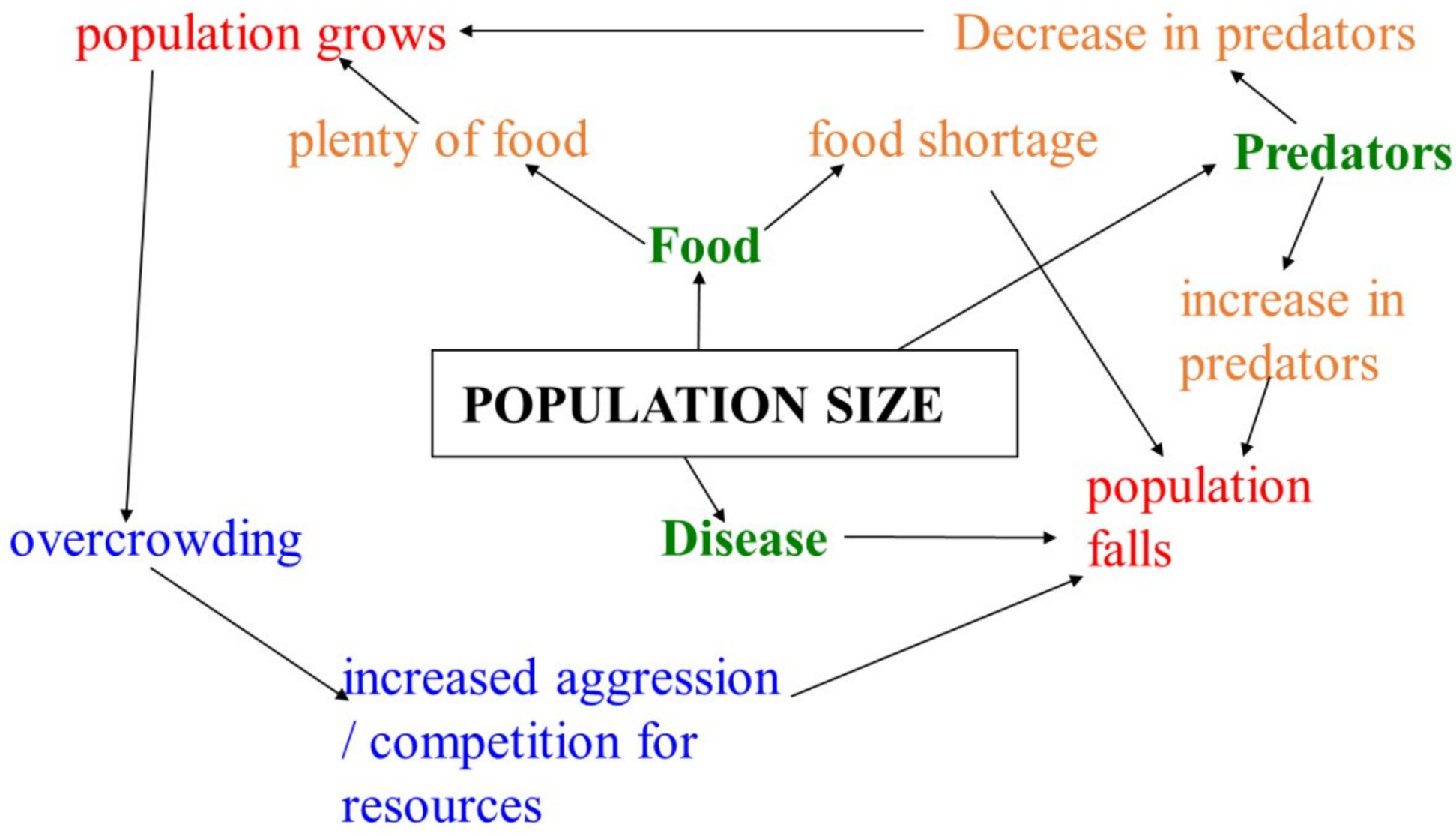
3. Food web – Interdependence

… and that was a simplified food web. Imagine the interactions in a less simplified web. Do you think the bobcat would be affected if the mosquito larvae were destroyed as a malaria risk?



4. Population Size

To do: Draw a concept map of what effects the size of a population.

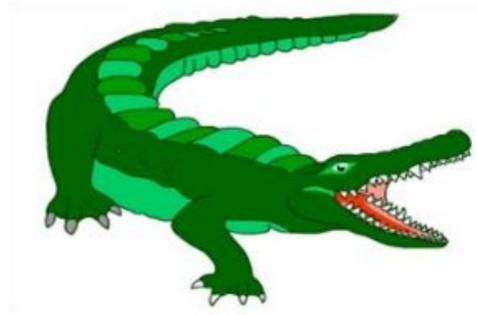


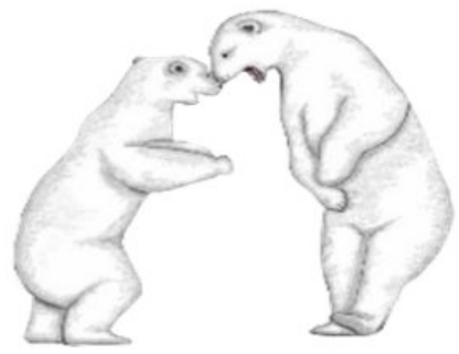
91-Predator-prey interactions

5. Animals that eat other animals

- Populations of animals are often limited by the amount of food.
- What are animals that eat other animals called?

predators





- What are the animals that predators eat called?

prey

To do: Define the words Predator and Prey and give an example of each.

Predator-prey interactions

- Mathematical models of predator-prey interactions by Lotka and Volterra suggest that exploitative interactions themselves can produce population cycles without any influences from outside forces such as weather.
- Predator-prey cycles have also been observed in a few laboratory populations under restricted circumstances.
- To persist in the face of exploitation, hosts and prey need refuges.
- The refuges that promote the persistence of hosts and prey include secure places to which the exploiter has limited access.
- However, living in large groups can be considered as a kind of refuge since it reduces the probability that an individual host or prey will be attacked.
- It appears that predator satiation is a defensive tactic used by a wide variety of organisms from rain forest trees to temperate insects.
- Growing to large size can also represent a kind of refuge when the prey species is faced by size-selective predators,
- Size is used as a refuge by prey species ranging from stream insects and intertidal invertebrates to rhinoceros.

Population control

- Predators and parasites have been used to control populations of insects that attack crops or to control invasive weeds.
- Recent research in Kenya has shown that a crayfish, *Procambarus clarkii*, controls the snails that act as intermediate hosts for Schistosoma, a highly pathogenic human parasite.

- Preliminary results indicate that crayfish successfully control host snails in the artificial impoundments used for livestock watering and domestic water, important sources of infection by Schistosoma.

92-Features of protists

Characteristics of Protists

- ✓ Eukaryotes
- ✓ unicellular

3 Main groups of protists;

- ✓ Protozoans
- ✓ algae
- ✓ fungus-like

1. Protozoans

Animal like protists

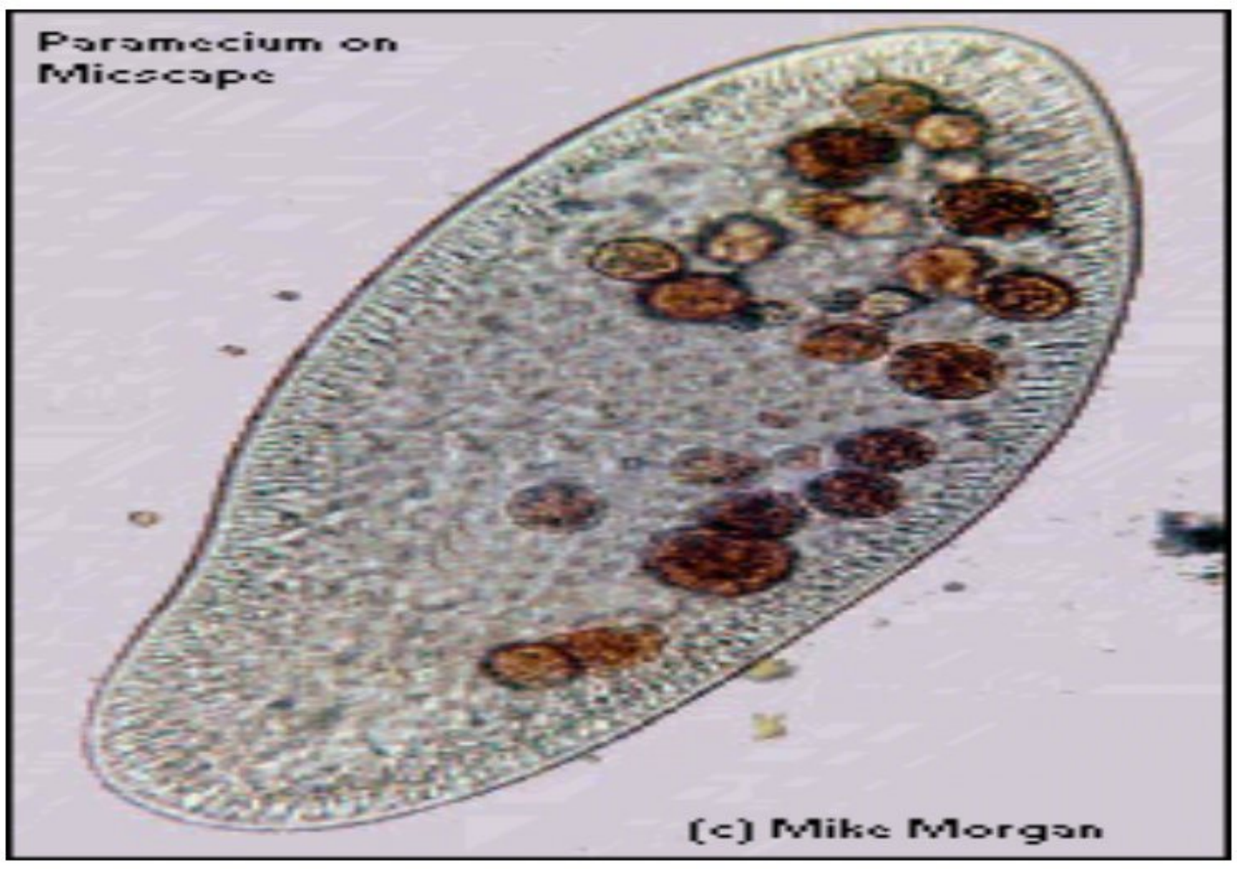
Protozoans - Animal like

Grouped by movement

- ✓ pseudopods
- ✓ cilia
- ✓ flagella
- ✓ parastites - no movement

Protozoan Characteristics

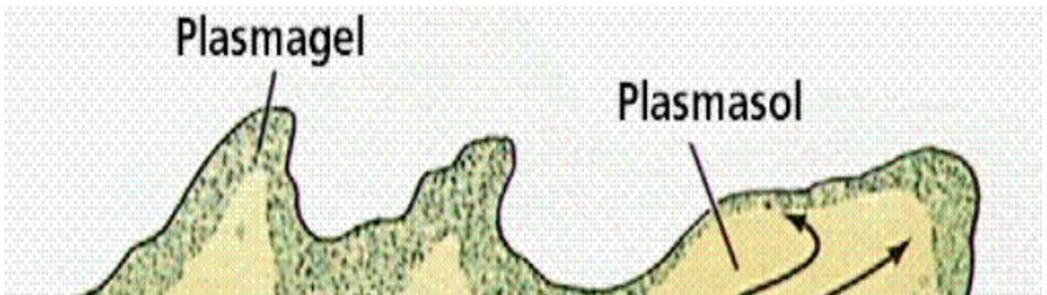
- ✓ No cell wall
- ✓ Free-living or parasitic
- ✓ Consumers
- ✓ Habitat – aquatic (ponds,lakes, oceans)
- ✓ Examples: amoeba, paramecium, dinoflagellates



Example 1: Protozoans – amoebas

- Amoebas
  - shapeless, aquatic

Pseudopods in Amoebas



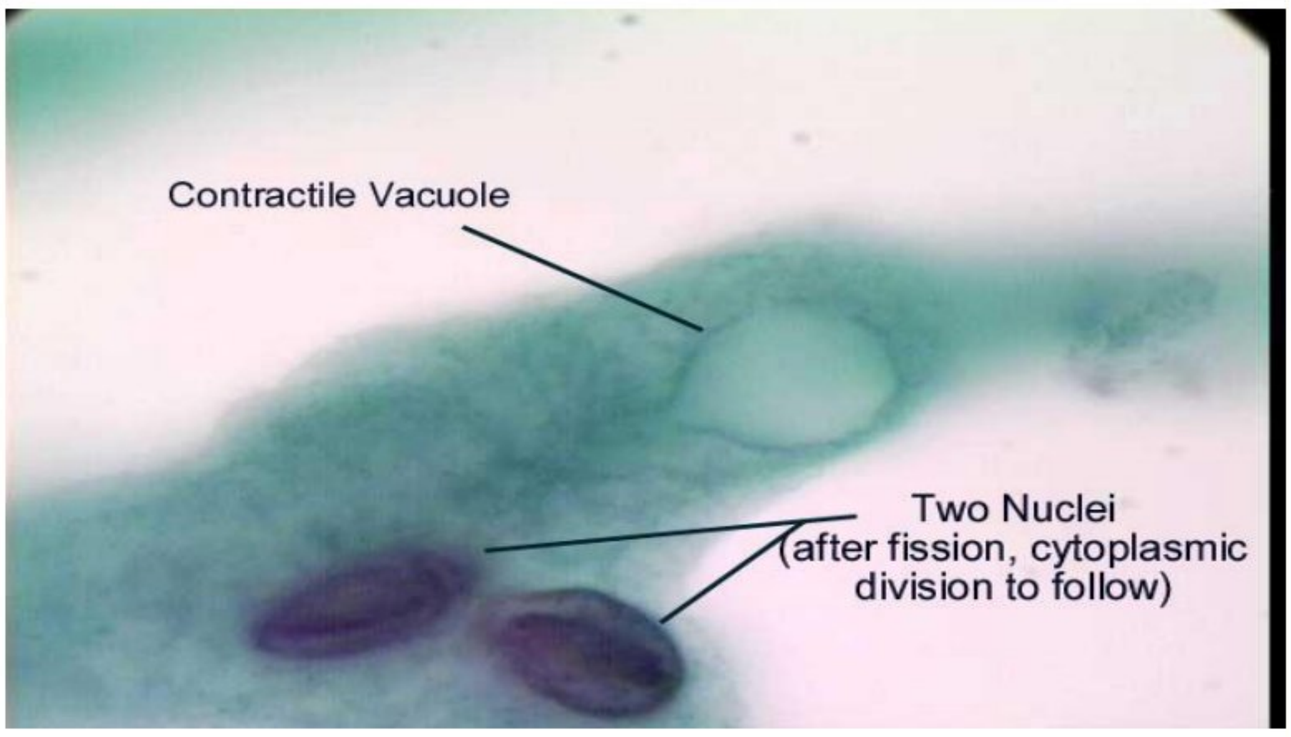
How Amoebas Eat ?

- feeding structures-

- food vacuole
- digestive enzymes
- diffusion

Getting Rid of Waste

- Contracting vacole – collect excess water from inside the cell and gets rid of it.



Reproduction of Amoeba

- asexual reproduction
  - Binary fission

93-Groups of protists

Dysentery (an illness)

- Inflammation of the large intestine
- Diarreha and vomiting
- Caused by an amoeba

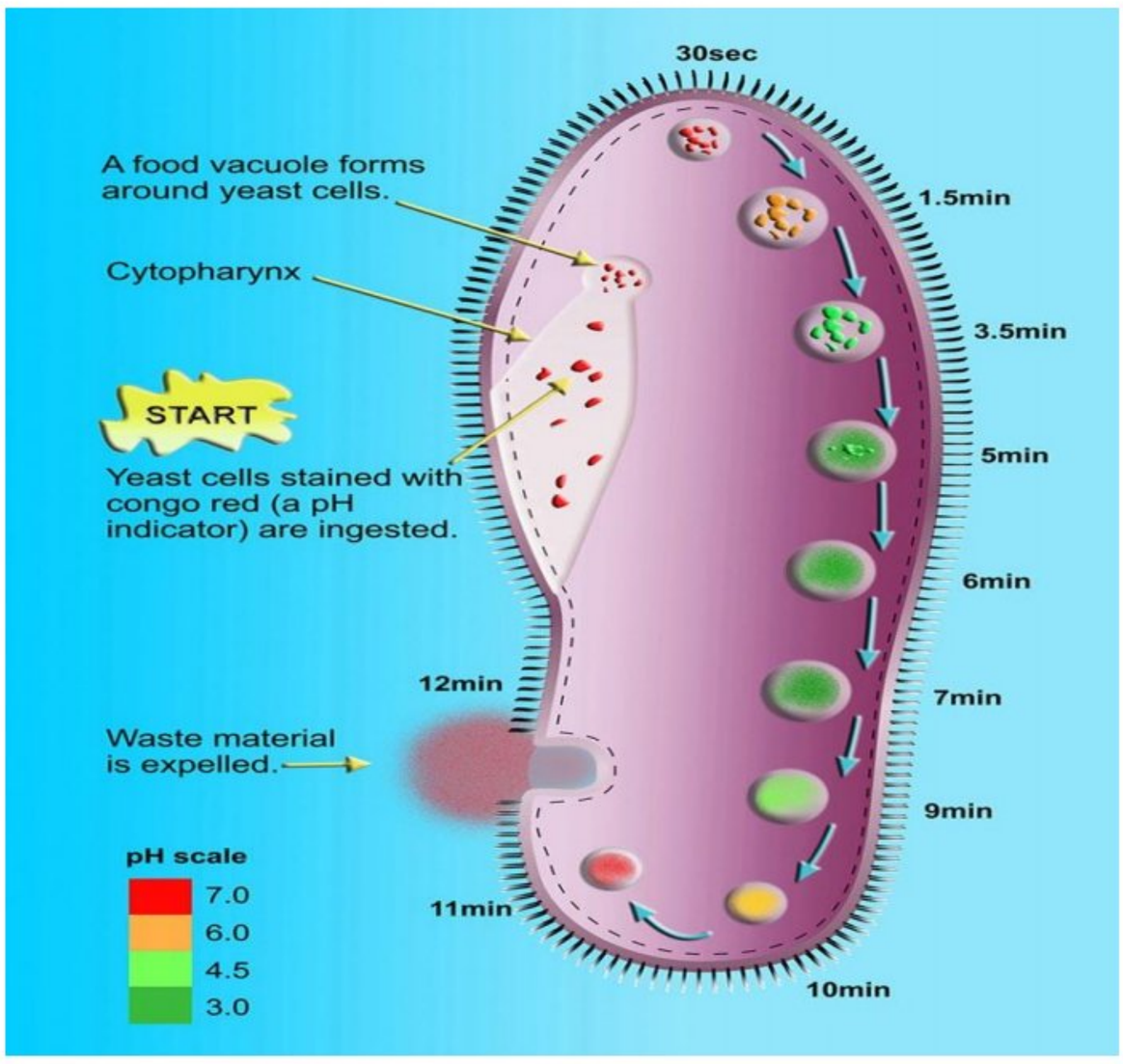
2. Example 2: Protozoans: Ciliates

- Locomotion - Cilia

1. Paracmecium - fresh water ciliate

How do Paramecium eat?

- Feed on bacteria
  - Food enters oral groove,
  - moves to the food vacoule,
  - anal pore



Paramecium Reproduction

- **Asexual**- create identical organisms
- **Conjucation** - sexual reproduction
  - ***exchange of genetic information***
  - ***Not identical - adds diversity***

3. Example 3: Protozoans - Flagellates

- Locomotion - use flagella
- Exmaples:

2. Trypanosoma - causes African Sleeping Sickness

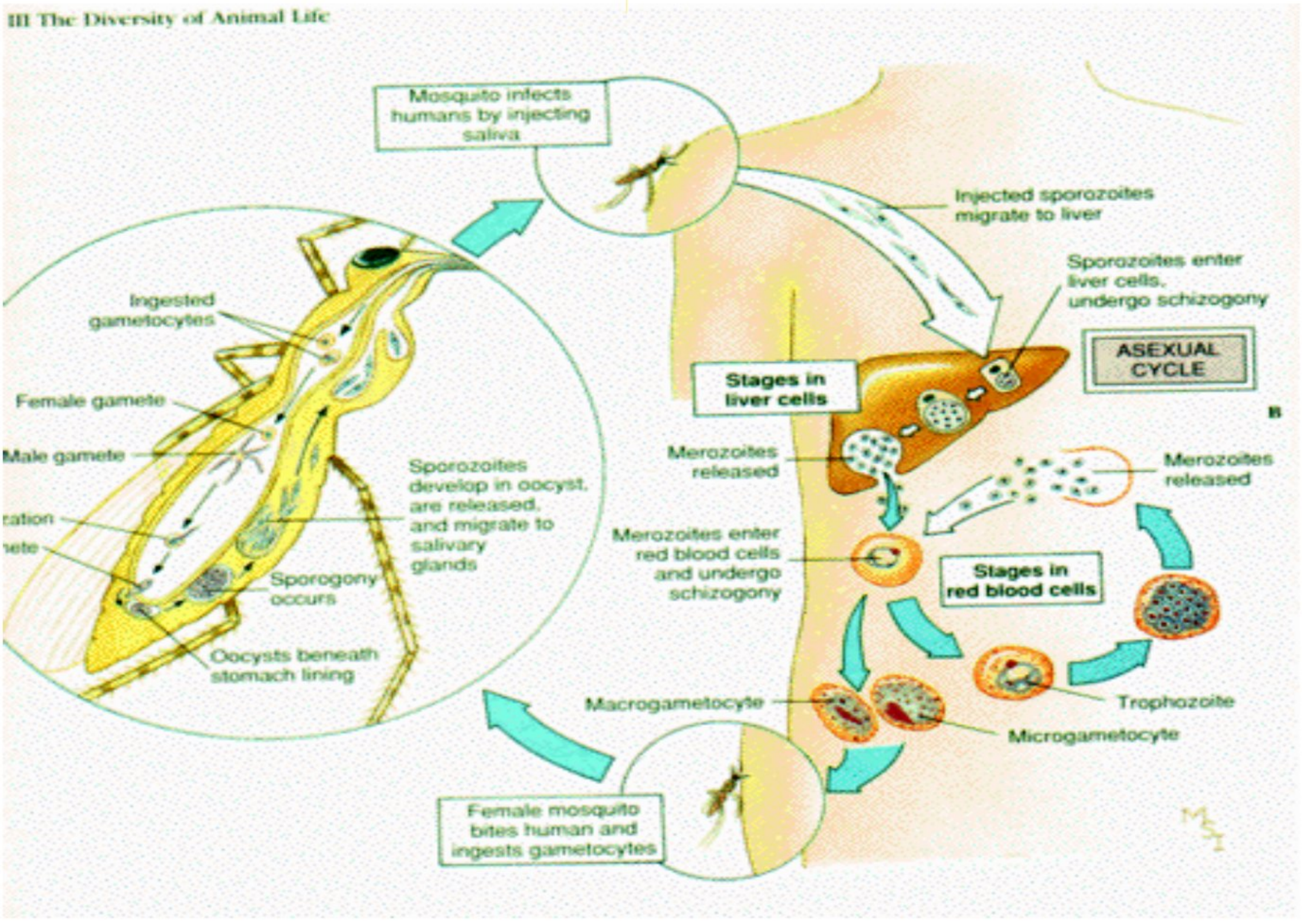
- spread by tsetse fly

4. Example 4: Protozoan - sporozoans

- Locomotion - parasitic
- reproduction
  - spores

Plasmodium - Causes Malaria

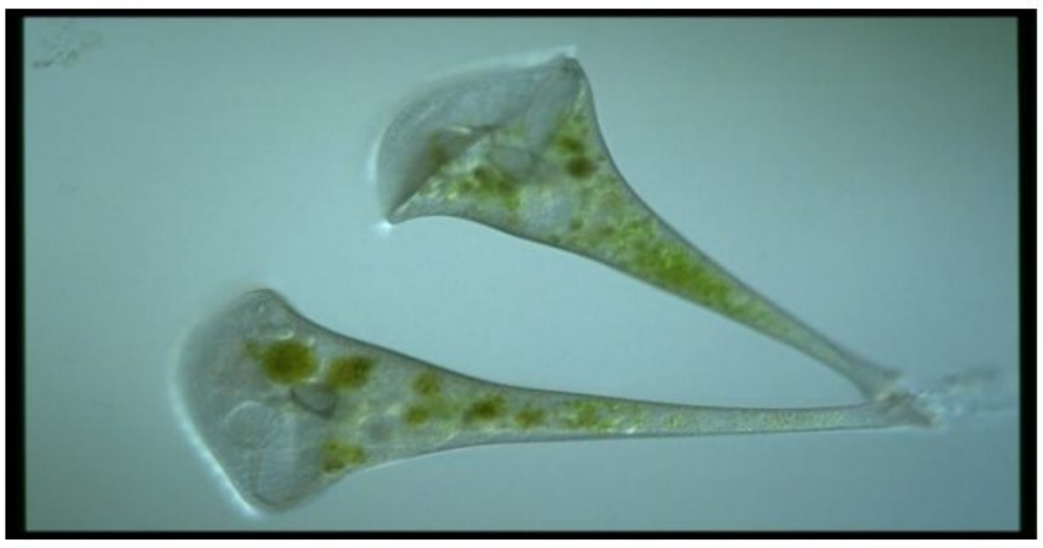
- Spread by mosquito
- Quinine – drug used to treat malaria



Vorticella

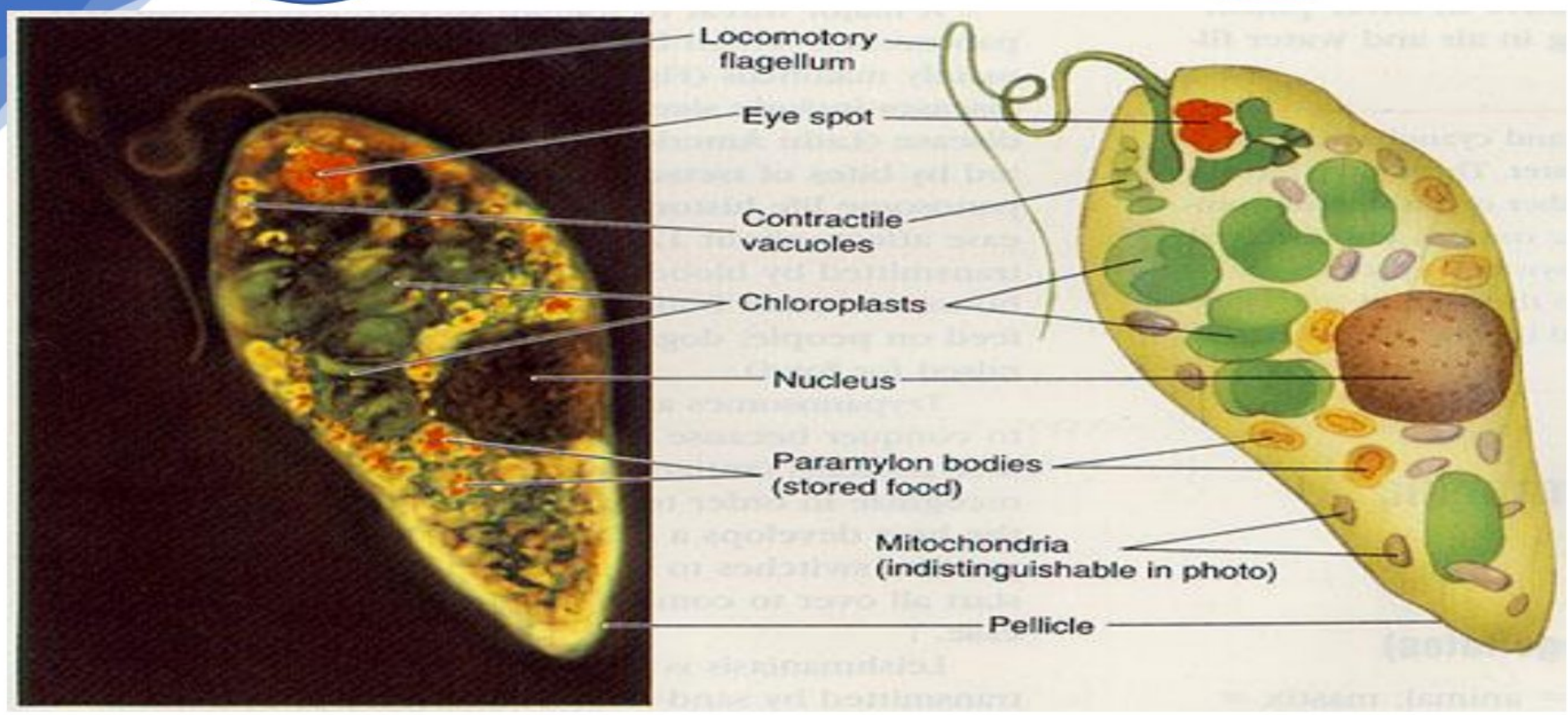
- Attaches to and spins its cilia to create a vortex (water movement"

Stentor



94-Groups of protists

- Both plant-like and animal-like



Euglenoids

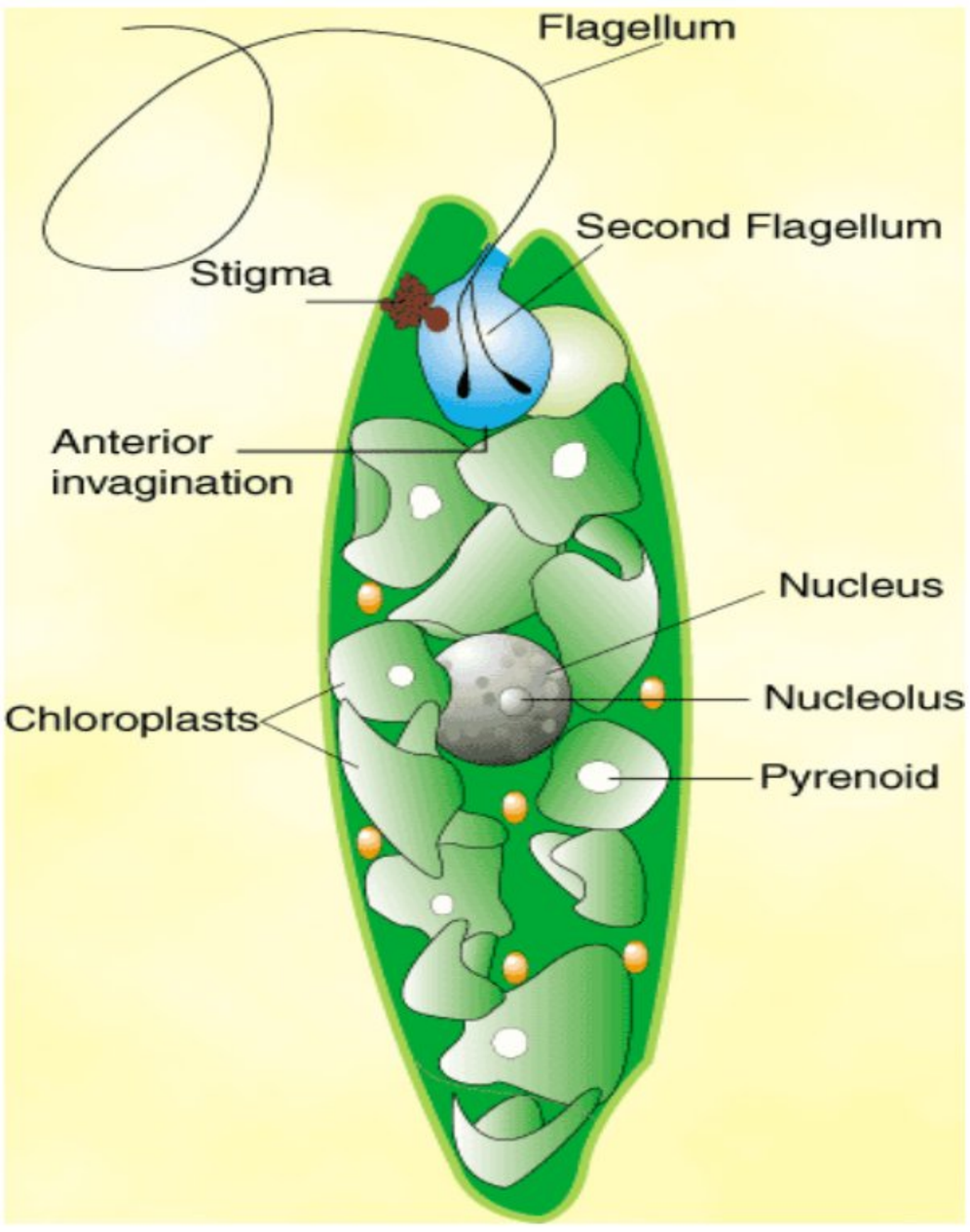
- - contain chlorophyll
- - move by flagella
- - Euglena (unique)-
  - eyespot,
  - chloroplast,
  - flagellum

Algae

Plant like protists

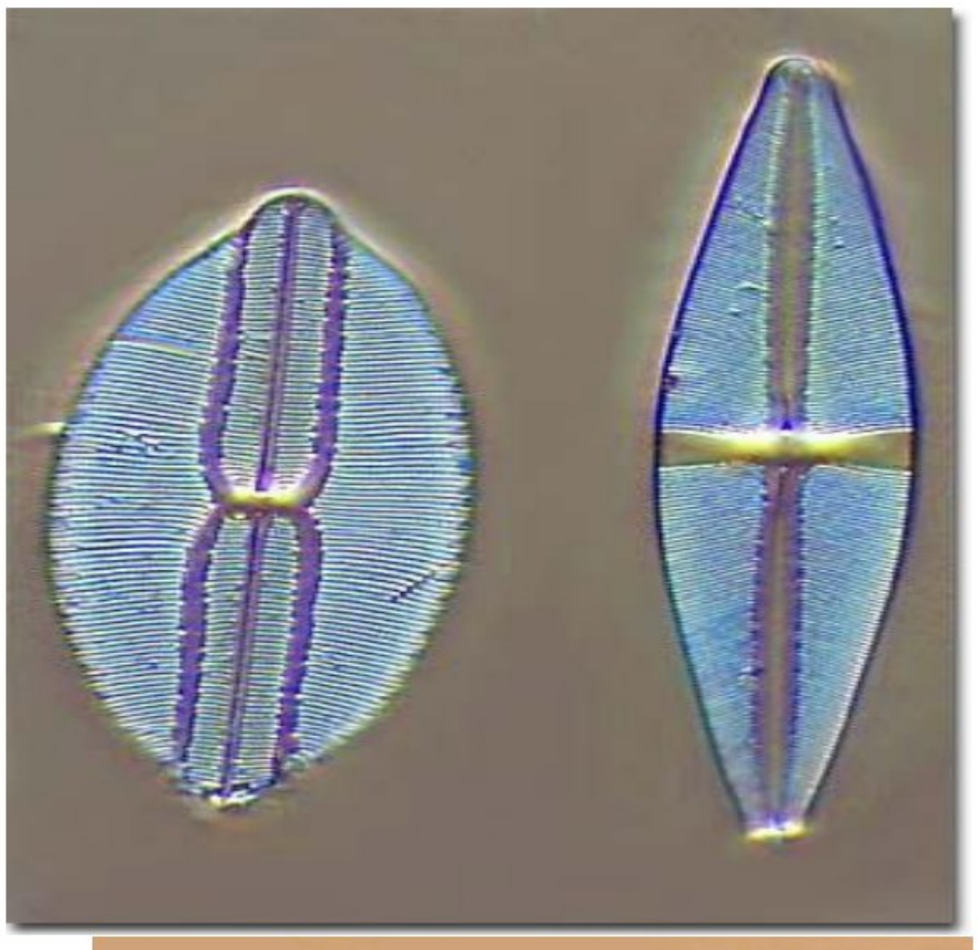
B. Algae - plant-like

- Photosynthetic
- Multi-cellular and unicellular
- Classified by pigments (Types of Algae)
  - red
  - brown
  - Green+
  - Examples; Volvox, diatoms, spirogyra



1. Diatoms

- - float in water
- - photosynthesis
- - unicellular
- - silica shells
- Commercial value: Insulating materials, Abrasives , Ceramics, Filtering

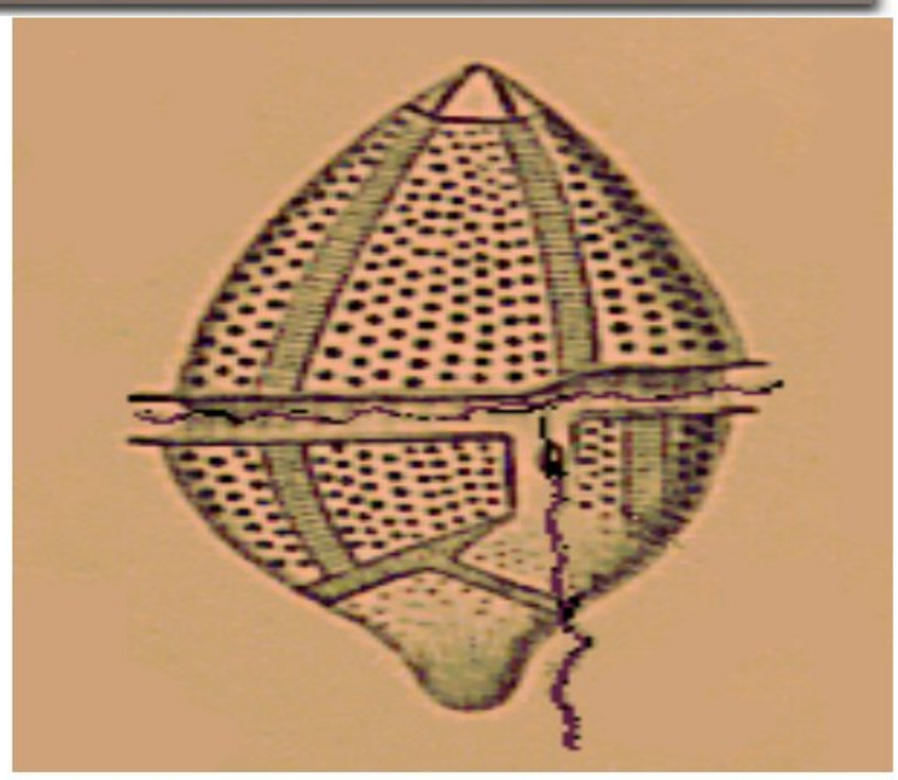


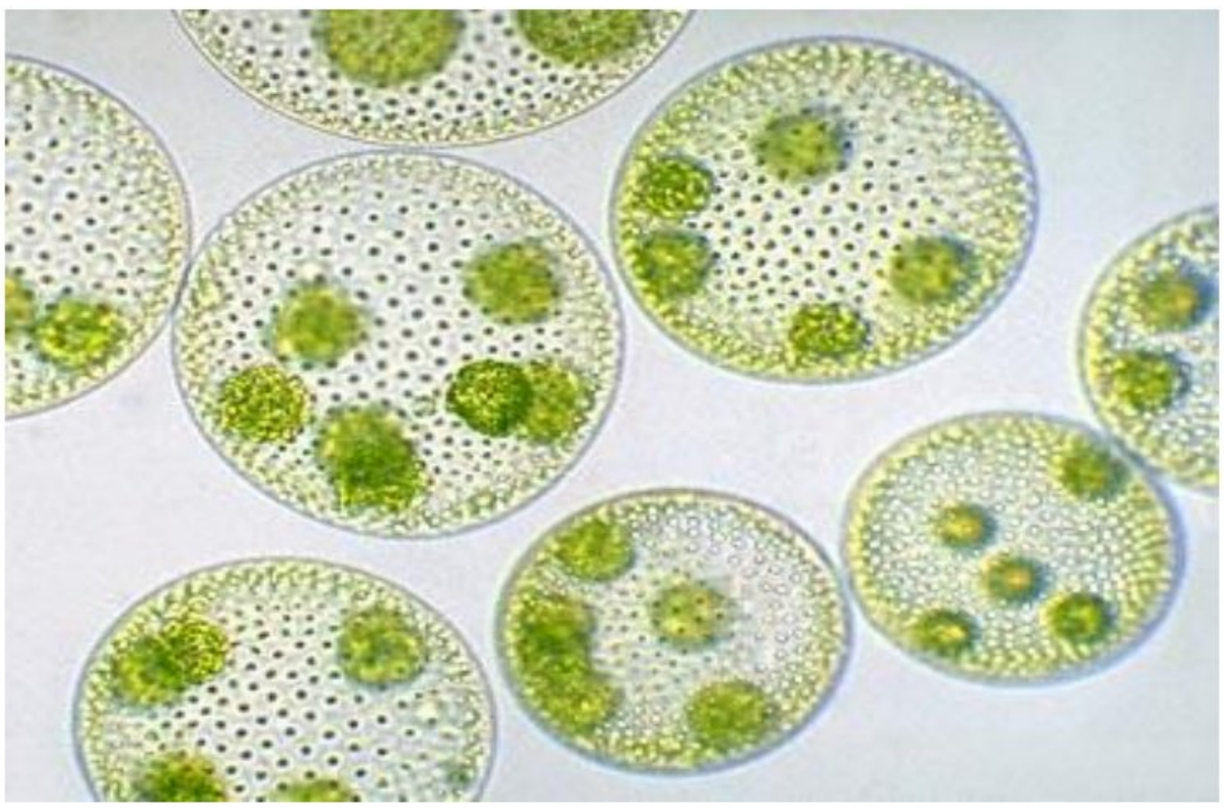
2. Dinoflagellates

- - move by two flagella
- - autotrophs
- - green glow and red tides

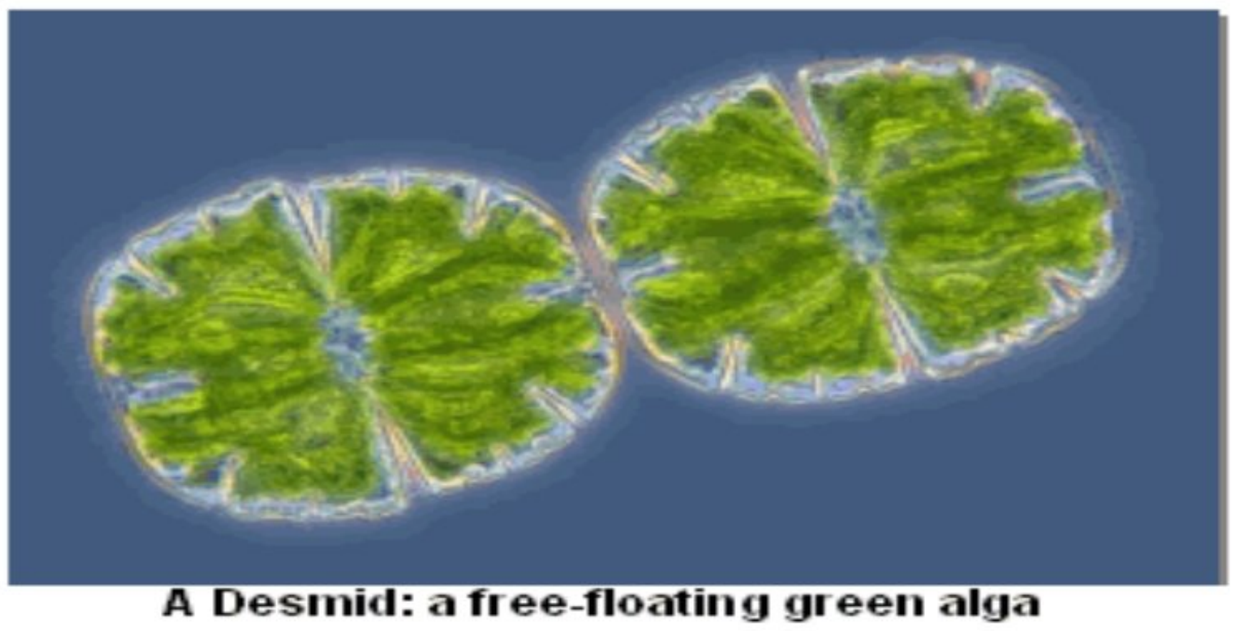
3. Green Algae (Lab 37)

- Most freshwater
- Food Source – called Plankton
- Some are:
  - Free-living
  - Colony
  - filament





Free-Floating –



A Desmid: a free-floating green alga

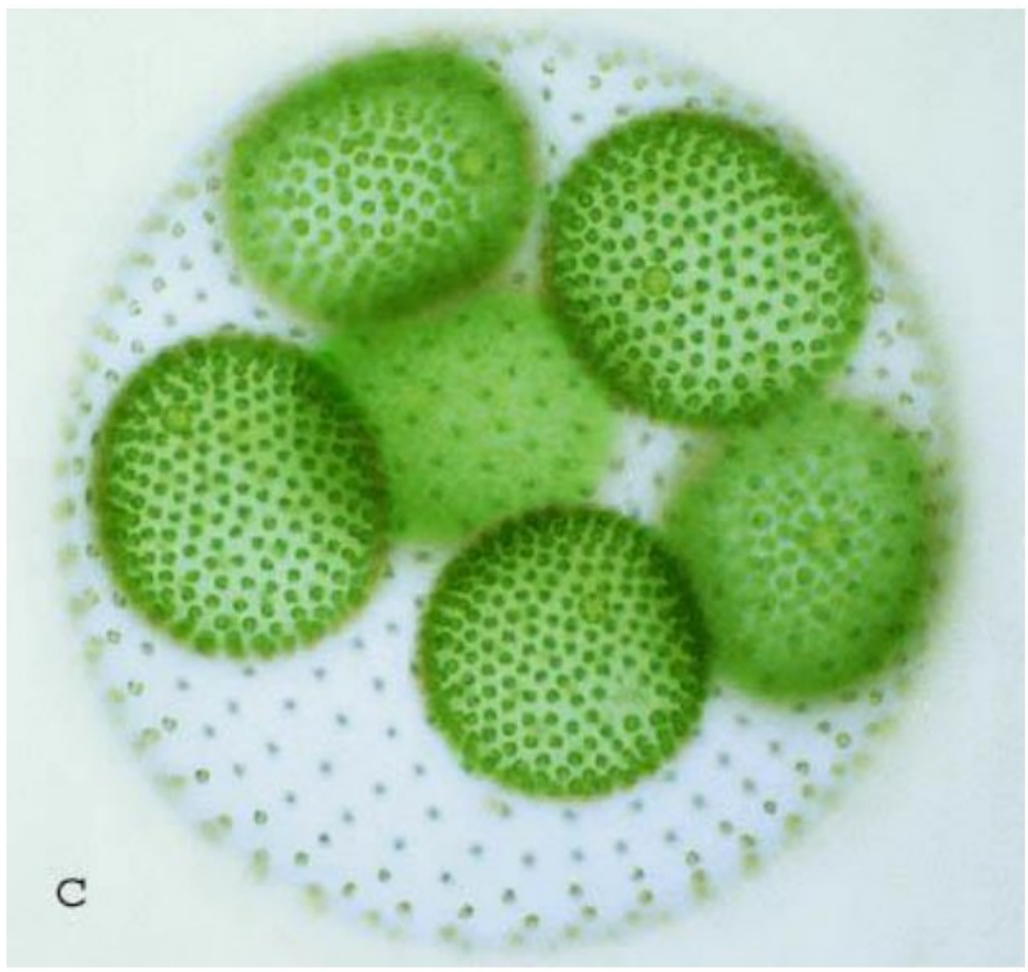
Green alga -*Pediastrum*

- Filament type – exist as a thread

Colony –

Label and Draw

- ✓ Ulothrix
- ✓ Zygnema
- ✓ Spirogyra
- ✓ Hydrodictyon
- ✓ Peridinium
- ✓ chlamydomonas



95-Features of protists

Red and Brown Algae

- ✓ All multicellular
- ✓ Marine
- ✓ Kelp – a form of brown algae
- ✓ Red algae – used in food.

Fungus Protists

Slime Molds

- - plasmodium- visible slime mass
- - moves
- - no cell membrane

- - many nuclei
- - grow on damp organic matter - decompose
- - form a stalk, release spores
- - Reproduce - asexual reproduction, making spores

Fungus Like Protists

- Classified by how they reproduce
  - slime molds
  - mold and mildew - disease causing

Mildew and Water Molds

- - fuzzy growths
- - found on bathroom tiles
- - live off of dead material
- - potato famine- 2 million people died in Ireland
- - caused immigration to the United States

Origins and Importance of Protists

- Green algae

Importance of :

- green algae - oxygen source
- food source -
  - plankton
  - zooplankton
  - phytoplankton

96-Food Webs

- A food web summarizes the feeding relations in a community.
- The earliest work on food webs concentrated on simplified communities in areas such as the Arctic islands.
- However, researchers such as Charles Elton (1927) soon found that even these so-called simple communities included very complex feeding relations.
- The level of food web complexity increased substantially, however, as researchers began to study complex communities.
- Studies of the food webs of tropical freshwater fish communities revealed highly complex networks of trophic interaction that persisted even in the face of various simplifications.
- A focus on strong interactions can simplify food web structure and identify those interactions responsible for most of the energy flow in communities.

97-Food web and structure of communities

- Manipulative studies of predaceous species have identified many keystone species, including starfish and snails in the marine intertidal zone and fish in rivers.
- On land, birds exert substantial influences on communities of their arthropod prey.

Jane Lubchenko (1978)

- Jane Lubchenko (1978) demonstrated that the influence of consumers on community structure depends upon their feeding preferences, their local population density, and the relative competitive abilities of prey species.
- Keystone species are those that, despite low biomass, exert strong effects on the structure of the communities they inhabit.

Exotic predators and the structure of food webs

- Exotic predators can collapse and simplify the structure of food webs.
- Introduced fishes have devastated the native fishes of Lake Atitlan and Gatun Lake in Central America.

Examples

- The influence of the Nile perch on the fish community of Lake Victoria is tangled with massive changes in the lake's ecosystem.
- Introduction of the Nile perch is rapidly reducing the species-rich fish fauna of Lake Victoria to a community dominated by a handful of species.

Humans

- Humans have acted as keystone species in communities. People have long manipulated food webs both as a consequence of their own feeding activities and by introducing or deleting species from existing food webs.
- In addition, many of these manipulations have been focused on keystone species.
- Hunters in tropical rain forests have been responsible for removing keystone animal species from large areas of the rain forests of Central and South America. Chinese farmers have used ants as keystone predators to control pests in citrus orchards for over 1,700 years.

98-Foundation and keystone species

- Some species have unusually strong impacts on community structure, preserving the balance of the community or even making its existence possible.
- These "special" species include foundation and keystone species.

Foundation species

- A foundation species plays a unique, essential role in creating and defining a community. Often, foundation species act by modifying the environment so that it can support the other organisms that form the community
- Kelp (brown algae) is a foundation species that forms the basis of the kelp forests off the coast of california. Kelps create environments that allow the survival of other organisms that make up the kelp forest community.
- The corals of a coral reef are another foundation species.
- The exoskeletons of living and dead coral make up most of the reef structure, which protects other species from waves and ocean currents. Beavers, which modify their environment by building dams, can also be seen as a foundation species.

Keystone species

- A **keystone species** is a species that has a disproportionately large effect on community structure relative to its biomass or abundance.
- Keystone species differ from foundation species in two main ways: they are more likely to belong to higher trophic levels (to be top predators), and they act in more diverse ways than foundation species, which tend to modify their environment.
- Different sources use different definitions for keystone and foundation species. In some sources, foundation species are viewed as a subcategory of keystone species.

99-From gene product to gene

In some cases the primary structure of a gene product (a protein) may be the starting point of gene discovery. This holds especially for proteins that can be isolated relatively easily by some marker or bioassay, or proteins that are highly induced by some experimental treatment.

- Working backwards from protein structure to gene characterization is very laborious and may only be applicable in cases where ecological functions can be associated a priori with a suspected protein
- The laborious Edman Degradation Technique has now been replaced mostly by sequencing using mass spectrometry, in which masses are estimated for peptides generated from proteolytic digests of the protein, while the fragment pattern obtained is compared with entries in a database that include sizes predicted from the genomic sequence.
- Working backwards from protein structure to gene characterization, and in some ecological applications this may be the only way to begin genomic explorations.

What Is a Gene?

- The idea of the gene has evolved through the history of genetics
- We have considered a gene as
  - A discrete unit of inheritance
  - A region of specific nucleotide sequence in a chromosome
  - A DNA sequence that codes for a specific polypeptide chain
  - In summary, a gene can be defined as a region of DNA that can be expressed to produce a final functional product, either a polypeptide or an RNA molecule

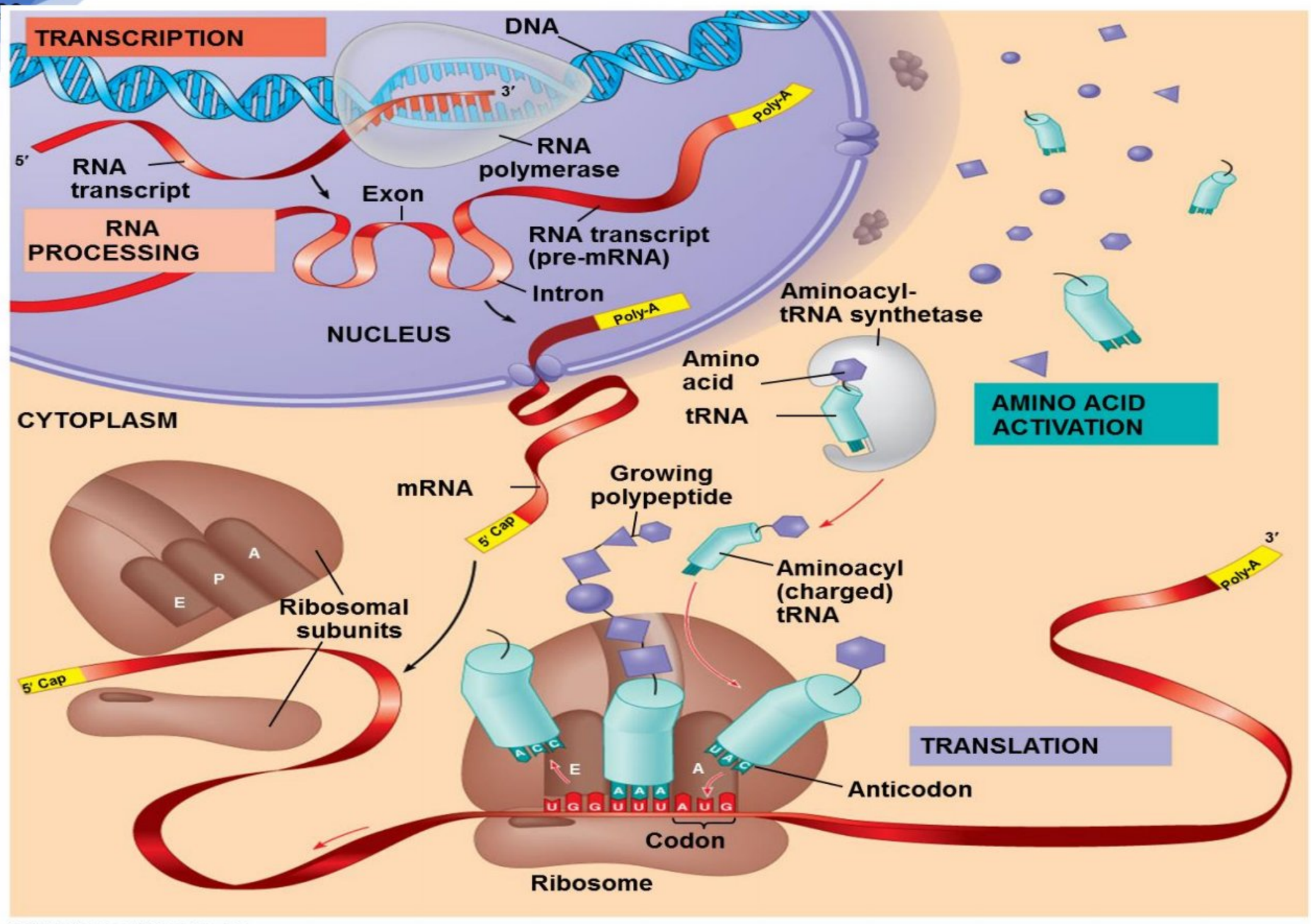


Figure 17.UN02

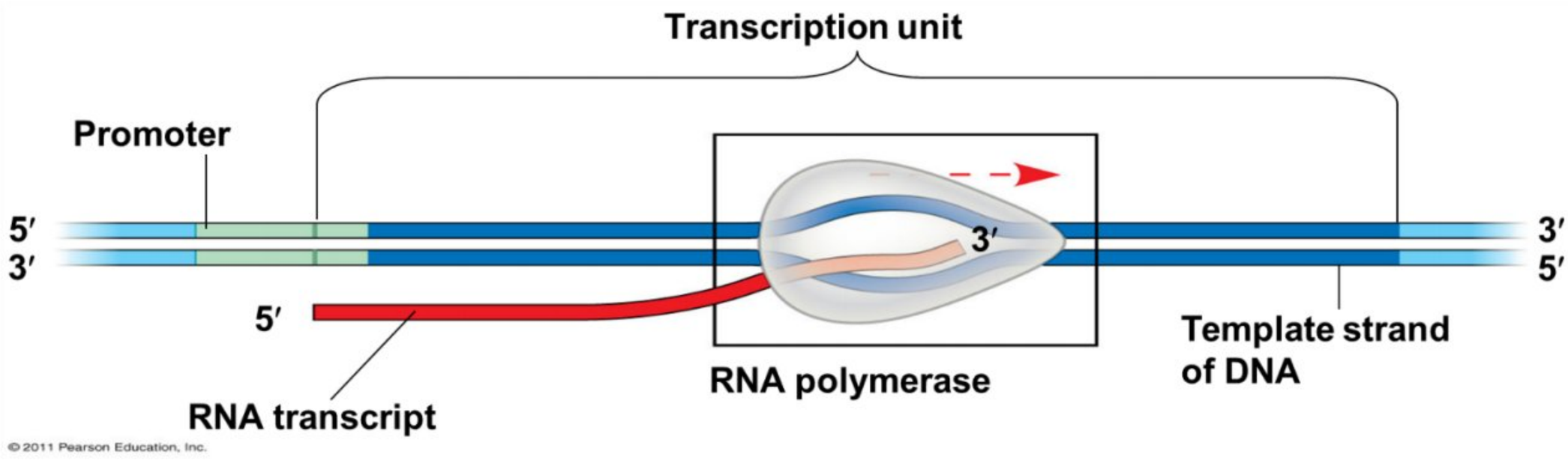
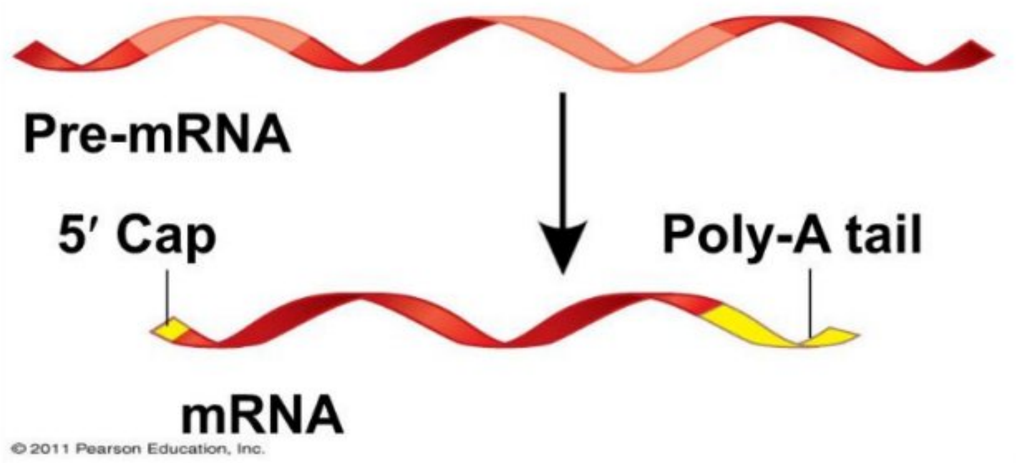
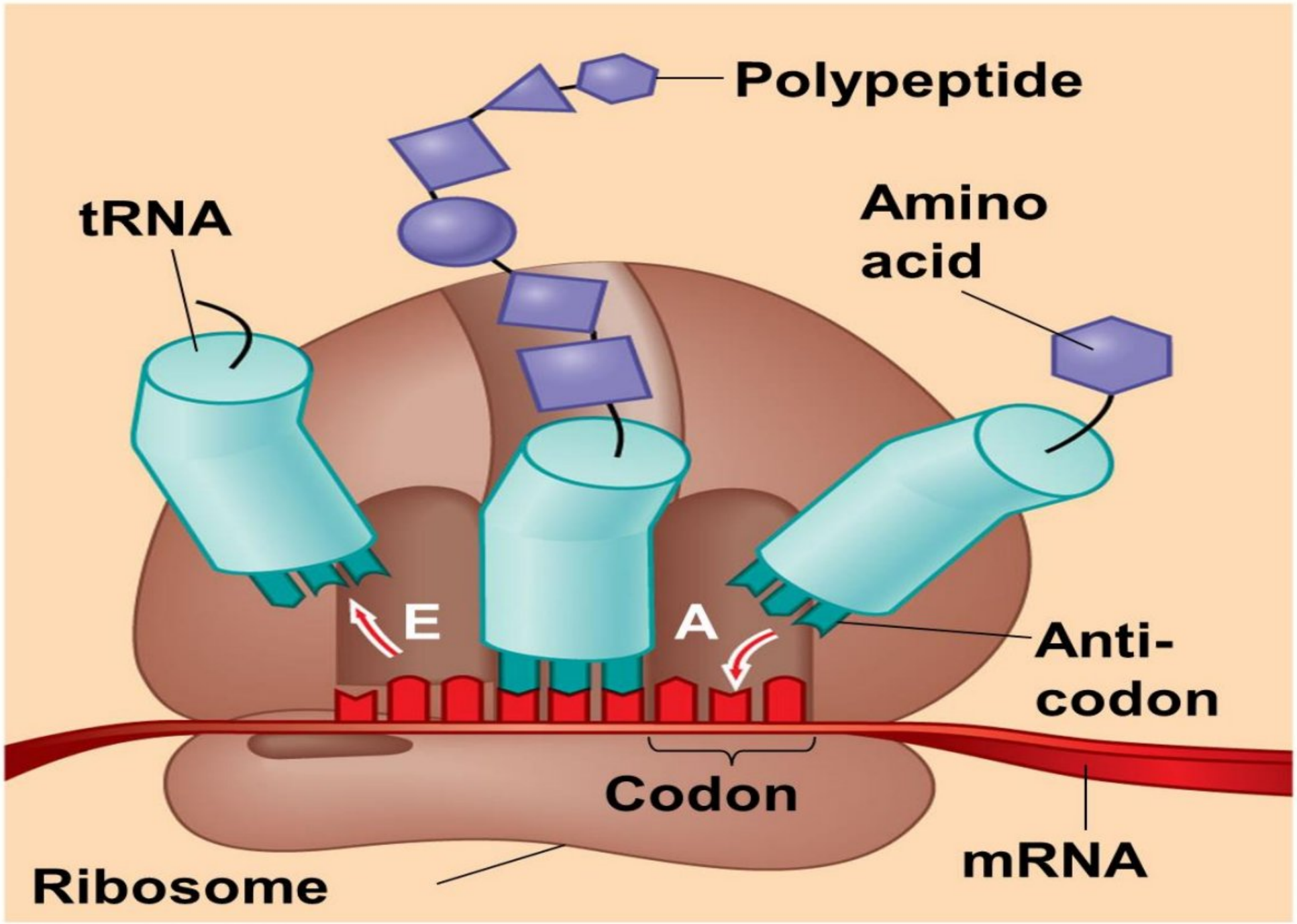


Figure 17.UN03





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Colinearity of Gene and Protein

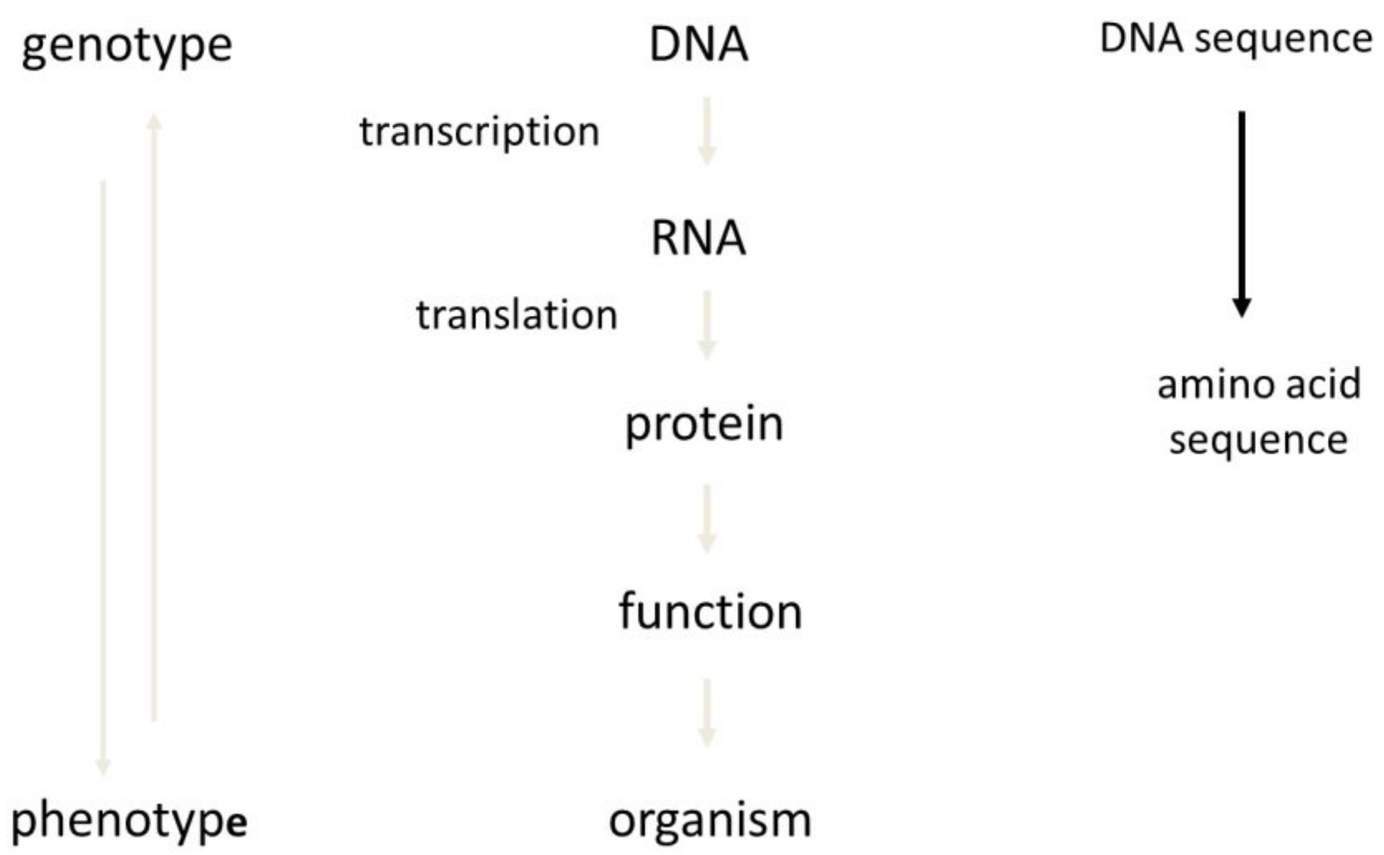

Figure 17.UN05

Type of RNA	Functions
Messenger RNA (mRNA)	
Transfer RNA (tRNA)	
	Plays catalytic (ribozyme) roles and structural roles in ribosomes
Primary transcript	
Small nuclear RNA (snRNA)	

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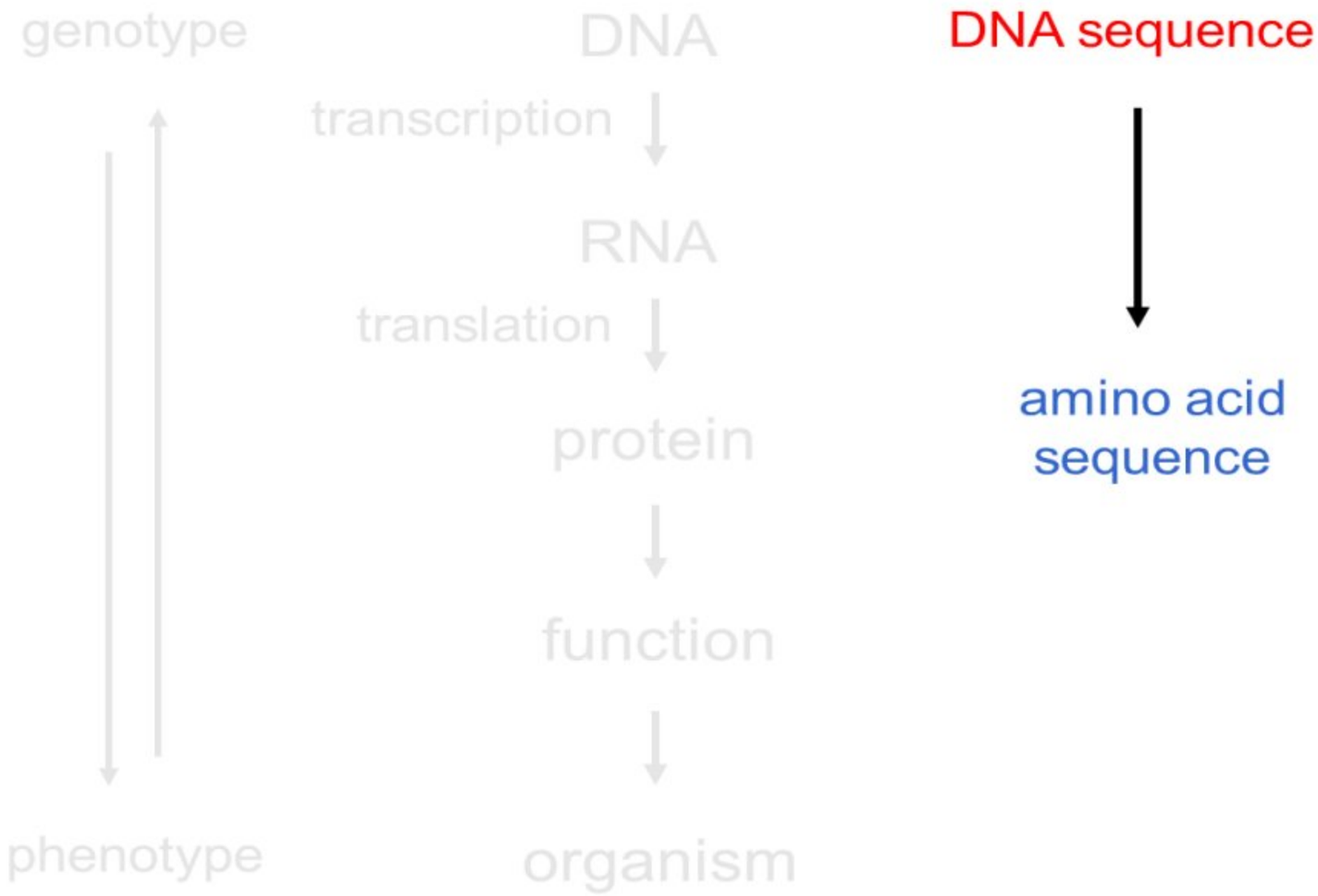
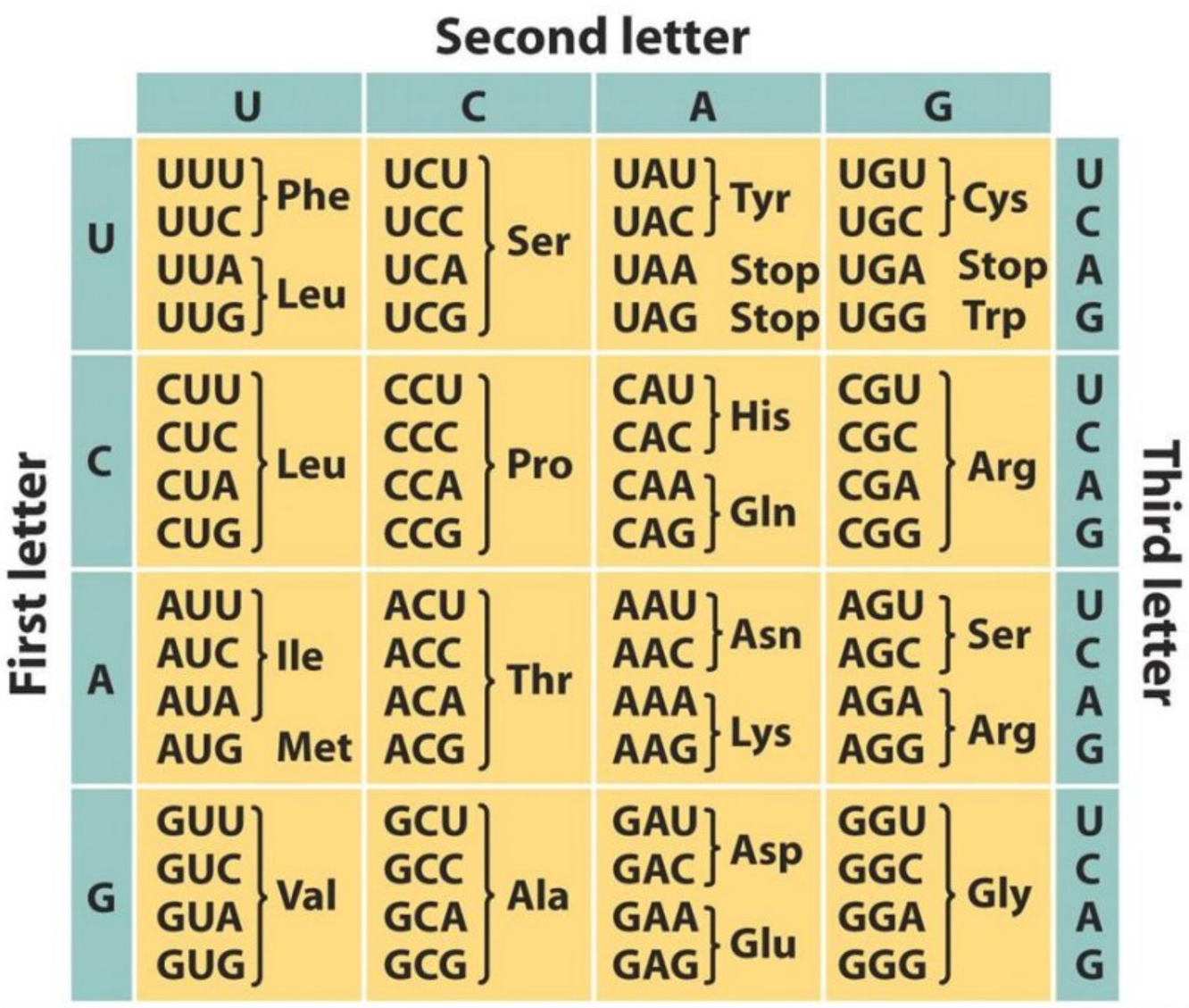
100-Colinearity of Gene and Protein

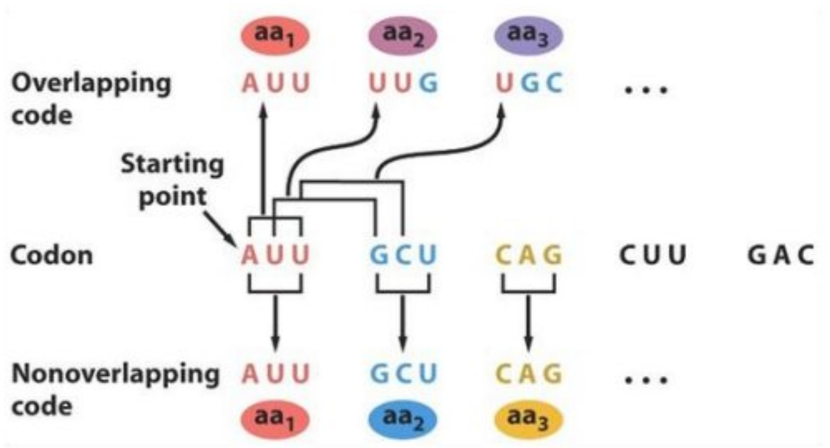
Colinearity of Gene and Protein



Genetic Code



Genetic Code is nonoverlapping.



A codon (three bases or triplet) encodes an amino acid.

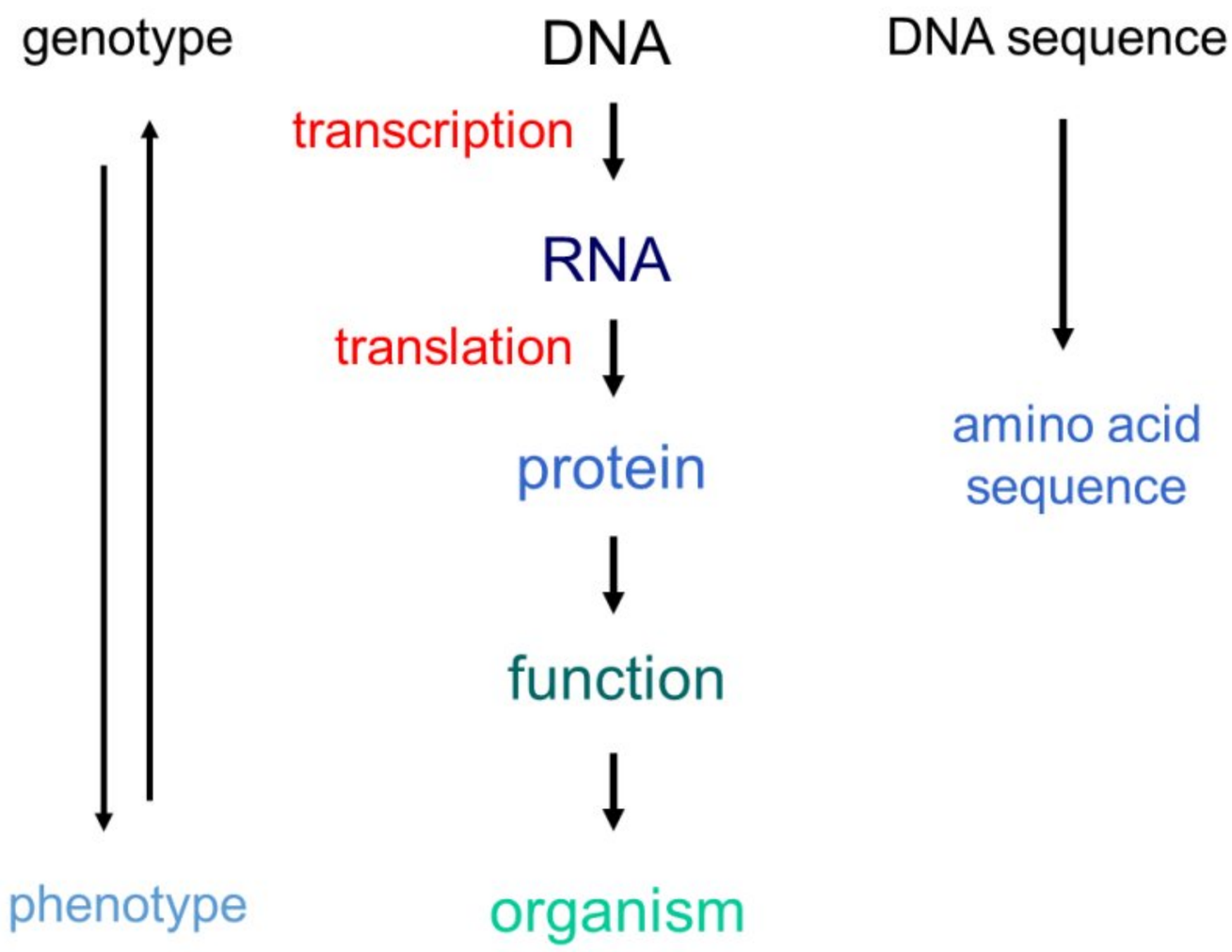
Genetic Code is read continuously from a fixed starting point.

There is a start codon (AUG).

There are three stop (termination) codons. They are often called nonsense codons.

Genetic Code is degenerate. Some amino acids are encoded by more than one codon.

Molecular Basis for
Relationship between Genotype and Phenotype



Eukaryotic RNA

Three RNA Polymerases

RNA Polymerase	Synthesis of
I	rRNA (except 5S rRNA)
II	mRNA*, some snRNA
III	tRNA, some snRNA, 5S rRNA

* eukaryotic RNA is monocistronic
prokaryotic RNA can be polycistronic